

# Locally Robust Commensurate BART for Borrowing External Controls

## Abstract

Real-world data (RWD) can supplement the control arm of a randomized controlled trial (RCT), but the comparability of external and trial controls may vary with patient characteristics. Study-level meta-analytic predictive, commensurate, and power priors apply a common borrowing parameter across patients. We propose locally robust commensurate Bayesian additive regression trees (LRC-BART) to model the RCT control surface as a pooled BART surface plus a sparse discrepancy ensemble. A spike-and-slab commensurate prior on each discrepancy leaf allows borrowing to vary across covariate regions. The spike represents commensurability and the slab serves as the vague component of a robust meta-analytic predictive prior. Tree partitions determine the regions over which each component applies. All leaf marginals are exact, so the sampler targets the exact posterior. A variance-based prior effective sample size is defined at the RCT-standardized control mean and calibrated against the RWD posterior with an explicit ceiling. We prove that the RWD-induced shift of a leaf estimate is bounded and vanishes at a Gaussian rate under conflict, that the posterior probability of a small local discrepancy is consistent under fixed partitions and representability of the discrepancy, and that the calibration is monotone and well posed. A covariate-resolved map reports posterior compatibility at each patient profile; realized regional effective sizes summarize borrowing. Simulations with Gaussian and survival outcomes and a single-arm myeloma trial with a registry control illustrate the method.

## 1 Introduction

Real-world data can supplement a randomized trial when collecting a sufficiently large control arm is difficult. Their usefulness depends on whether external outcomes represent the outcomes trial patients would have experienced under control. Trial selection, treatment practice and outcome ascertainment may vary with age, disease stage and prior therapy. External and trial controls may therefore be comparable for some patient profiles and differ substantially for others.

A common borrowing parameter may overborrow where sources differ or discard useful information where they agree. We propose a Bayesian tree model with borrowing that varies over learned covariate regions. We accompany the treatment-effect estimate with profile-specific summaries of compatibility and external information.

Established borrowing priors act on a study-level parameter. The MAP prior of Neuenschwander

et al. [17] and Schmidli et al. [23] treats historical and current control parameters as exchangeable and uses the current parameter’s posterior predictive distribution as its prior. Robust MAP mixes it with a vague component to accommodate conflict. The commensurate prior of Hobbs et al. [9, 10],  $\theta | \theta_0 \sim \mathcal{N}(\theta_0, 1/\tau)$ , learns the precision  $\tau$  under a spike-and-slab hyperprior. The power prior of Ibrahim and Chen [11] discounts the historical likelihood by  $a_0$ ; the normalizations of Duan et al. [7] and Neuenschwander et al. [16] allow a prior on  $a_0$ . With a single historical study, the normal MAP, commensurate and power-prior forms nearly coincide [21, 22]. Their common heterogeneity or discount scale does not depend on patient covariates.

Locally robust commensurate BART (LRC-BART) models both control sources with a pooled Bayesian additive regression tree (BART) surface  $f$  and a sparse discrepancy ensemble  $g$ . The RWD follow  $f$  and RCT controls follow  $f + g$ . Each leaf of  $g$  has a small-scale commensurate spike and an outcome-scale vague slab. Its posterior state allows local agreement or a discrepancy over regions learned by the tree partitions.

The decomposition follows Bayesian causal forests [8], using source membership in place of treatment. The leaf prior combines commensurate shrinkage [9] with a vague component [23]; its relation to single-study MAP [22] is shared with those established priors. Calibration averages variance-based precision ratios, motivated by the normal-prior ESS of Neuenschwander et al. [18].

The leaf mixture lets borrowing vary over learned regions, with separate priors for the external mean and discrepancy (Web Appendix G.6). Closed-form leaf marginals permit exact tree updates. We calibrate a variance-based ESS at the RCT-standardized control mean, with targets stated relative to its RWD-only ceiling. The conditional theory establishes bounded leaf influence that vanishes at a Gaussian rate under conflict, map consistency under fixed partitions and representability, and monotone calibration. The posterior of  $g$  supplies a profile-specific compatibility probability alongside information summaries. Accelerated failure time (AFT) data augmentation accommodates right censoring; a single-arm variant represents uncertainty about an unobserved trial control surface.

## 1.1 Related work

A second group of methods lets the amount of borrowing vary with covariates, through propensity-score strata [4, 25], a commensurate precision that depends on the covariates through kernel local regression [13], exchangeability over sources or over latent patient classes [1, 14, 15], a global

congruence statistic [12], or clustering of the patients on the covariates before the outcome model is fitted [2, 3]; Web Appendix G.0 reviews them, and the closest kernel-local relative of our borrowing map is the covariate-adaptive historical control borrowing of Jin et al. [13].

Among tree methods, Zhou and Ji [27] is the most directly related approach. They fit one BART ensemble [5] to RCT and external controls with the data source as a splitting covariate, so that trees pool the sources wherever they do not split on the source; borrowing is implicit and has no parameter, and their discussion proposes as future work a prior discouraging source splits together with a source offset whose scale controls borrowing. We model the proposed offset nonparametrically and give it a calibratable prior. Pan et al. [19] apply BART to patient-level borrowing across subgroups. HE-BART [26] places group-specific random effects inside each terminal node with a shared precision, the structural precedent for a hierarchical parameter inside a leaf, with the aim of mixed-effects prediction rather than controlling historical borrowing for a current study. flexBART [6] lets a tree assign several levels of a categorical predictor to one branch, which with the source as predictor gives the implicit borrowing of Zhou and Ji [27] in a richer split space. None of these places a commensurate prior on a leaf, and among the covariate-adaptive methods only Jin et al. [13] and Alt et al. [1] report borrowing at the resolution of a patient profile, with CAHB using nonparametric kernel estimates. LRC-BART instead uses joint BART surfaces and spike-and-slab discrepancy leaves.

Section 2 develops the method, Section 3 evaluates it in simulations, Section 4 illustrates a single-arm myeloma comparison, and Section 5 discusses limitations. Proofs are in the Web Appendix.

## 2 Methods

### 2.1 Setting and estimands

We observe an RCT with  $N$  subjects, of whom  $n_1$  are controls and  $N - n_1$  are treated, and an external source of  $n_2$  RWD controls. For subject  $i$  we record the outcome  $y_i$ , the covariate vector  $x_i \in \mathbb{R}^p$ , and the source  $g_i \in \{1, 2\}$ , with  $g_i = 1$  for an RCT control and  $g_i = 2$  for an RWD control;  $D_i = \mathbf{1}\{g_i = 1\}$  indicates an RCT control. Continuous outcomes are modelled as Gaussian given the covariates. Survival outcomes are modelled on the log scale under a log-normal AFT model

with right censoring, and  $y_i$  then denotes the log survival time, observed or imputed as described in Section 2.3.

The estimands are standardized to the RCT covariate distribution. Let  $f_T(x)$  be the mean outcome under treatment and  $f_1(x)$  the mean outcome of an RCT control at covariate  $x$ . For Gaussian outcomes the average treatment effect is  $\text{ATE} = N^{-1} \sum_{i \leq N} \{f_T(x_i) - f_1(x_i)\}$ , the average over all  $N$  RCT profiles, treated and control. For survival outcomes we report the ratio of population median survival times  $q_{0.5}(\bar{S}_T)/q_{0.5}(\bar{S}_1)$ , where  $\bar{S}_a(t) = N^{-1} \sum_i S_a(t | x_i)$  is the arm- $a$  survival function standardized to the RCT profiles, and the ratio of restricted mean survival times  $\int_0^{t^*} \bar{S}_T(t) dt / \int_0^{t^*} \bar{S}_1(t) dt$  at a horizon  $t^*$  inside the follow-up. The treatment surface  $f_T$  is estimated from the RCT treated subjects alone with standard BART or AFT-BART. External observations inform the control surface  $f_1$ . We define effective sample size through the RCT-standardized control mean  $\mu = N^{-1} \sum_{i \leq N} f_1(x_i)$ .

Both ensembles below are BART models [5]. A BART surface is a sum of  $H$  regression trees,  $\sum_{h=1}^H m(x; T_h, \theta_h)$ , where  $T_h$  is a binary tree whose internal nodes split on one covariate at one cutpoint and whose terminal nodes (leaves)  $\ell = 1, \dots, L_h$  carry means  $\theta_{h\ell}$ , and  $m(x; T_h, \theta_h)$  returns the mean of the leaf to which  $x$  is routed. The tree prior splits a node at depth  $d$  with probability  $\alpha(1+d)^{-\beta}$ , chooses the splitting variable and cutpoint uniformly, and the leaf means are independent  $\mathcal{N}(0, \lambda^2)$  with  $\lambda = (\max y - \min y)/(2k\sqrt{H})$  and  $k = 2$ , the range rule, so that the sum of  $H$  trees spans the outcome range with high prior probability.

## 2.2 The model

The two control sources are modelled jointly as

$$y_i = f(x_i) + D_i g(x_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_{g_i}^2), \quad (1)$$

Here  $f(x)$  is the external-control response mean and  $f_1(x) = f(x) + g(x)$  is the RCT-control response mean, with a separate residual variance for each source. A profile  $x$  is a vector of patient characteristics. The surfaces describe conditional means at that profile; individual outcomes also contain the residual variation in (1). For survival outcomes these are means of log survival time. Both surfaces are estimated jointly.

We write the pooled BART surface explicitly as

$$f(x) = \sum_{h=1}^{H_f} \mu_{h, \ell_h^f(x)}^f, \quad \mu_{h\ell}^f | T_h^f \stackrel{\text{indep}}{\sim} \mathcal{N}(0, \lambda_f^2), \quad \lambda_f = \frac{\max y - \min y}{2k\sqrt{H_f}}, \quad k = 2,$$

where  $T_h^f$  is the  $h$ th tree,  $\ell_h^f(x)$  is its leaf containing  $x$ , and  $\mu_{h\ell}^f$  is that leaf's contribution to the mean. The trees have independent split priors with  $(\alpha_f, \beta_f) = (0.95, 2)$ . We use  $H_f = 50$  in the simulations; the application uses  $H_f = 10$ . Observations sharing leaves contribute to estimating  $f(x)$ , so its precision depends on information across profiles.

The discrepancy surface  $g$  has  $H_g = 5$  trees with split prior  $(\alpha_g, \beta_g) = (0.5, 3)$ : a tree is a stump a priori with probability one half and rarely deeper than two levels. Let  $\theta_{h\ell}$  denote the constant discrepancy contribution in leaf  $\ell$  of tree  $h$ . The leaf contributions have the independent mixture prior

$$\theta_{h\ell} | w, \tau_0^2, \tau_1^2 \stackrel{\text{iid}}{\sim} w \mathcal{N}(0, \tau_0^2) + (1 - w) \mathcal{N}(0, \tau_1^2). \quad (2)$$

Equivalently, let  $z_{h\ell} | w \stackrel{\text{iid}}{\sim} \text{Bernoulli}(w)$ . Given these indicators and the scales, the leaf contributions are independent normals with mean zero and variance  $\tau_{1-z_{h\ell}}^2$ :  $z = 1$  selects the narrow spike  $\tau_0^2$  and  $z = 0$  the wider slab  $\tau_1^2$ .

Let  $T_g$  denote the discrepancy partitions and  $\ell_h(x)$  the leaf reached by  $x$  in tree  $h$ , so that  $g(x) = \sum_{h=1}^{H_g} \theta_{h, \ell_h(x)}$ . At a fixed  $x$ , summing these contributions and integrating their indicators induces the mixture prior

$$f_1(x) | f, T_g, w, \tau_0^2, \tau_1^2 \sim \sum_{m=0}^{H_g} \binom{H_g}{m} w^m (1 - w)^{H_g - m} \mathcal{N}(f(x), m\tau_0^2 + (H_g - m)\tau_1^2),$$

where  $m$  is the number of reached leaves in the spike. The commensurate component  $m = H_g$  has variance  $H_g\tau_0^2$ ; the all-slab component  $m = 0$  has variance  $H_g\tau_1^2$ . Switching one reached leaf from  $z = 1$  to  $z = 0$  increases the component variance by  $\tau_1^2 - \tau_0^2$ , permitting a larger local departure. Every component remains centered at the external mean  $f(x)$ . This prior concerns the latent control mean; observation variance enters through (1).

We set  $w \sim \text{Beta}(1, 1)$  and  $\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2)$  with  $\nu_0 = 3$ . The range rule fixes  $\tau_1^2 = (\max y - \min y)^2 / (4k^2 H_g)$ . We require  $\tau_0^2 < \tau_1^2$  (Assumption A1 of the Web Appendix); the

sampler draws  $\tau_0^2$  from its conditional truncated to  $(0, \tau_1^2)$ . The residual variances carry independent inverse-gamma priors. Section 2.4 calibrates the spike scale  $\tau_0$  through the full ensemble’s effective sample size.

Under an untruncated scale prior, integrating  $z$  and  $\tau_0^2$  gives the leaf marginal  $w t_{\nu_0}(\theta; 0, s_0^2) + (1 - w)\mathcal{N}(\theta; 0, \tau_1^2)$ . The sampler uses the truncated scale prior, whose spike marginal is not exactly Student- $t$ .

Three special cases place the model among its comparators. With  $w = 1$  every leaf is in the spike and the model is a commensurate BART (C-BART) with a single commensurability scale  $\tau_0^2$ ; with  $w = 1$  and  $\tau_0^2 \rightarrow 0$  the discrepancy vanishes and the model is BART fitted to the pooled controls (BART-CP); with  $w = 0$  every leaf is in the slab and  $g$  is a free nonparametric source offset regularized by the range rule, the tree analogue of BART with the source indicator as a covariate (BART-PP). LRC-BART with  $w$  learned from the data moves between these per leaf.

Separate ensembles allow the complexity of the external mean and that of the source discrepancy to be controlled independently. A small  $g$  ensemble with shallow trees expresses our prior preference for discrepancies that are simpler than the outcome surface. The fixed-partition calculation in Web Appendix G.6 describes how the number of discrepancy contributions affects their prior cost; it does not establish a detection threshold for the fitted model or rule out regional borrowing with shared leaves. At the population level,  $f$  and  $f + g$  are identified on their respective source supports, and their difference is identified on the intersection. Outside that intersection, decomposition or prediction requires extrapolation. A  $g$  leaf with no RCT controls has its prior conditional on the partition and current hyperparameters, but shared leaves and learned hyperparameters can transmit trial information beyond trial support (Web Appendix E).

### 2.3 Posterior computation

We use Markov chain Monte Carlo with the backfitting scheme of Chipman et al. [5], one tree at a time with the others held fixed. The tree-move factors are closed form: conditional on  $(\tau_0^2, \tau_1^2, w)$  the leaf prior (2) is a finite Gaussian mixture. For a leaf of an  $f$  tree the partial residuals  $r_i = y_i - f_{-h}(x_i) - D_i g(x_i)$  come from both sources with their own variances; with  $A = n_1/\sigma_1^2 + n_2/\sigma_2^2$  and  $B = S_1/\sigma_1^2 + S_2/\sigma_2^2$ , where  $n_g$  and  $S_g$  are the count and sum of the source- $g$  residuals in the leaf,

the observation-level factors  $\prod_i \phi(r_i; 0, \sigma_{g_i}^2)$  cancel from every tree-move ratio and the leaf factor is

$$M_f = (1 + \lambda_f^2 A)^{-1/2} \exp\left\{\frac{\lambda_f^2 B^2}{2(1 + \lambda_f^2 A)}\right\}, \quad (3)$$

the heteroscedastic BART marginal. A  $g$  leaf uses residuals  $r_i = y_i - f(x_i) - g_{-h}(x_i)$  from RCT controls in that leaf, with count  $n_1$  and mean  $\bar{y}_1$ , and writing  $v_1 = \sigma_1^2/n_1$ ,

$$M_g = \frac{w \phi(\bar{y}_1; 0, \tau_0^2 + v_1) + (1 - w) \phi(\bar{y}_1; 0, \tau_1^2 + v_1)}{\phi(\bar{y}_1; 0, v_1)}, \quad (4)$$

with  $M_g = 1$  for a leaf with no RCT controls. Grow, prune and change moves multiply the tree-prior and proposal ratios by the affected leaf factors; a grow move contributes  $M_L M_R / M_P$ . Each collapsed tree move is followed immediately by conditional leaf and indicator draws. This ordering preserves the joint posterior (Lemma A1 and Proposition A1 of Web Appendix A).

Conditional on the trees, we update the leaf parameters conjugately. An  $f$  leaf draws  $\mu_{hl}^f \sim \mathcal{N}(\lambda_f^2 B / (1 + \lambda_f^2 A), \lambda_f^2 / (1 + \lambda_f^2 A))$ . A  $g$  leaf is updated as a block: first

$$P(z = 1 \mid \bar{y}_1) = \frac{w \phi(\bar{y}_1; 0, \tau_0^2 + v_1)}{w \phi(\bar{y}_1; 0, \tau_0^2 + v_1) + (1 - w) \phi(\bar{y}_1; 0, \tau_1^2 + v_1)}, \quad (5)$$

then  $\theta \mid z \sim \mathcal{N}(\tau_{1-z}^2 \bar{y}_1 / (\tau_{1-z}^2 + v_1), \tau_{1-z}^2 v_1 / (\tau_{1-z}^2 + v_1))$ . With  $K_0$  spike leaves among the  $L_g$  leaves of  $g$  and  $S_0$  the sum of their squared means, the hyperparameters update as  $\tau_0^2 \mid \cdot \sim \text{scaled-Inv-}\chi^2(\nu_0 + K_0, (\nu_0 s_0^2 + S_0) / (\nu_0 + K_0))$  truncated to  $(0, \tau_1^2)$  and  $w \mid \cdot \sim \text{Beta}(1 + K_0, 1 + L_g - K_0)$ , and the residual variances update from their inverse-gamma conditionals using the residuals  $y_i - f(x_i) - D_i g(x_i)$  of each source. The range rule fixes  $\tau_1^2$  on the outcome scale. An  $f$  leaf with one source is an ordinary BART leaf. The minimum leaf size is five, using pooled data for  $f$  and RCT controls for  $g$ . A sweep costs a standard BART on  $n_1 + n_2$  observations plus a small ensemble on  $n_1$ .

For survival outcomes,  $y_i = \log T_i$  denotes log time. At each iteration censored values are imputed from  $\mathcal{N}(f(x_i) + D_i g(x_i), \sigma_{g_i}^2)$  truncated to  $(\log C_i, \infty)$ , where  $C_i$  is the censoring time, and used in the subsequent tree and parameter updates. The treatment-arm AFT-BART uses the same augmentation.

## 2.4 Prior effective sample size and calibration

We calibrate the spike scale through a variance-based effective sample size (ESS) for the standardized control mean. Stage 1 fits  $f$  to the RWD alone and estimates  $V_\mu^f = \text{Var}(N^{-1} \sum_i f(x_i) | \text{RWD})$ . For a normal prior of variance  $V$  and a reference observation with distribution  $\mathcal{N}(\mu, \sigma_1^2)$ ,  $\sigma_1^2/V$  equals the expected local-information-ratio (ELIR) ESS of Neuenschwander et al. [18]. The marginal BART and scale-mixture priors here are generally nonnormal. We therefore use the precision ratios below as working information summaries; they are not exact marginal ELIR values. For survival outcomes,  $\sigma_1^2$  sets uncensored log-time reference units, not the information in one censored observation.

For fixed discrepancy partitions  $\mathcal{T}$ , let  $a_{h\ell}$  be the proportion of the  $N$  standardization profiles in leaf  $\ell$  of tree  $h$ . Independence of the leaf priors gives

$$\mu_g = \sum_{h,\ell} a_{h\ell} \theta_{h\ell}, \quad V_g(\mathcal{T}, z, \tau_0^2) = \sum_{h,\ell} a_{h\ell}^2 \{z_{h\ell} \tau_0^2 + (1 - z_{h\ell}) \tau_1^2\}.$$

A standardized mean can involve several leaves in each tree. Its prior average must therefore integrate over all their indicators; a binomial count of  $H_g$  indicators suffices only when one leaf per tree has positive standardization weight, including a pointwise profile. Web Appendix D gives the general leaf-state average.

For calibration we set every discrepancy leaf to the spike as an informative reference. Define  $c_g = \mathbb{E}_{T_h}[\sum_\ell a_{h\ell}^2]$ , estimated by Monte Carlo under the tree prior;  $c_g = 1$  for stumps. We replace the random sum  $\sum_{h,\ell} a_{h\ell}^2$  by  $c_g H_g$  and define

$$\text{ESS}_0(s_0^2) = \sigma_1^2 \mathbb{E}_{\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2)} \left[ \frac{1}{V_\mu^f + c_g H_g \tau_0^2} \right]. \quad (6)$$

This averages inverse conditional variances over the untruncated scale law, with the mean partition coefficient substituted before inversion. The fitting prior truncates the scale at  $\tau_1^2$ ; equation (6) is the calibration convention used for all reported results. It differs from averaging over the full partition prior or computing the information of the marginal mixture. We calibrate this informative reference and report  $w$  separately. Conditional on  $w$ , the probability that all leaves reached by a profile are in the spike is  $w^{H_g}$ ; for a standardized mean involving  $L_+$  distinct leaves it is  $w^{L_+}$ .

The ceiling of (6) is  $\sigma_1^2/V_\mu^f$ . It includes the regularization of the RWD-only BART. For a single

fixed tree with a flat leaf prior and RWD support in every target leaf, it equals  $n_2\sigma_1^2/\sigma_2^2$  when source leaf proportions match and decreases by the factor  $\sum_\ell a_\ell^2/\pi_\ell$  under covariate shift (Theorem 3(d)). This equality and direction of change need not hold under finite leaf-prior regularization. We recommend a sensitivity ladder of fractions of the fitted ceiling when overlap is poor.

We divide the Stage-1 draws into  $B$  blocks and rescale their variances  $V^{(b)}$  to have mean  $V_\mu^f$ . On a grid  $\mathcal{S}$ , the Pr-rule is

$$\hat{s}_{0*}^2 = \inf\{s_0^2 \in \mathcal{S} : \hat{G}(s_0^2) \geq q\}, \quad \hat{G}(s_0^2) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\text{ESS}_0(V^{(b)}; s_0^2) \leq N_{\text{target}}\}, \quad (7)$$

with  $q = 0.95$ . If the requested target is at or above the whole-chain ceiling, the implementation instead returns the smallest grid scale and flags the fit as capped; this is a separate override of (7). If no grid value satisfies the rule, it returns and flags the largest scale.

The reported realized ESS is  $\mathbb{E}_{\text{post}}[\sigma_1^2/\{V_\mu^f + V_g(\mathcal{T}, z, \tau_0^2)\}]$ , with exact squared leaf proportions at each posterior state. Here  $V_g$  is a prior variance evaluated at learned states, not the posterior variance of  $\mu_g$  or a measured reduction in posterior uncertainty. Regional summaries standardize separately within each region and are not additive.

The application uses the pointwise summary  $\text{ESS}(x) = \hat{\sigma}_1^2/\{V_f(x) + H_g\bar{\tau}_0^2\}$ , where  $V_f(x) = \text{Var}\{f(x) \mid \text{RWD}\}$  is the Stage-1 uncertainty in the conditional mean and  $\bar{\tau}_0^2$  is the fitted mean spike variance. The denominator combines uncertainty in that mean with the allowed source discrepancy. An ESS of  $m$  equates this working variance to  $\hat{\sigma}_1^2/m$ , the variance of a mean of  $m$  independent reference controls with the same profile and residual variance  $\hat{\sigma}_1^2$ . For survival these reference observations are uncensored log times. This expresses precision in estimating a mean at  $x$ , rather than uncertainty in an individual outcome or a count of matched registry patients. In the single-arm application it describes prior external support; compatibility cannot be checked against concurrent trial controls. Appendix D distinguishes this plug-in from averaging pointwise precision over the scale prior.

## 2.5 The borrowing map and its guarantees

We summarize compatibility over covariate space by the posterior probability that the local discrepancy is smaller than a threshold  $c > 0$  fixed in advance on the outcome scale,

$$\mathcal{B}_c(x) = P(|g(x)| < c \mid \text{data}), \quad (8)$$

computed as the fraction of posterior draws, together with the posterior mean and interval of  $g(x)$  and the pointwise ESS of Section 2.4. We also retain the all-spike probability  $B(x)$ , the posterior probability that every leaf covering  $x$  belongs to the spike, as a diagnostic; its prior scale is  $w^{H_g}$  and, as Theorem 2 explains, its limiting value need not identify the true compatibility state, whereas  $\mathcal{B}_c$  is consistent under the stated conditions. In the simulation study  $c = 0.5$ , one third of the RCT residual standard deviation for Gaussian outcomes and 0.5 on the log-time scale for survival outcomes.

We give three theoretical results. The first two concern leaf-level inference and condition on the pooled surface  $f$ , on the partition of the  $g$  trees, and on  $(\sigma_1^2, \tau_0^2, \tau_1^2, w)$ , and consider a  $g$  leaf with  $n$  RCT controls, residual mean  $\bar{y}$ , and  $v = \sigma_1^2/n$ . Write  $\rho_j = \tau_j^2/(\tau_j^2 + v)$  for  $j \in \{0, 1\}$ , so that  $\hat{\theta}_1 = \rho_1 \bar{y}$  is the slab estimate, close to the RCT-only residual mean, and  $\hat{\theta}_0 = \rho_0 \bar{y}$  the spike-shrunk estimate, close to zero, that is, to the pooled surface. Let  $p = P(z = 0 \mid \bar{y})$  be the posterior slab probability and

$$\kappa = \frac{1}{2} \left( \frac{1}{\tau_0^2 + v} - \frac{1}{\tau_1^2 + v} \right), \quad a = \log \frac{1-w}{w} + \frac{1}{2} \log \frac{\tau_0^2 + v}{\tau_1^2 + v}, \quad C = e^{-a}. \quad (9)$$

**Theorem 1** (Bounded local influence). *Under Assumption A1,  $0 < w < 1$ , and the conditioning above,  $\theta \mid \bar{y}$  is the mixture  $p\mathcal{N}(\hat{\theta}_1, \rho_1 v) + (1-p)\mathcal{N}(\hat{\theta}_0, \rho_0 v)$  with  $\text{logit} p = a + \kappa \bar{y}^2$ , and the shift of the posterior mean relative to the slab estimate,  $I(\bar{y}) = \mathbb{E}[\theta \mid \bar{y}] - \hat{\theta}_1 = -(1-p)(\rho_1 - \rho_0)\bar{y}$ , satisfies*

$$\sup_{\bar{y}} |I(\bar{y})| \leq \sqrt{2v \max\{\log C, \frac{1}{2}\}}, \quad 1-p \leq C e^{-\kappa \bar{y}^2},$$

so  $I(\bar{y}) \rightarrow 0$  at a Gaussian rate as  $|\bar{y}| \rightarrow \infty$ . With  $w = 1$  (C-BART) the shift is  $-(\rho_1 - \rho_0)\bar{y}$ , linear and unbounded, and with  $w = 1, \tau_0^2 \rightarrow 0$  (BART-CP) it is  $-\rho_1 \bar{y}$ .

Theorem 1 is Theorem A1 of Web Appendix B. It bounds the spike-induced shift relative to the slab estimate, on the scale of an RCT leaf standard error. The slab itself shrinks toward the pooled surface by  $\rho_1$ , the range-rule regularization applied to a BART leaf. C-BART and BART-CP have unbounded conditional shifts. Corollary A1 extends the bound to the standardized control mean at fixed  $g$  partitions; the corresponding C-BART and BART-CP shifts are linear and unbounded unless the linear functional vanishes.

**Theorem 2** (Detection and the borrowing map). *Under Assumption A1,  $0 < w < 1$ , and the conditioning above, let the RCT residuals in the leaf be  $d + \epsilon_i$  with  $\epsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_1^2)$ , where  $d$  is the true local discrepancy.*

- (a) *The spike probability  $P(z = 1 | \bar{y}) = \{1 + \exp(a + \kappa \bar{y}^2)\}^{-1}$  is decreasing in  $|\bar{y}|$ , and when  $\log \frac{w}{1-w} + \frac{1}{2} \log \frac{\tau_1^2 + v}{\tau_0^2 + v} > 0$  it equals one half at*

$$d_{1/2} = \left[ \frac{2(\tau_0^2 + v)(\tau_1^2 + v)}{\tau_1^2 - \tau_0^2} \left\{ \log \frac{w}{1-w} + \frac{1}{2} \log \frac{\tau_1^2 + v}{\tau_0^2 + v} \right\} \right]^{1/2},$$

*so the probability that the leaf abstains at true shift  $d$  is  $\bar{\Phi}\{(d_{1/2} - d)/\sqrt{v}\} + \bar{\Phi}\{(d_{1/2} + d)/\sqrt{v}\}$ , with  $\bar{\Phi}$  the standard normal survival function.*

- (b) *For a single  $g$  tree,  $\mathcal{B}_c(x) = p \Psi_1(c) + (1 - p) \Psi_0(c)$  with  $\Psi_j(c) = P\{|\mathcal{N}(\rho_j \bar{y}, \rho_j v)| < c\}$ , and as  $n \rightarrow \infty$  with  $|d| \neq c$ ,  $\mathcal{B}_c(x) \rightarrow \mathbf{1}\{|d| < c\}$  almost surely.*

- (c) *As  $n \rightarrow \infty$  with  $\tau_0^2$  fixed, the spike probability converges to a limit in  $(0, 1)$  that is at most  $\frac{w\tau_1}{(1-w)\tau_0} \exp\{-\frac{d^2}{2}(\tau_0^{-2} - \tau_1^{-2})\}$  for  $d \neq 0$  and exceeds  $w$  for  $d = 0$ ; it tends to zero only as  $|d|/\tau_0 \rightarrow \infty$ , or for every fixed  $d \neq 0$  if  $\tau_0^2 \rightarrow 0$  with  $n$ .*

Theorem 2 combines Theorems A2 and A3 of Web Appendix C. The ensemble extension requires fixed partitions and a discrepancy representable on their additive span. The leaf results also condition on  $f$ ; they do not establish consistency under the full posterior over both ensembles. Further interpretation is given in Web Appendix G.3.

**Theorem 3** (Monotone, well-posed calibration). *Fix  $\sigma_1^2 > 0$ ,  $V > 0$ ,  $c_g H_g > 0$ ,  $N_{\text{target}} > 0$  and  $q \in (0, 1)$ , and write  $\text{ESS}_0(V; s)$  for (6) at  $V_\mu^f = V$  and  $s = s_0^2$ .*

- (a)  $s \mapsto \text{ESS}_0(V; s)$  is continuous, convex, and strictly decreasing on  $(0, \infty)$ , with limits  $\sigma_1^2/V$  as  $s \downarrow 0$  and 0 as  $s \uparrow \infty$ , under the untruncated calibration prior in (6). Under truncation to  $(0, \tau_1^2)$ , continuity and strict decrease remain, but the limit as  $s \uparrow \infty$  is  $\sigma_1^2/(V + c_g H_g \tau_1^2) > 0$ ; convexity is asserted only for the untruncated law.
- (b) Under the untruncated calibration law, for each  $V$  the sublevel set  $\{s : \text{ESS}_0(V; s) \leq N_{\text{target}}\}$  is a half-line  $[s^\dagger(V), \infty)$ , so the population rule  $G(s) = P\{s^\dagger(V) \leq s\}$  is a distribution function and  $\{s : G(s) \geq q\} = [s_\star, \infty)$  has a unique attained infimum;  $s_\star > 0$  if and only if  $P(N_{\text{target}} < \sigma_1^2/V) > 1 - q$ , and otherwise  $s_\star = 0$ , the zero-scale limit of the informative reference.
- (c) For a stationary, aperiodic Harris-ergodic full Stage-1 chain and a square-integrable standardized surface, fixed  $B$  and increasing block length select the smallest grid value satisfying the whole-chain rule, on a grid separated from its root. For fixed block length and increasing  $B$ , the rule converges to the selection based on the limiting rescaled block law, under the positivity, continuity and grid-separation conditions in Web Appendix D. These statements concern (7) with a nonempty admissible grid, not the ceiling override.
- (d) For a single  $f$  tree with fixed partition, RCT profile proportions  $a_\ell$  and RWD proportions  $\pi_\ell > 0$  on every target leaf, the ceiling is at least  $n_2 \sigma_1^2 / \sigma_2^2 / \sum_\ell a_\ell^2 / \pi_\ell$ . Equality holds in the flat-prior limit; the denominator is at least one, with equality if and only if  $a = \pi$  (zero-weight 0/0 terms contribute zero).

Theorem 3, proved in Web Appendix D, concerns the defined calibration summary. Under the untruncated law, each positive target below the ceiling has a unique positive root; truncation adds a positive lower limit. The block rule, finite grid and ceiling override are distinct parts of the implementation. These results establish neither predictive consistency of marginal ELIR nor a guarantee on frequentist operating characteristics.

## 2.6 The single-arm variant

When the trial has no control arm,  $n_1 = 0$ , every  $g$  leaf holds no RCT controls,  $M_g = 1$ , and the  $f$  moves are exactly those of standard BART on the RWD. The control surface is then  $f_1 = f + g$  with

$f$  the RWD surface and  $g$  a prior draw, a zero-mean discrepancy process with pointwise variance  $H_g\{w\tau_0^2 + (1-w)\tau_1^2\}$  conditional on the scales and  $w$ , and  $\tau_0^2$  and  $w$  stay at their priors. The indicators receive no likelihood information, and the slab contributes  $(1-w)H_g\tau_1^2$  to its conditional pointwise variance, so we report the  $w = 1$  model, the commensurate prior with the discrepancy carried by  $g$  at the calibrated scale, and use  $w < 1$  as a sensitivity analysis. Calibration uses the RWD-only Stage-1 fit. We assume  $\sigma_1^2 = \sigma_2^2$  for calibration and counterfactual survival prediction; single-arm data cannot check this equality. Targets are absolute control counts or fractions of the ceiling. With  $w = 1$  the single-arm model is a hierarchically centered commensurate AFT-BART,  $f_1(x) = f(x) + \sum_h e_h$  with  $e_h \sim \mathcal{N}(0, \tau_0^2)$ , with the discrepancy defined on the partition of  $g$  instead of that of  $f$ . The prior mean of the Gaussian control surface is preserved, but nonlinear survival-ratio means can change when discrepancy uncertainty is added. Because the compatibility map is constant across profiles in this setting, the covariate-resolved display for a single-arm analysis is the local ESS map  $\text{ESS}(x)$ , which describes variation in RWD support across trial profiles.

### 3 Simulation studies

#### 3.1 Design

Each replicate consists of an RCT of  $N = 300$  subjects and an external source of  $n_2 = 300$  RWD controls, with  $p = 10$  covariates  $X_j \sim \mathcal{N}(\mu_j, 1)$ ,  $\boldsymbol{\mu} = (1, -1, 0.5, -1, 2, -2, 2, -3, 1, 0)$ , and  $X_1$  replaced with probability 0.15 by a draw from  $\mathcal{N}(\pm 3, 0.5^2)$  so that the sources overlap imperfectly. The RCT randomizes 2:1 for Gaussian outcomes, giving about  $n_1 = 100$  controls, and 3:1 for survival outcomes, giving about 75, as in a hybrid design with a smaller control arm. The RCT control and RWD outcome surfaces are the same nonlinear function

$$\begin{aligned} f_0(\mathbf{W}) = & \alpha + \beta_1 W_1^2 + \beta_2 e^{-(W_2+1)^2/2} + \beta_3 |W_3 - 0.5| + \beta_4 |W_4 + 1| \\ & + \beta_5 \tanh(W_5 - 2) + \beta_6 \tanh(W_6 + 2) + 2\{b(W_5 - 2) + b(W_6 + 2)\} + \sum_{j=7}^{10} \beta_j W_j, \end{aligned} \quad (10)$$

with  $b(u) = u^2 / \{1 + (u/1.5)^2\}$  and

$$\boldsymbol{\beta} = (-0.50, -0.75, -0.50, -0.50, -1.35, -0.80, -0.01, -0.01, -0.01, -0.01),$$

and  $\mathbf{W} = \mathbf{X}$  except where stated; the treated RCT subjects add  $\delta = 1$  for Gaussian outcomes and  $\delta = \log 1.4$  on the log-time scale for survival outcomes. Residual standard deviations are  $\sigma_1 = 1.5$  and  $\sigma_2 = 1.8$  for Gaussian outcomes and 1.4 and 1.8 for survival outcomes, where the survival time is log-normal, dropout is Exponential(0.02), entry is Uniform(0, 1) and administrative censoring occurs at calendar time 3.

We consider five patterns of agreement between sources. Sc1 assigns the source at random. Sc2 assigns it through  $P(D = 1 | \mathbf{X}) = \text{logit}^{-1}\{(\ell - q_{0.5}(\ell))/0.5\}$  with  $\ell = -1.20(X_5 + X_6)$ , so that source membership depends on the two covariates with the largest outcome effects. Sc3 drives the same assignment by unmeasured  $(U_1, U_2)$  that also replace  $(X_5, X_6)$  in the outcome, with the observed  $(X_5, X_6)$  correlated with  $\mathbf{U}$  at  $\rho \in \{-0.5, 0, 0.5\}$ , so the RWD surface differs from the RCT surface everywhere in a way the covariates cannot remove; Web Appendix G lists the intercepts and the assignment constants.

Sc4 and Sc5 evaluate borrowing under regional and global outcome shifts. Sc4 keeps the assignment of Sc1 and raises the RWD surface by  $\delta_2$  outside a compatible region  $\mathcal{R} = \{X_5 > 2\}$ , so that the true discrepancy  $g = f_1 - f$  is  $-\delta_2$  there and 0 inside. The region holds half of the RCT population and lies on the covariate with the dominant nonlinear effect, so the shift has to be separated from a slope that  $f$  already models. We take  $\delta_2 \in \{0.5, 1, 2\}$ , one third to 1.3 RCT residual standard deviations, chosen around the leaf resolution  $d_{1/2}$  of Theorem 2(a), which at  $\sigma_1 = 1.5$ , fifty RCT controls per leaf and the spike scale calibrated in Sc1 is about 0.5. Sc5 raises the RWD surface by  $\delta_2 \in \{1, 2\}$  everywhere, so the compatible region is empty and borrowing should be suppressed throughout the covariate space. Survival outcomes use Sc4 with  $\delta_2 \in \{0.5, 1\}$  on the log scale and Sc5 with  $\delta_2 = 1$ .

The estimands are those of Section 2.1, standardized to the replicate’s RCT profiles: the ATE for Gaussian outcomes, and for survival outcomes the ratio of standardized population medians, whose true value is 1.4, and the ratio of standardized restricted mean survival times at  $t^* = 3$ . Over  $R = 500$  replicates we report the bias of the posterior mean, the mean posterior SD, the Monte Carlo RMSE of the posterior mean, the coverage of the central 95% credible interval, and power, the fraction of replicates whose 95% interval excludes the null value, 0 for the ATE and 1 for the ratios; the arm-specific means are summarized the same way. Monte Carlo standard errors are about 0.009 for a Gaussian ATE bias and 0.01 for a coverage or power, and the replicate data are

paired across methods and scenarios, so paired standard errors of a contrast between two methods are much smaller. For Sc4 and Sc5 we add the RMSE of the posterior mean control surface over all  $N$  RCT profiles inside and outside  $\mathcal{R}$ , the mean of the borrowing map  $\mathcal{B}_c(x)$  over the RCT control profiles in each region at  $c = 0.5$  on the outcome scale, and the mean of the posterior mean of  $g(x)$  in each region, whose true values are 0 in  $\mathcal{R}$  and  $-\delta_2$  outside it. For every LRC-BART fit we also record the prior  $\text{ESS}_0$ , the realized ESS, the ceiling, and whether the target was capped. Since  $\text{ESS}_0$  averages  $\sigma_1^2 / \{V^{(b)} + c_g H_g \tau_0^2\}$  over the Stage-1 blocks while the ceiling uses the whole-chain variance, Jensen’s inequality puts the printed  $\text{ESS}_0$  of a capped fit a few percent above the printed ceiling.

We compare LRC-BART with Bayesian linear and log-normal AFT regressions fitted to the RCT controls alone (LM-NP, AFT-NP), to the pooled controls (LM-CP, AFT-CP) and to the pooled controls with the source indicator as a covariate (LM-PP, AFT-PP); standard BART and AFT-BART [5, 24] in the same three configurations (BART-NP, BART-CP, BART-PP), the last being the implicit borrowing of Zhou and Ji [27]; the propensity-score composite likelihood PSCL of Chen et al. [4]; the MAP prior for the control mean without covariate adjustment; hierarchical linear and AFT models (HierLM, HierAFT); and the two nested versions of the proposed model, C-BART ( $w = 1$ ) and the free-offset LRC-BART ( $w = 0$ ) of Section 2.2. Web Appendix G.2 gives their priors, samplers and implementation details. We also transfer the analysis component of CAHB [13] to Sc1, Sc4 and Sc5 in  $R = 200$  Gaussian replicates paired with the first 200 full-study replicates. Our choices for the historical smoother, covariate selection, bandwidth, precision scales and joint draws are described in Web Appendix G.2, together with the ten-covariate and literal-constant variants.

LRC-BART is fitted as in Section 2.2, with  $H_f = 50$ ,  $H_g = 5$ ,  $(\alpha_g, \beta_g) = (0.5, 3)$ , the range-rule slab,  $\nu_0 = 3$ ,  $w \sim \text{Beta}(1, 1)$ , and a treated-arm BART with 50 trees. In Stage 1, we fit the calibration model to the RWD with 1,000 retained draws cut into  $B = 20$  blocks, take  $\sigma_1^2$  from a plain BART fitted to the RCT controls, compute  $c_g$  once by Monte Carlo under the  $g$ -tree prior, and search a geometric grid of 141 spike scales from  $10^{-6}$  to 10. We report an absolute ladder  $N_{\text{target}} \in \{50, 75, 100\}$ , with the ceiling override applied when the requested target reaches it, and a fractional ladder at 0.25, 0.5 and 0.9 of the ceiling, with LRC-BART (100) as the reference configuration and LRC-BART (0.9) as the fractional reference; LRC-BART ( $H_g = 10$ ) doubles the discrepancy ensemble. Averaged over scenarios and replicates in the full study, run on 16 forked workers, a full LRC-BART fit with calibration and treated arm took 2.3 seconds for Gaussian

and 2.5 seconds for survival outcomes, against 1.6 and 2.1 seconds for BART-PP and 1.1 and 1.6 for BART-NP, and the whole study ran in 5.6 hours. The configuration of  $g$  was chosen on a development run of  $R = 200$  paired with the first 200 replicates here (Web Appendix G.3).

### 3.2 Gaussian outcomes

Table 1 reports the two-arm Gaussian results for selected scenarios and methods, and Web Table S1 the full set.

Under agreement between sources (Sc1), the RMSE of LRC-BART (100) is 0.172 against 0.205 for BART-PP and 0.220 for BART-NP, with coverage 0.94; the paired difference in mean squared error from BART-PP is  $-0.012$  with standard error 0.001. BART-CP has RMSE 0.142, followed by C-BART at 0.150, LRC-BART at 0.172 and BART-PP at 0.205. LRC-BART (100) has mean prior  $ESS_0$  of 82, ceiling 127 and realized ESS 53, using posterior indicators and  $\tau_0^2$ . The target exceeds the ceiling in 20% of replicates, triggering the smallest-scale override. Tree-model power is 1.00 at  $\delta = 1$ , also with a 90% interval (Web Appendix G).

Under measured confounding (Sc2), the covariate-adjusted methods have small biases: LRC-BART (100) with bias 0.023 and RMSE 0.192 against 0.007 and 0.204 for BART-PP, whereas BART-CP, which pools the sources, and C-BART, which has a common discrepancy scale without a slab, carry biases of 0.058 and 0.042 with coverage 0.91. The ESS ceiling is substantially lower in Sc2, consistent with the covariate-overlap calculation in Theorem 3(d): the RCT profiles concentrate in regions with limited RWD support, the ceiling falls from 127 in Sc1 to 23, every absolute target is capped in every replicate and the three absolute targets give the same fit (Web Table S3), whereas the fractional ladder still spans prior ESS 5, 10 and 16 with biases 0.012, 0.016 and 0.019. We therefore recommend reporting fractional targets when covariate overlap is poor.

Under unmeasured confounding, Sc3, the RWD surface differs from the RCT surface everywhere by an amount the covariates cannot explain, and the two tree models with fixed commensurate components have substantial bias, as do their linear counterparts: BART-CP has bias 0.98 at  $\rho = 0$  with coverage 0, and C-BART, with one commensurability scale and no slab, has bias 0.23 with coverage 0.87, consistent with the unbounded influence under fixed components in Theorem 1.

LRC-BART (100) largely suppresses borrowing. Its realized ESS is 1.6, its map averages 0.07, and its bias is 0.041 with coverage 0.94 and RMSE 0.280, against 0.029 and 0.273 for BART-PP

and  $-0.002$  and  $0.289$  for BART-NP, with similar results at the three values of  $\rho$ . The  $0.04$  residual bias exceeds BART-PP's by  $0.012$  (paired Monte Carlo standard error  $0.001$ ). The spike provides a possible explanation: a leaf whose RCT residual mean is not far beyond  $d_{1/2}$  keeps a small but nonzero posterior spike probability, and Theorem 1 bounds the resulting shift by about one RCT standard error of the leaf. Doubling the discrepancy ensemble raises the bias to  $0.055$ , as  $H_g\tau_0^2$  increases the variation permitted under the spike. This favors a small  $H_g$  in this study.

Table 2 and Figure 1 report Sc4 and Sc5. At  $\delta_2 = 2$  the map separates the regions:  $\bar{\mathcal{B}}_c$  is  $0.64$  over the RCT control profiles in  $\mathcal{R}$  and  $0.05$  outside it, and the posterior mean of  $g$  averages  $-0.26$  in  $\mathcal{R}$  and  $-1.71$  outside it against true values of  $0$  and  $-2$ .

Figure 1 shows how discrepancy estimates vary across the boundary. Away from the boundary at  $X_5 = 2$  the posterior mean of  $g$  is  $-1.8$  outside  $\mathcal{R}$  and  $-0.15$  inside it, short of the truth by about a tenth of the shift on each side, and the map is  $0.02$  outside and  $0.7$  inside; the transition occupies the unit around the boundary, with  $g$  averaging  $-1.66$  in the half-unit bin below it and  $-0.44$  in the half-unit bin above it. The boundary is learned from the RCT controls alone, about four of them per  $0.1$  of  $X_5$ , and the pooled surface  $f$ , which is estimated from both sources, smooths the RWD step across it, leaving the shallow discrepancy ensemble  $g$  to estimate the remaining difference. The discrepancy underestimation persisted with  $f$  fixed at the truth and under every  $g$  prior we tried (Web Appendix G.3), so the configurations examined did not resolve it.

The empirical ATE bias at  $\delta_2 = 2$  is  $0.000$  (coverage  $0.92$ ), because the errors on the two sides of the boundary have opposite signs and the estimand averages over both. Control-surface estimation is less accurate: its RMSE over the RCT profiles in  $\mathcal{R}$  is  $0.824$  against  $0.785$  for BART-PP, and outside  $\mathcal{R}$  it is  $0.857$  against  $0.822$ , so at  $\delta_2 = 2$  LRC-BART does not beat the free-offset model on the surface, while C-BART, at  $1.087$  outside  $\mathcal{R}$ , has a larger error than either method. A secondary version of Sc4 places the shift on  $X_7$ , a covariate with negligible effect, and the map and ATE results are similar to those on  $X_5$  at both  $\delta_2 = 2$  and  $\delta_2 = 1$  (Web Appendix G.3).

At  $\delta_2 = 1$ , a shift of about twice  $d_{1/2}$ , the map separates only partially ( $0.62$  against  $0.52$ ), the posterior mean of  $g$  is  $-0.6$  outside  $\mathcal{R}$  and  $-0.25$  inside it (Figure 1), and LRC-BART borrows too much outside  $\mathcal{R}$ : the ATE bias is  $-0.041$  (paired difference from BART-PP  $-0.039$ , standard error  $0.003$ ), the RMSE  $0.223$  against  $0.207$ , and the coverage  $0.89$ , whereas C-BART and BART-CP have biases of  $-0.21$  and  $-0.34$ .

At  $\delta_2 = 0.5$ , below the stated resolution, the map is flat at 0.84,  $g$  averages  $-0.15$  in  $\mathcal{R}$  and  $-0.18$  outside it, and the bias is  $-0.043$  with RMSE 0.200, below BART-PP’s 0.203 because the variance reduction offsets the bias in this scenario. Lower fractional targets reduce bias under the moderate shift,  $-0.032$  at a quarter of the ceiling against  $-0.041$ , at a cost of 0.01 in RMSE in Sc1 (Web Table S3).

Under the global shift of Sc5, LRC-BART suppresses borrowing: at  $\delta_2 = 2$  the map averages 0.01, the realized ESS is 0.2, the bias is  $-0.001$  and the RMSE 0.203, against BART-NP’s 0.220 and BART-PP’s 0.205, whereas C-BART and BART-CP have biases of  $-0.32$  and  $-1.39$ ; at  $\delta_2 = 1$  the map is 0.10 and the bias  $-0.005$ .

The realized regional ESS summarizes learned local precision (Table 2). Across the first 200 replicates, it is 32.7 and 31.9 on the two sides of the Sc4 partition under Sc1. Under Sc4 it is 15.1 in  $\mathcal{R}$  and 12.7 outside at  $\delta_2 = 1$ , and 9.5 and 0.44 at  $\delta_2 = 2$ . Under Sc5 with  $\delta_2 = 2$ , it is about 0.2 on both sides. These quantities use the control mean standardized within each region and are not additive. The prior pointwise ESS uses the RWD fit and cannot establish that trial outcomes support regional borrowing.

The surface loss persists away from the boundary. On RCT control profiles with  $X_5 > 2.5$ , the Sc4  $\delta_2 = 2$  RMSE is 0.747 for LRC-BART and 0.723 for BART-PP, a paired difference of 0.024 with standard error 0.005 over 200 replicates. Using all RCT profiles in  $\mathcal{R}$  over those same replicates gives a difference of 0.039 with standard error 0.006. Web Appendix G.3 reports both profile sets. Thus the evidence for LRC-BART is its efficiency under Sc1 and its regional compatibility and ESS summaries, with a surface-estimation loss under Sc4.

The transferred CAHB analysis has Sc1 RMSE 0.258, compared with 0.175 for LRC-BART and 0.223 for BART-NP over the same 200 replicates. At Sc4  $\delta_2 = 2$ , its bias is  $-0.333$  and its historical precision shares average 0.49 and 0.41 in the two regions. At Sc5  $\delta_2 = 1$ , its bias is  $-0.437$ . These transfer-specific results do not establish a general ranking of CAHB and tree-based borrowing.

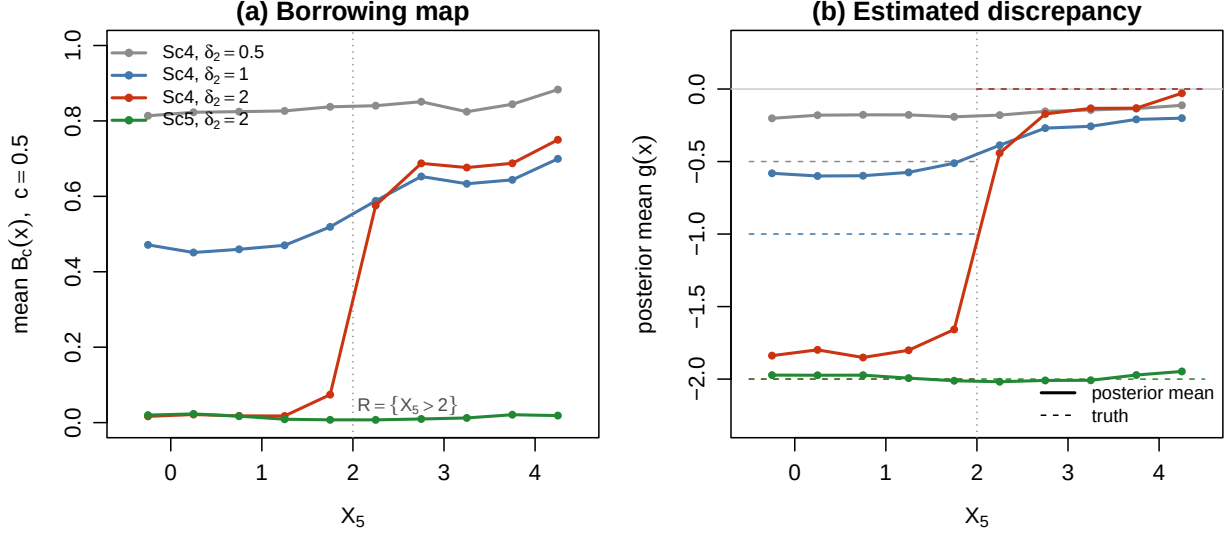


Figure 1: Regional and global shifts, Gaussian outcomes, LRC-BART (100): (a) the borrowing map  $B_c(x) = P(|g(x)| < 0.5 | \text{data})$  and (b) the posterior mean of  $g(x)$ , both averaged over the RCT control profiles in bins of width 0.5 on  $X_5$  across the first 40 replicates. The RWD surface is raised by  $\delta_2$  where  $X_5 \leq 2$  (Sc4) or everywhere (Sc5), so the true discrepancy, dashed in (b), is  $-\delta_2$  there.

Table 1: Gaussian outcomes, two-arm design,  $R = 500$  replicates: average treatment effect (ATE) bias, mean posterior SD, Monte Carlo RMSE, coverage of the 95% credible interval, and power (95% interval excludes zero at  $\delta = 1$ ), with bias and RMSE of the RCT-standardized control mean. Sc4 raises the RWD control surface by  $\delta_2$  outside  $R = \{X_5 > 2\}$ ; Sc5 raises it everywhere. LRC-BART (100) targets a prior ESS of 100 RCT controls and LRC-BART (0.9) targets 0.9 of the ESS ceiling; CAHB uses  $R = 200$  paired replicates and our transfer described in Web Appendix G.2; other rows use  $R = 500$ . C-BART is LRC-BART with  $w = 1$ . PSCL reports no arm-specific means. The full table with every scenario and method is Web Table S1.

| Scenario | Method         | ATE    |       |       |      |      | Control mean |       |
|----------|----------------|--------|-------|-------|------|------|--------------|-------|
|          |                | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  |
| Sc1      | LM-PP          | -0.002 | 0.347 | 0.337 | 0.94 | 0.85 | 0.001        | 0.243 |
|          | BART-NP        | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.001        | 0.185 |
|          | BART-CP        | 0.008  | 0.145 | 0.142 | 0.96 | 1.00 | -0.003       | 0.096 |
|          | BART-PP        | 0.006  | 0.207 | 0.205 | 0.94 | 1.00 | -0.001       | 0.170 |
|          | PSCL           | 0.005  | 0.269 | 0.261 | 0.95 | 0.97 | -0.005       | 0.171 |
|          | MAP (100)      | 0.006  | 0.292 | 0.265 | 0.97 | 0.95 | -0.005       | 0.174 |
|          | HierLM         | -0.003 | 0.338 | 0.341 | 0.93 | 0.85 | 0.003        | 0.249 |
|          | C-BART         | 0.007  | 0.155 | 0.150 | 0.95 | 1.00 | -0.002       | 0.106 |
|          | LRC-BART (100) | 0.009  | 0.171 | 0.172 | 0.94 | 1.00 | -0.004       | 0.134 |
|          | LRC-BART (0.9) | 0.011  | 0.171 | 0.171 | 0.95 | 1.00 | -0.005       | 0.132 |
|          | CAHB           | -0.003 | 0.410 | 0.258 | 1.00 | 0.78 | -0.023       | 0.184 |

Table 1, continued

| Scenario                 | Method         | ATE    |       |       |      |      | Control mean |       |
|--------------------------|----------------|--------|-------|-------|------|------|--------------|-------|
|                          |                | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  |
| Sc2                      | LM-PP          | 0.010  | 0.340 | 0.331 | 0.95 | 0.84 | -0.010       | 0.248 |
|                          | BART-NP        | -0.008 | 0.201 | 0.219 | 0.93 | 1.00 | 0.009        | 0.180 |
|                          | BART-CP        | 0.058  | 0.177 | 0.195 | 0.91 | 1.00 | -0.056       | 0.151 |
|                          | BART-PP        | 0.007  | 0.208 | 0.204 | 0.96 | 1.00 | -0.005       | 0.163 |
|                          | PSCL           | 0.024  | 0.321 | 0.292 | 0.97 | 0.91 | -0.029       | 0.205 |
|                          | MAP (100)      | 0.332  | 0.360 | 0.491 | 0.83 | 0.97 | -0.337       | 0.424 |
|                          | HierLM         | -0.011 | 0.318 | 0.321 | 0.95 | 0.88 | 0.011        | 0.238 |
|                          | C-BART         | 0.042  | 0.168 | 0.186 | 0.91 | 1.00 | -0.041       | 0.141 |
|                          | LRC-BART (100) | 0.023  | 0.181 | 0.192 | 0.94 | 1.00 | -0.021       | 0.148 |
|                          | LRC-BART (0.9) | 0.019  | 0.183 | 0.193 | 0.93 | 1.00 | -0.017       | 0.151 |
| Sc3, $\rho = 0$          | LM-PP          | 0.043  | 0.356 | 0.346 | 0.95 | 0.85 | -0.027       | 0.241 |
|                          | BART-NP        | -0.002 | 0.292 | 0.289 | 0.94 | 0.92 | 0.024        | 0.219 |
|                          | BART-CP        | 0.984  | 0.202 | 1.003 | 0.00 | 1.00 | -0.962       | 0.971 |
|                          | BART-PP        | 0.029  | 0.275 | 0.273 | 0.94 | 0.96 | -0.007       | 0.197 |
|                          | PSCL           | 0.704  | 0.269 | 0.753 | 0.26 | 1.00 | -0.693       | 0.713 |
|                          | MAP (100)      | 0.363  | 0.360 | 0.504 | 0.82 | 0.97 | -0.352       | 0.431 |
|                          | HierLM         | 0.039  | 0.362 | 0.354 | 0.96 | 0.82 | -0.023       | 0.253 |
|                          | C-BART         | 0.226  | 0.300 | 0.380 | 0.87 | 0.98 | -0.204       | 0.311 |
|                          | LRC-BART (100) | 0.041  | 0.284 | 0.280 | 0.94 | 0.96 | -0.018       | 0.206 |
|                          | LRC-BART (0.9) | 0.037  | 0.282 | 0.277 | 0.94 | 0.96 | -0.015       | 0.203 |
| Sc4, $X_5, \delta_2 = 1$ | LM-PP          | -0.012 | 0.349 | 0.339 | 0.95 | 0.83 | 0.011        | 0.245 |
|                          | BART-NP        | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.001        | 0.185 |
|                          | BART-CP        | -0.338 | 0.146 | 0.368 | 0.35 | 1.00 | 0.343        | 0.357 |
|                          | BART-PP        | -0.002 | 0.209 | 0.207 | 0.95 | 1.00 | 0.007        | 0.173 |
|                          | PSCL           | -0.246 | 0.271 | 0.358 | 0.87 | 0.80 | 0.247        | 0.300 |
|                          | MAP (100)      | -0.224 | 0.306 | 0.361 | 0.91 | 0.75 | 0.224        | 0.292 |
|                          | HierLM         | -0.025 | 0.338 | 0.343 | 0.94 | 0.83 | 0.024        | 0.250 |
|                          | C-BART         | -0.207 | 0.173 | 0.273 | 0.73 | 1.00 | 0.212        | 0.258 |
|                          | LRC-BART (100) | -0.041 | 0.195 | 0.223 | 0.89 | 1.00 | 0.046        | 0.194 |
|                          | LRC-BART (0.9) | -0.041 | 0.196 | 0.224 | 0.90 | 1.00 | 0.047        | 0.194 |
| Sc4, $X_5, \delta_2 = 2$ | CAHB           | -0.208 | 0.413 | 0.333 | 0.98 | 0.42 | 0.181        | 0.262 |
|                          | LM-PP          | -0.023 | 0.354 | 0.343 | 0.95 | 0.81 | 0.022        | 0.249 |
|                          | BART-NP        | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.001        | 0.185 |
|                          | BART-CP        | -0.685 | 0.149 | 0.701 | 0.00 | 0.56 | 0.690        | 0.698 |
|                          | BART-PP        | -0.006 | 0.209 | 0.205 | 0.96 | 1.00 | 0.011        | 0.170 |
|                          | PSCL           | -0.498 | 0.273 | 0.562 | 0.54 | 0.44 | 0.498        | 0.528 |
|                          | MAP (100)      | -0.377 | 0.333 | 0.495 | 0.78 | 0.46 | 0.378        | 0.437 |
|                          | HierLM         | -0.044 | 0.339 | 0.345 | 0.94 | 0.81 | 0.043        | 0.253 |
|                          | C-BART         | -0.227 | 0.197 | 0.391 | 0.66 | 0.90 | 0.232        | 0.383 |
|                          | LRC-BART (100) | 0.000  | 0.189 | 0.202 | 0.92 | 1.00 | 0.005        | 0.168 |
| Sc5, $\delta_2 = 2$      | LRC-BART (0.9) | -0.000 | 0.189 | 0.202 | 0.93 | 1.00 | 0.006        | 0.170 |
|                          | CAHB           | -0.333 | 0.422 | 0.453 | 0.95 | 0.25 | 0.309        | 0.398 |
|                          | LM-PP          | -0.042 | 0.347 | 0.339 | 0.94 | 0.80 | 0.042        | 0.246 |
|                          | BART-NP        | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.001        | 0.185 |
|                          | BART-CP        | -1.390 | 0.152 | 1.398 | 0.00 | 0.72 | 1.395        | 1.399 |
|                          | BART-PP        | -0.012 | 0.208 | 0.205 | 0.95 | 1.00 | 0.017        | 0.171 |
|                          | PSCL           | -1.002 | 0.269 | 1.035 | 0.04 | 0.04 | 1.002        | 1.017 |
|                          | MAP (100)      | -0.395 | 0.348 | 0.525 | 0.78 | 0.41 | 0.395        | 0.465 |
|                          | HierLM         | -0.038 | 0.338 | 0.344 | 0.94 | 0.82 | 0.037        | 0.251 |
|                          | C-BART         | -0.316 | 0.194 | 0.650 | 0.76 | 0.93 | 0.322        | 0.648 |
|                          | LRC-BART (100) | -0.001 | 0.190 | 0.203 | 0.92 | 1.00 | 0.005        | 0.171 |
|                          | LRC-BART (0.9) | -0.001 | 0.190 | 0.205 | 0.92 | 1.00 | 0.006        | 0.171 |
|                          | CAHB           | -0.067 | 0.479 | 0.400 | 0.95 | 0.52 | 0.041        | 0.326 |

Table 2: Regional and global shifts, Gaussian outcomes,  $R = 500$ : ATE bias and coverage; RMSE of the posterior mean control surface over all  $N$  RCT profiles (treated and control) inside the compatible region  $\mathcal{R}$  and outside it; the borrowing map  $\mathcal{B}_c(x) = P(|g(x)| < 0.5 \mid \text{data})$  averaged over the RCT control profiles in each region; the posterior mean of  $g(x)$  averaged over each region (true value 0 in  $\mathcal{R}$  and  $-\delta_2$  in  $\mathcal{R}^c$ ); and the realized ESS. In Sc5 the compatible region is empty. Sc4' places the shift on the null covariate  $X_7$ . LRC-BART ( $w = 0$ ) has no spike, so its map reflects the slab alone. CAHB rows use  $R = 200$ ; all other rows use  $R = 500$ . CAHB precision shares and effective sizes have different definitions and are reported separately in Web Appendix G.2. For Sc1, the two columns use the Sc4 partition although both regions are compatible.

| Scenario  | Method               | ATE    |      | Surface RMSE  |                 | $\tilde{\mathcal{B}}_c$ |                 | Mean $\hat{g}$ |                 | ESS  |
|-----------|----------------------|--------|------|---------------|-----------------|-------------------------|-----------------|----------------|-----------------|------|
|           |                      | Bias   | Cov  | $\mathcal{R}$ | $\mathcal{R}^c$ | $\mathcal{R}$           | $\mathcal{R}^c$ | $\mathcal{R}$  | $\mathcal{R}^c$ |      |
| Sc1       | BART-PP              | 0.006  | 0.94 | 0.750         | 0.780           | —                       | —               | —              | —               | —    |
|           | C-BART               | 0.007  | 0.95 | 0.728         | 0.756           | 1.00                    | 1.00            | 0.00           | 0.00            | 81.0 |
|           | LRC-BART ( $w = 0$ ) | 0.009  | 0.92 | 0.755         | 0.783           | 0.79                    | 0.79            | -0.00          | 0.01            | 0.1  |
|           | LRC-BART (100)       | 0.009  | 0.94 | 0.736         | 0.763           | 0.92                    | 0.92            | -0.00          | 0.00            | 52.5 |
|           | CAHB                 | -0.003 | 1.00 | 2.071         | 2.273           | —                       | —               | —              | —               | —    |
| Sc4 (0.5) | BART-PP              | 0.000  | 0.95 | 0.758         | 0.795           | —                       | —               | —              | —               | —    |
|           | C-BART               | -0.109 | 0.90 | 0.736         | 0.799           | 0.98                    | 0.98            | -0.06          | -0.06           | 79.4 |
|           | LRC-BART ( $w = 0$ ) | 0.007  | 0.93 | 0.768         | 0.795           | 0.71                    | 0.69            | -0.21          | -0.27           | 0.1  |
|           | LRC-BART (100)       | -0.043 | 0.92 | 0.751         | 0.796           | 0.84                    | 0.83            | -0.15          | -0.18           | 42.9 |
| Sc4 (1)   | BART-PP              | -0.002 | 0.95 | 0.784         | 0.820           | —                       | —               | —              | —               | —    |
|           | C-BART               | -0.207 | 0.73 | 0.749         | 0.915           | 0.91                    | 0.90            | -0.14          | -0.16           | 73.6 |
|           | LRC-BART ( $w = 0$ ) | 0.006  | 0.93 | 0.801         | 0.827           | 0.51                    | 0.40            | -0.35          | -0.63           | 0.1  |
|           | LRC-BART (100)       | -0.041 | 0.89 | 0.783         | 0.843           | 0.62                    | 0.52            | -0.32          | -0.52           | 20.8 |
|           | CAHB                 | -0.208 | 0.98 | 2.133         | 2.207           | —                       | —               | —              | —               | —    |
| Sc4 (2)   | BART-PP              | -0.006 | 0.96 | 0.785         | 0.822           | —                       | —               | —              | —               | —    |
|           | C-BART               | -0.227 | 0.66 | 0.801         | 1.087           | 0.66                    | 0.39            | -0.30          | -0.97           | 45.7 |
|           | LRC-BART ( $w = 0$ ) | 0.004  | 0.92 | 0.836         | 0.858           | 0.57                    | 0.05            | -0.25          | -1.73           | 0.1  |
|           | LRC-BART (100)       | 0.000  | 0.92 | 0.824         | 0.857           | 0.64                    | 0.05            | -0.26          | -1.71           | 0.8  |
|           | CAHB                 | -0.333 | 0.95 | 2.172         | 2.157           | —                       | —               | —              | —               | —    |
| Sc4' (1)  | BART-PP              | -0.001 | 0.96 | 0.807         | 0.810           | —                       | —               | —              | —               | —    |
|           | C-BART               | -0.206 | 0.73 | 0.767         | 0.899           | 0.91                    | 0.90            | -0.15          | -0.16           | 73.3 |
|           | LRC-BART ( $w = 0$ ) | 0.009  | 0.92 | 0.820         | 0.817           | 0.52                    | 0.41            | -0.37          | -0.63           | 0.1  |
|           | LRC-BART (100)       | -0.038 | 0.90 | 0.805         | 0.833           | 0.62                    | 0.52            | -0.33          | -0.52           | 20.5 |
| Sc4' (2)  | BART-PP              | -0.002 | 0.95 | 0.805         | 0.806           | —                       | —               | —              | —               | —    |
|           | C-BART               | -0.218 | 0.69 | 0.828         | 1.075           | 0.64                    | 0.38            | -0.33          | -0.97           | 44.7 |
|           | LRC-BART ( $w = 0$ ) | 0.007  | 0.92 | 0.859         | 0.848           | 0.57                    | 0.05            | -0.25          | -1.74           | 0.1  |
|           | LRC-BART (100)       | 0.001  | 0.93 | 0.850         | 0.848           | 0.64                    | 0.05            | -0.26          | -1.71           | 0.8  |
| Sc5 (1)   | BART-PP              | -0.004 | 0.95 | —             | 0.769           | —                       | —               | —              | —               | —    |
|           | C-BART               | -0.244 | 0.68 | —             | 0.821           | —                       | 0.43            | —              | -0.60           | 41.5 |
|           | LRC-BART ( $w = 0$ ) | 0.007  | 0.92 | —             | 0.769           | —                       | 0.10            | —              | -1.00           | 0.1  |
|           | LRC-BART (100)       | -0.005 | 0.91 | —             | 0.768           | —                       | 0.10            | —              | -0.98           | 1.2  |
|           | CAHB                 | -0.437 | 0.91 | —             | 2.203           | —                       | —               | —              | —               | —    |
| Sc5 (2)   | BART-PP              | -0.012 | 0.95 | —             | 0.771           | —                       | —               | —              | —               | —    |
|           | C-BART               | -0.316 | 0.76 | —             | 0.932           | —                       | 0.19            | —              | -1.54           | 45.4 |
|           | LRC-BART ( $w = 0$ ) | 0.006  | 0.93 | —             | 0.771           | —                       | 0.01            | —              | -2.00           | 0.1  |
|           | LRC-BART (100)       | -0.001 | 0.92 | —             | 0.771           | —                       | 0.01            | —              | -1.99           | 0.2  |
|           | CAHB                 | -0.067 | 0.95 | —             | 1.983           | —                       | —               | —              | —               | —    |

| Scenario | Regional ESS, $X_5 > 2$ | Regional ESS, $X_5 \leq 2$ |
|----------|-------------------------|----------------------------|
| Sc1      | 32.66                   | 31.86                      |
| Sc4 (1)  | 15.14                   | 12.74                      |
| Sc4 (2)  | 9.51                    | 0.44                       |
| Sc5 (2)  | 0.24                    | 0.24                       |

Regional realized ESS for LRC-BART (100), averaged over the first 200 paired replicates. Each estimand is the control mean standardized to all RCT profiles in that region. The partition is held fixed for Sc1 and Sc5; both regions are compatible in Sc1 and neither is compatible in Sc5. Regional effective sizes are not additive.

### 3.3 Survival outcomes

Table 3 reports the survival study for the median ratio and the RMST ratio side by side, and Web Table S2 the full set. With 75 RCT controls and about a quarter of the times censored, the median ratio has a wide and skewed posterior under every method, and its posterior mean carries an upward bias that grows with the posterior spread: BART-NP has bias 0.26 and RMSE 0.51 in Sc1, where its RMST-ratio bias is 0.13, and AFT-NP is wider still. Borrowing reduces this uncertainty at the studied sample size.

In Sc1 LRC-BART (100) has median-ratio bias 0.045 and RMSE 0.312 against 0.032 and 0.345 for BART-PP, with power 0.34 against 0.25, and its RMST-ratio RMSE is 0.097 against 0.106; in Sc2, where the ceiling falls to 40 and the absolute targets are capped, the two are within 0.03 of each other on both estimands.

Under unmeasured confounding (Sc3), the tree models with a fixed commensurate component have large biases, as in the Gaussian study: BART-CP with median-ratio bias 2.3 and C-BART with 0.58 at  $\rho = 0$ , and LRC-BART abstains with a realized ESS of 2.3. Its median-ratio bias is then 0.15 against 0.04 for BART-PP and 0.19 for BART-NP, with RMSE 0.558 against 0.451 and coverage 0.94; the paired difference in bias from BART-PP is 0.11 with standard error 0.005, the largest loss to BART-PP in the study. The control log-median is biased by only 0.02, against  $-0.006$  for BART-PP, so most of the difference on the ratio scale comes from the wider posterior LRC-BART has once it has switched borrowing off with 75 controls, with posterior SD 0.52 against 0.39, together with the nonlinear transformation to the ratio scale. On the RMST ratio the biases are 0.031 and  $-0.005$  and the RMSEs 0.146 and 0.137, a paired difference in mean squared error of 0.003 with standard error 0.001. We report both estimands because the median ratio amplifies a small control bias and a wide posterior in a way the RMST ratio does not, and the RMST ratio is

the estimand of the application.

Under the regional shift, Sc4 at  $\delta_2 = 1$  on the log scale with a residual SD of 1.4, the map separates weakly, 0.64 in  $\mathcal{R}$  against 0.58 outside, and the treatment-effect biases are  $-0.011$  on the median ratio and  $-0.037$  on the RMST ratio where C-BART and BART-CP have biases of  $-0.20$  and  $-0.37$ ; under the global shift of Sc5 the map is 0.16, the median-ratio bias 0.03, and C-BART's ladder behaves as in the Gaussian study.

Table 3: Survival outcomes, two-arm design,  $R = 500$ : bias, Monte Carlo RMSE, coverage and power (95% interval excludes one) for the ratio of standardized population medians (true value 1.4) and the ratio of standardized restricted mean survival times at  $t^* = 3$ . Sc4 raises the RWD log-time surface by  $\delta_2 = 1$  outside  $\mathcal{R} = \{X_5 > 2\}$  and Sc5 by 1 everywhere. Point estimates are posterior means. The full table is Web Table S2.

| Scenario                 | Method         | Median ratio |       |      |      | RMST ratio, $t^* = 3$ |       |      |      |
|--------------------------|----------------|--------------|-------|------|------|-----------------------|-------|------|------|
|                          |                | Bias         | RMSE  | Cov  | Pow  | Bias                  | RMSE  | Cov  | Pow  |
| Sc1                      | AFT-PP         | 0.261        | 0.737 | 0.95 | 0.12 | -0.009                | 0.142 | 0.94 | 0.10 |
|                          | BART-NP        | 0.260        | 0.511 | 0.92 | 0.46 | 0.131                 | 0.206 | 0.87 | 0.47 |
|                          | BART-CP        | 0.064        | 0.232 | 0.96 | 0.63 | -0.011                | 0.072 | 0.95 | 0.35 |
|                          | BART-PP        | 0.032        | 0.345 | 0.95 | 0.25 | -0.023                | 0.106 | 0.94 | 0.17 |
|                          | HierAFT        | 0.246        | 0.733 | 0.94 | 0.12 | -0.008                | 0.152 | 0.92 | 0.11 |
|                          | C-BART         | 0.034        | 0.248 | 0.97 | 0.42 | 0.003                 | 0.079 | 0.95 | 0.32 |
|                          | LRC-BART (100) | 0.045        | 0.312 | 0.95 | 0.34 | 0.003                 | 0.097 | 0.94 | 0.28 |
|                          | LRC-BART (0.9) | 0.042        | 0.304 | 0.96 | 0.38 | 0.002                 | 0.095 | 0.95 | 0.30 |
| Sc2                      | AFT-PP         | 0.389        | 0.818 | 0.95 | 0.19 | 0.050                 | 0.154 | 0.95 | 0.20 |
|                          | BART-NP        | 0.251        | 0.495 | 0.91 | 0.43 | 0.110                 | 0.177 | 0.88 | 0.45 |
|                          | BART-CP        | 0.194        | 0.383 | 0.93 | 0.56 | 0.037                 | 0.100 | 0.94 | 0.43 |
|                          | BART-PP        | 0.036        | 0.335 | 0.95 | 0.24 | -0.002                | 0.097 | 0.95 | 0.21 |
|                          | HierAFT        | 0.233        | 0.695 | 0.97 | 0.14 | 0.001                 | 0.134 | 0.95 | 0.12 |
|                          | C-BART         | 0.138        | 0.331 | 0.93 | 0.55 | 0.043                 | 0.104 | 0.92 | 0.49 |
|                          | LRC-BART (100) | 0.061        | 0.322 | 0.94 | 0.34 | 0.018                 | 0.100 | 0.93 | 0.31 |
|                          | LRC-BART (0.9) | 0.058        | 0.325 | 0.94 | 0.33 | 0.017                 | 0.100 | 0.94 | 0.30 |
| Sc3, $\rho = 0$          | AFT-PP         | 0.440        | 0.919 | 0.94 | 0.19 | 0.054                 | 0.158 | 0.95 | 0.19 |
|                          | BART-NP        | 0.186        | 0.571 | 0.90 | 0.30 | 0.058                 | 0.173 | 0.92 | 0.21 |
|                          | BART-CP        | 2.308        | 2.412 | 0.00 | 1.00 | 0.588                 | 0.607 | 0.01 | 1.00 |
|                          | BART-PP        | 0.041        | 0.451 | 0.91 | 0.20 | -0.005                | 0.137 | 0.92 | 0.15 |
|                          | HierAFT        | 0.337        | 0.890 | 0.95 | 0.13 | 0.006                 | 0.147 | 0.95 | 0.11 |
|                          | C-BART         | 0.576        | 0.909 | 0.88 | 0.40 | 0.139                 | 0.222 | 0.85 | 0.38 |
|                          | LRC-BART (100) | 0.155        | 0.558 | 0.94 | 0.17 | 0.031                 | 0.146 | 0.94 | 0.18 |
|                          | LRC-BART (0.9) | 0.148        | 0.554 | 0.94 | 0.17 | 0.029                 | 0.144 | 0.93 | 0.18 |
| Sc4, $X_5, \delta_2 = 1$ | AFT-PP         | 0.191        | 0.691 | 0.95 | 0.10 | -0.039                | 0.139 | 0.92 | 0.07 |
|                          | BART-NP        | 0.260        | 0.511 | 0.92 | 0.46 | 0.131                 | 0.206 | 0.87 | 0.47 |
|                          | BART-CP        | -0.367       | 0.401 | 0.50 | 0.04 | -0.156                | 0.168 | 0.32 | 0.07 |
|                          | BART-PP        | -0.012       | 0.339 | 0.97 | 0.17 | -0.059                | 0.113 | 0.91 | 0.07 |
|                          | HierAFT        | 0.193        | 0.700 | 0.94 | 0.10 | -0.023                | 0.149 | 0.91 | 0.09 |
|                          | C-BART         | -0.202       | 0.305 | 0.84 | 0.08 | -0.093                | 0.120 | 0.79 | 0.04 |
|                          | LRC-BART (100) | -0.011       | 0.340 | 0.93 | 0.24 | -0.037                | 0.108 | 0.90 | 0.16 |
|                          | LRC-BART (0.9) | -0.018       | 0.342 | 0.93 | 0.24 | -0.040                | 0.110 | 0.90 | 0.16 |
| Sc5, $\delta_2 = 1$      | AFT-PP         | 0.193        | 0.695 | 0.94 | 0.09 | -0.029                | 0.141 | 0.93 | 0.08 |
|                          | BART-NP        | 0.260        | 0.511 | 0.92 | 0.46 | 0.131                 | 0.206 | 0.87 | 0.47 |
|                          | BART-CP        | -0.703       | 0.712 | 0.00 | 0.64 | -0.256                | 0.262 | 0.01 | 0.65 |
|                          | BART-PP        | -0.029       | 0.333 | 0.95 | 0.17 | -0.041                | 0.110 | 0.93 | 0.11 |
|                          | HierAFT        | 0.204        | 0.701 | 0.94 | 0.10 | -0.018                | 0.150 | 0.92 | 0.09 |
|                          | C-BART         | -0.233       | 0.405 | 0.77 | 0.13 | -0.085                | 0.143 | 0.77 | 0.11 |
|                          | LRC-BART (100) | 0.032        | 0.353 | 0.93 | 0.30 | 0.001                 | 0.112 | 0.93 | 0.26 |
|                          | LRC-BART (0.9) | 0.029        | 0.350 | 0.94 | 0.31 | 0.001                 | 0.111 | 0.94 | 0.26 |

### 3.4 The single-arm design

To assess the single-arm setting of Section 4, we use the same mechanisms with no RCT control arm: the trial contributes  $n_T$  treated subjects only, 200 for Gaussian outcomes and 30 or 200 for survival outcomes, the 300 RWD controls are the sole control source, and LRC-BART runs in the single-arm mode of Section 2.6 with  $w = 1$  against the complete-pooling models, the hierarchical models and BART-CP; Sc3 is not identifiable without an internal control and is omitted. Table 4 reports the results and Web Appendix G.7 discusses them in detail. Under Sc1 the tree models have Gaussian RMSE 0.16 with coverage 0.94 to 1.00.

Under Sc2, all methods have substantial bias because there are no internal controls to identify a discrepancy and RWD overlap is limited: on the Gaussian ATE, BART-CP has bias 0.41 with coverage 0.75 and LRC-BART 0.42. Adding a zero-mean discrepancy preserves the expected Gaussian control mean, but nonlinear survival-ratio means can change. The  $w = 0.9$  sensitivity substantially increases their uncertainty and bias (Web Appendix G.7), supporting  $w = 1$  for the primary single-arm analysis.

Table 4: Single-arm design,  $R = 500$ : the trial contributes  $n_T$  treated subjects only and the  $n_2 = 300$  RWD controls are the sole control source, under Sc1 (no confounding) and Sc2 (measured confounding). LRC-BART (pooled) is  $w = 1$  with the spike scale at the grid minimum, LRC-BART (0.9) is  $w = 1$  at 0.9 of the ESS ceiling, and LRC-BART ( $w = 0.9$ ) is the slab sensitivity at the absolute target 100. HierLM and HierAFT use the stated hyperprior  $\tau_j^2 \sim \text{Inv-Gamma}(1.5, 0.75)$  with no RCT controls, under which the control surface is drawn from the hierarchy; values above  $10^3$  are printed as such. Control columns are on the outcome scale (Gaussian) or the log-median scale (survival). Web Table S6 adds the absolute-target rows and the median ratio at  $n_T = 200$ .

| Design                              | Method                 | Treatment effect |       |       |      |      |        | Control |       |     |
|-------------------------------------|------------------------|------------------|-------|-------|------|------|--------|---------|-------|-----|
|                                     |                        | Bias             | SD    | RMSE  | Cov  | Pow  | Bias   | RMSE    | Prior | ESS |
| Gaussian, $n_T = 200$<br>ATE<br>Sc1 | LM-CP                  | 0.007            | 0.259 | 0.254 | 0.95 | 0.97 | -0.003 | 0.233   | —     |     |
|                                     | HierLM                 | -0.014           | 8.239 | 0.497 | 1.00 | 0.00 | 0.018  | 0.487   | —     |     |
|                                     | BART-CP                | 0.010            | 0.163 | 0.162 | 0.94 | 1.00 | -0.003 | 0.127   | —     |     |
|                                     | LRC-BART (pooled)      | 0.007            | 0.165 | 0.162 | 0.94 | 1.00 | 0.001  | 0.127   | 172.3 |     |
|                                     | LRC-BART (0.9)         | 0.007            | 0.202 | 0.163 | 0.98 | 1.00 | 0.000  | 0.128   | 115.8 |     |
|                                     | LRC-BART ( $w = 0.9$ ) | 0.010            | 1.383 | 0.169 | 1.00 | 0.00 | -0.003 | 0.136   | 81.9  |     |
| Gaussian, $n_T = 200$<br>ATE<br>Sc2 | LM-CP                  | 1.682            | 0.417 | 1.734 | 0.01 | 1.00 | -1.676 | 1.726   | —     |     |
|                                     | HierLM                 | 1.642            | 8.317 | 1.754 | 1.00 | 0.00 | -1.636 | 1.747   | —     |     |
|                                     | BART-CP                | 0.408            | 0.310 | 0.506 | 0.75 | 1.00 | -0.399 | 0.492   | —     |     |
|                                     | LRC-BART (pooled)      | 0.420            | 0.316 | 0.515 | 0.75 | 1.00 | -0.412 | 0.501   | 34.8  |     |
|                                     | LRC-BART (0.9)         | 0.417            | 0.416 | 0.511 | 0.89 | 0.98 | -0.409 | 0.498   | 23.1  |     |
|                                     | LRC-BART ( $w = 0.9$ ) | 0.416            | 1.329 | 0.511 | 1.00 | 0.00 | -0.408 | 0.498   | 34.8  |     |

Table 4, continued

| Design                                      | Method                 | Treatment effect |          |          |      |      | Control  |          |           |
|---------------------------------------------|------------------------|------------------|----------|----------|------|------|----------|----------|-----------|
|                                             |                        | Bias             | SD       | RMSE     | Cov  | Pow  | Bias     | RMSE     | Prior ESS |
| Survival, $n_T = 30$<br>median ratio<br>Sc1 | AFT-CP                 | 1.540            | 4.678    | 4.444    | 0.92 | 0.15 | -0.006   | 0.204    | —         |
|                                             | HierAFT                | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | $> 10^3$ | $> 10^3$ | —         |
|                                             | BART-CP                | -0.013           | 0.580    | 0.482    | 0.97 | 0.05 | -0.026   | 0.103    | —         |
|                                             | LRC-BART (pooled)      | 0.182            | 0.918    | 0.686    | 0.98 | 0.02 | -0.017   | 0.102    | 80.2      |
|                                             | LRC-BART (0.9)         | 0.222            | 1.174    | 0.711    | 0.99 | 0.01 | -0.005   | 0.133    | 51.8      |
|                                             | LRC-BART ( $w = 0.9$ ) | 1.423            | 11.199   | 1.959    | 1.00 | 0.00 | 0.318    | 0.433    | 79.3      |
| Survival, $n_T = 30$<br>median ratio<br>Sc2 | AFT-CP                 | 20.944           | 46.014   | 51.645   | 0.19 | 0.89 | -0.441   | 0.526    | —         |
|                                             | HierAFT                | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | $> 10^3$ | $> 10^3$ | —         |
|                                             | BART-CP                | 0.814            | 1.109    | 1.231    | 0.92 | 0.25 | -0.199   | 0.277    | —         |
|                                             | LRC-BART (pooled)      | 1.324            | 1.859    | 1.864    | 0.95 | 0.21 | -0.207   | 0.284    | 29.2      |
|                                             | LRC-BART (0.9)         | 1.511            | 3.461    | 2.055    | 0.97 | 0.12 | -0.180   | 0.279    | 19.3      |
|                                             | LRC-BART ( $w = 0.9$ ) | 4.243            | 28.669   | 5.305    | 1.00 | 0.00 | 0.179    | 0.537    | 28.9      |
| Survival, $n_T = 30$<br>RMST ratio<br>Sc1   | AFT-CP                 | 0.036            | 0.178    | 0.202    | 0.90 | 0.16 | 0.079    | 0.187    | —         |
|                                             | HierAFT                | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.311    | 0.355    | —         |
|                                             | BART-CP                | -0.111           | 0.176    | 0.186    | 0.92 | 0.02 | -0.021   | 0.089    | —         |
|                                             | LRC-BART (pooled)      | -0.046           | 0.192    | 0.157    | 0.97 | 0.02 | 0.002    | 0.085    | 80.2      |
|                                             | LRC-BART (0.9)         | -0.040           | 0.218    | 0.156    | 0.98 | 0.01 | 0.002    | 0.085    | 51.8      |
|                                             | LRC-BART ( $w = 0.9$ ) | 0.080            | 0.998    | 0.200    | 1.00 | 0.00 | 0.020    | 0.089    | 79.3      |
| Survival, $n_T = 30$<br>RMST ratio<br>Sc2   | AFT-CP                 | 1.216            | 0.522    | 1.336    | 0.13 | 0.94 | -0.518   | 0.559    | —         |
|                                             | HierAFT                | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.017    | 0.201    | —         |
|                                             | BART-CP                | 0.149            | 0.271    | 0.286    | 0.95 | 0.14 | -0.219   | 0.260    | —         |
|                                             | LRC-BART (pooled)      | 0.234            | 0.297    | 0.339    | 0.94 | 0.19 | -0.210   | 0.251    | 29.2      |
|                                             | LRC-BART (0.9)         | 0.256            | 0.452    | 0.358    | 0.97 | 0.13 | -0.207   | 0.248    | 19.3      |
|                                             | LRC-BART ( $w = 0.9$ ) | 0.527            | 2.689    | 0.643    | 1.00 | 0.00 | -0.185   | 0.228    | 28.9      |
| Survival, $n_T = 200$<br>RMST ratio<br>Sc1  | AFT-CP                 | -0.023           | 0.101    | 0.098    | 0.94 | 0.15 | 0.074    | 0.114    | —         |
|                                             | HierAFT                | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.305    | 0.317    | —         |
|                                             | BART-CP                | -0.001           | 0.084    | 0.082    | 0.97 | 0.32 | -0.021   | 0.061    | —         |
|                                             | LRC-BART (pooled)      | -0.024           | 0.078    | 0.076    | 0.95 | 0.25 | 0.001    | 0.054    | 169.5     |
|                                             | LRC-BART (0.9)         | -0.022           | 0.094    | 0.076    | 0.99 | 0.17 | 0.002    | 0.054    | 114.0     |
|                                             | LRC-BART ( $w = 0.9$ ) | 0.109            | 1.179    | 0.275    | 1.00 | 0.00 | 0.021    | 0.057    | 81.8      |
| Survival, $n_T = 200$<br>RMST ratio<br>Sc2  | AFT-CP                 | 1.164            | 0.428    | 1.244    | 0.01 | 1.00 | -0.535   | 0.550    | —         |
|                                             | HierAFT                | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.001    | 0.101    | —         |
|                                             | BART-CP                | 0.305            | 0.206    | 0.361    | 0.61 | 0.77 | -0.233   | 0.261    | —         |
|                                             | LRC-BART (pooled)      | 0.284            | 0.206    | 0.336    | 0.70 | 0.74 | -0.224   | 0.250    | 35.9      |
|                                             | LRC-BART (0.9)         | 0.301            | 0.350    | 0.353    | 0.85 | 0.50 | -0.222   | 0.248    | 23.9      |
|                                             | LRC-BART ( $w = 0.9$ ) | 0.562            | 2.419    | 0.633    | 1.00 | 0.00 | -0.200   | 0.228    | 36.2      |

## 4 Application: a single-arm myeloma trial with a registry control

### 4.1 Data and harmonization

EloKRd is a single-arm phase II trial of elotuzumab with carfilzomib, lenalidomide and dexamethasone in 30 patients with newly diagnosed multiple myeloma, with progression-free survival (PFS) and overall survival (OS) measured from the first day of treatment; 13 patients progressed or died and 10 died over a maximum follow-up of 6.75 years. The external control is the University of Chicago Multiple Myeloma (UCMM) registry, an institutional database of 797 patients from which we retain, in order, those without negative survival times, with a recorded induction start,

no transplant on or before induction, and a regimen other than elotuzumab with lenalidomide and dexamethasone, leaving 253 patients on standard induction regimens, mostly VRd, Vd, KRd and CyBorD, with survival measured from the start of induction. Five baseline covariates are complete in both sources: age, sex, race, ethnicity and high-risk cytogenetics. We also adjust for recorded autologous stem cell transplantation (ASCT) after induction (Web Table S5). The trial cohort is enriched for high-risk cytogenetics (50% against 23%) and male sex (70% against 51%), and far fewer trial patients received ASCT (30% against 80%). The unadjusted Kaplan–Meier curves give a 5-year RMST ratio of 1.015 for PFS and 0.90 for OS, with log-rank  $p$ -values of 0.51 and 0.048.

We harmonize race and ethnicity before fitting the models and retain the original coding as a sensitivity analysis (Web Appendix G.5). The ESS ceilings under the original coding are 12.6 for PFS and 14.6 for OS, compared with 63.7 and 36.7 after harmonization; a target of 30 is capped under the original coding.

## 4.2 Estimand and calibration

Neither trial median is reached, and the registry median PFS of 5.1 years falls near the end of the trial follow-up, so we take the standardized 5-year RMST ratio as the primary estimand and the median ratio as a secondary summary. Writing  $\bar{S}_a(t)$  for the arm- $a$  log-normal survival function averaged over the 30 trial profiles,  $\text{RMST}_a(5) = \int_0^5 \bar{S}_a(t) dt$  has a closed form under the AFT model and is evaluated at every posterior draw. The treated arm is AFT-BART with 50 trees fitted to the 30 trial patients, and the control arm is LRC-BART in single-arm mode with  $f$  an AFT-BART of  $H_f = 10$  trees on the registry,  $H_g = 5$ , the split prior  $(0.5, 3)$ ,  $\nu_0 = 3$  and  $w = 1$ , with  $w = 0.9$  and  $H_f = 50$  as sensitivity analyses. Four chains of 2,000 burn-in and 2,000 retained iterations give a split- $\hat{R}$  of 1.000 and bulk effective sizes above 6,800 of 8,000 draws for the log RMST ratio.

The application is a descriptive model-based comparison standardized to the recorded trial profiles, including post-induction ASCT. This adjustment does not identify a total causal treatment effect, and the single-arm data cannot verify compatibility with an unobserved trial control surface. Calibration uses the registry alone;  $\sigma_1 = \sigma_2$  is also assumed for counterfactual survival prediction. The resulting variance-based ceiling concerns the standardized control log-time mean at the 30 trial profiles: 63.7 RCT-equivalent controls for PFS and 36.7 for OS, far below the 253 registry patients, because the trial profiles, half of them high-risk and 70% without ASCT, lie where registry support

is limited. A target of 30 is therefore about half the ceiling for PFS and most of the ceiling for OS, and the Pr-rule returns  $s_0^2 = 0.0056$  and  $0.0050$  with  $\text{ESS}_0$  of 28.1 and 24.4. We report the absolute ladder  $N_{\text{target}} \in \{30, 22.5, 15\}$  and the fractional ladder at 0.9, 0.5 and 0.25 of the ceiling; the realized ESS equals  $\text{ESS}_0$  within one unit in every row, consistent with  $g$  remaining a prior draw.

### 4.3 Results

Table 5 reports the standardized 5-year RMST ratio for both endpoints. At  $N_{\text{target}} = 30$  LRC-BART gives 0.97 (0.77, 1.23) for PFS and 0.90 (0.77, 1.04) for OS, with  $P(\text{ratio} > 1)$  of 0.37 and 0.08; lowering the target to 15 widens the intervals to (0.75, 1.40) and (0.76, 1.11) and moves the means by at most 0.02, and the fractional ladder does the same.

The harmonized reference analyses give similar estimates: AFT-BART on the registry with no discrepancy gives 0.97 and 0.90, and the complete-pooling AFT gives 0.87 (0.69, 1.06) and 0.80 (0.65, 0.94), the only interval in the table that excludes one, while the hierarchical AFT stays near 1.2 for both endpoints (Web Appendix G.5). Under the harmonized coding, the nonparametric 95% intervals include one for both endpoints; the OS point estimates are below one. These are adjusted descriptive comparisons conditional on the recorded profiles and discrepancy prior.

The earlier MAP-AFT-BART rows in Table 5 had 97.5% limits of 16 to 33 for PFS and 4.8 to 7.2 for OS; the current fits have shorter observed tails (Web Appendix G.5). The  $w = 0.9$  sensitivity widens the PFS interval to (0.74, 1.48) and changes its posterior mean from 0.97 to 1.00, illustrating the nonlinear effect of discrepancy uncertainty, and  $H_f = 50$  changes the means by less than 0.015.

For this single-arm analysis, Figure 2 reports the local prior ESS  $\text{ESS}(x)$  at the 30 trial profiles. It ranges from 5 to 17 for PFS (median 9.5) and 6 to 15 for OS, highest for White patients with ASCT and standard-risk cytogenetics, where registry support is strongest, and lowest for the one Hispanic patient, the Other/Unknown-race patients, and the high-risk profiles without ASCT. The compatibility probability  $\mathcal{B}_c$  is constant across profiles in this single-arm model, since  $g$  is a prior draw here, mean zero with SD 0.26 on the log-time scale at  $N_{\text{target}} = 30$  at every profile (Web Figure S1); the patient-specific summary therefore describes registry precision under the stated model. It does not allocate the target across patients. A target of 30 does not imply equally strong external support at every trial profile.

Table 5: EloKRd versus UCMM: standardized 5-year RMST ratio, posterior mean [posterior median] and 95% credible interval, for progression-free survival (PFS) and overall survival (OS) at the 30 EloKRd profiles. The ESS ceilings are 63.7 (PFS) and 36.7 (OS). Reference rows are rerun under the harmonized covariate coding of Section 4.1; the Kaplan–Meier row is unadjusted and reports the estimate with its large-sample interval from the survRM2 package. The published rows are those of the earlier analysis under the original coding, reported there as posterior medians.

| Method                                                                                         | PFS                      | OS                       | ESS <sub>0</sub> (PFS / OS) |
|------------------------------------------------------------------------------------------------|--------------------------|--------------------------|-----------------------------|
| <i>Reference methods, harmonized coding</i>                                                    |                          |                          |                             |
| Kaplan–Meier, unadjusted                                                                       | 1.01 (0.84, 1.22)        | 0.90 (0.79, 1.03)        |                             |
| AFT, complete pooling                                                                          | 0.87 [0.87] (0.69, 1.06) | 0.80 [0.80] (0.65, 0.94) |                             |
| AFT, hierarchical ( $s^2 = 0.5$ )                                                              | 1.23 [1.21] (0.74, 1.90) | 1.21 [1.19] (0.77, 1.70) |                             |
| AFT-BART on UCMM, no discrepancy                                                               | 0.97 [0.97] (0.79, 1.16) | 0.90 [0.91] (0.78, 1.02) |                             |
| <i>LRC-BART, single-arm mode, harmonized coding, <math>H_f = 10</math>, <math>w = 1</math></i> |                          |                          |                             |
| $N_{\text{target}} = 30$                                                                       | 0.97 [0.96] (0.77, 1.23) | 0.90 [0.90] (0.77, 1.04) | 28.1 / 24.4                 |
| $N_{\text{target}} = 22.5$                                                                     | 0.98 [0.96] (0.76, 1.29) | 0.91 [0.90] (0.76, 1.06) | 20.6 / 19.0                 |
| $N_{\text{target}} = 15$                                                                       | 0.99 [0.96] (0.75, 1.40) | 0.91 [0.90] (0.76, 1.11) | 14.2 / 12.7                 |
| 0.9 $\times$ ceiling                                                                           | 0.97 [0.96] (0.79, 1.15) | 0.90 [0.90] (0.77, 1.04) | 52.4 / 26.8                 |
| 0.5 $\times$ ceiling                                                                           | 0.97 [0.96] (0.78, 1.22) | 0.91 [0.90] (0.76, 1.09) | 29.6 / 15.3                 |
| 0.25 $\times$ ceiling                                                                          | 0.99 [0.96] (0.75, 1.38) | 0.92 [0.91] (0.75, 1.16) | 15.4 / 8.2                  |
| <i>Sensitivity analyses at <math>N_{\text{target}} = 30</math></i>                             |                          |                          |                             |
| $w = 0.9$                                                                                      | 1.00 [0.97] (0.74, 1.48) | 0.91 [0.90] (0.76, 1.10) | 28.1 / 24.4                 |
| $H_f = 50$                                                                                     | 0.98 [0.97] (0.78, 1.23) | 0.91 [0.91] (0.77, 1.05) | 27.1 / 25.3                 |
| Original covariate coding                                                                      | 1.04 [1.01] (0.79, 1.43) | 0.91 [0.90] (0.76, 1.07) | 12.9 / 15.0                 |
| <i>Published rows (original coding, posterior medians)</i>                                     |                          |                          |                             |
| AFT, complete pooling                                                                          | 1.03 (0.71, 1.61)        | 0.90 (0.71, 1.17)        |                             |
| AFT, hierarchical ( $s^2 = 0.5$ )                                                              | 1.21 (0.73, 2.68)        | 1.18 (0.77, 1.89)        |                             |
| Standard BART                                                                                  | 1.06 (0.79, 1.51)        | 0.90 (0.76, 1.08)        |                             |
| MAP-AFT-BART, $N_{\text{target}} = 30$                                                         | 1.08 (0.66, 16.40)       | 0.97 (0.74, 4.77)        |                             |
| MAP-AFT-BART, $N_{\text{target}} = 15$                                                         | 1.10 (0.65, 32.80)       | 1.00 (0.73, 7.22)        |                             |

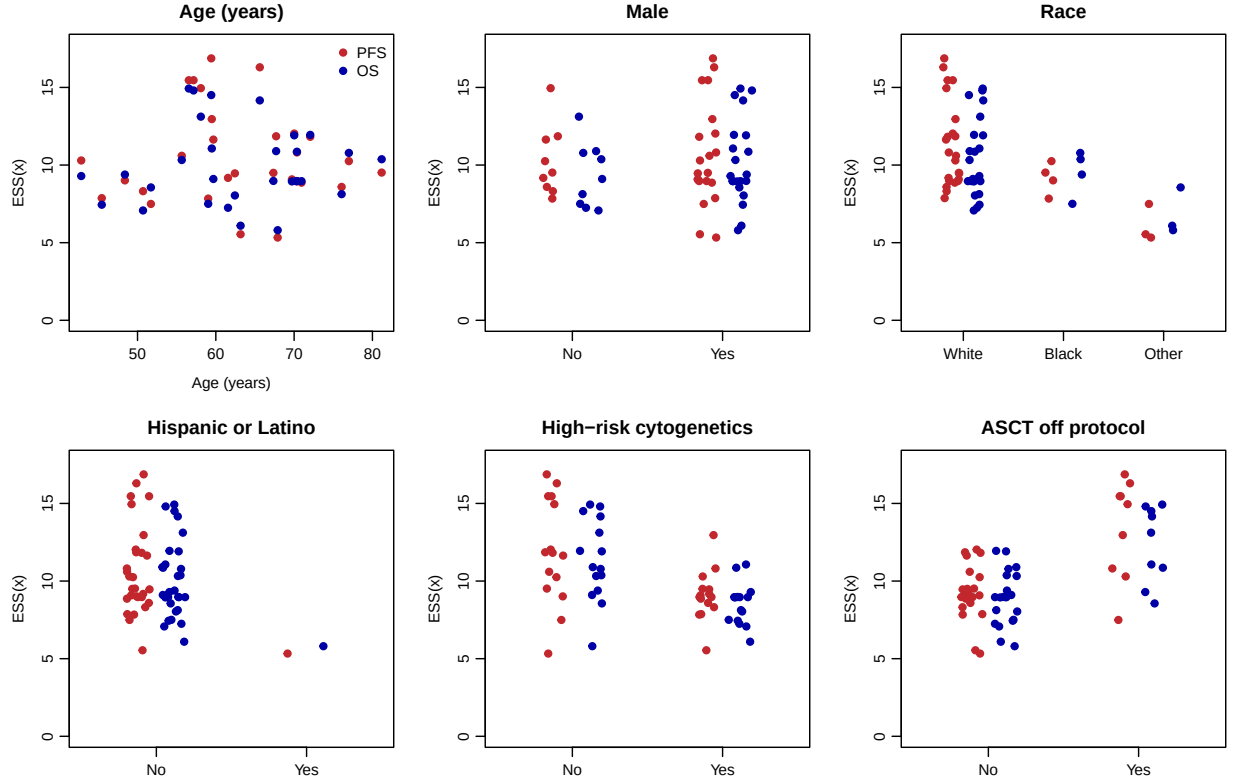


Figure 2: Local prior effective sample size  $ESS(x) = \hat{\sigma}_1^2 / \{V_f(x) + H_g \bar{\tau}_0^2\}$  at the 30 EloKRd profiles for PFS (red) and OS (blue) at  $N_{\text{target}} = 30$ , plotted against each harmonized covariate;  $V_f(x)$  is the UCMM posterior variance of the log-time surface at  $x$ .

## 5 Discussion

LRC-BART combines an efficiency gain under agreement with a covariate-resolved assessment of compatibility. In the Gaussian simulations, its RMSE is about 16% lower than BART-PP’s under Sc1. Under global disagreement, its residual bias is about 0.04. For a regional shift of 1.3 residual SDs, the compatibility map and realized regional ESS distinguish the two regions, although the transition spans about one unit of the splitting covariate. The control-surface RMSE remains higher than BART-PP’s, including away from the boundary. At a regional shift of two thirds of a residual SD, excessive borrowing produces bias of 0.04 and coverage of 0.89. The leaf calculation in Theorem 2(a) describes the dependence of detection on the spike scale and local sample size; it does not establish the performance of the full fitted ensemble.

The CAHB comparison illustrates the difficulty of kernel-local borrowing in our nonlinear ten-covariate design. It also depends on our choices for smoothing, precision scaling and joint posterior draws. We therefore interpret it as an assessment of this transfer of CAHB, with the variants reported alongside it. The survival study shows a further limitation: with 75 controls and censoring, the median ratio can amplify control-surface bias and posterior uncertainty. We report the RMST ratio with it. In a single-arm study, trial data cannot identify a control discrepancy. The ceiling and local prior ESS then describe external support at trial profiles, but cannot demonstrate compatibility with an unobserved trial control surface.

The theoretical results condition on features that are learned in the fitted model. Map consistency requires fixed discrepancy partitions and a true discrepancy in their additive span. We have no corresponding result for the posterior over partitions. The leaf-level influence bound also treats  $f$  as fixed; uncertainty in the fitted pooled surface perturbs the residuals used to estimate  $g$ . These limitations prevent a direct extension of the conditional results to the full posterior. The observed surface loss under Sc4 remains an empirical limitation even when profiles near the boundary are excluded.

The spike scale is global, with locality determined by the indicators. Leaf-specific scales could accommodate different degrees of compatibility, at the cost of the  $\text{Inv-}\chi^2$  conjugacy. Multiple external sources would require source-specific discrepancies and a joint calibration that accounts for their information at the trial profiles; additivity of the ceilings has not been established. Binary

and count outcomes require an alternative likelihood, and a nonparametric survival model would require a different RMST calculation. Prior-ESS calibration alone does not control Type I error or power. A confirmatory design would need to evaluate those operating characteristics under specified scenarios and pre-specify how the compatibility map would be used.

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## Web Appendices

### Web Appendix A: Model, notation, and exact leaf marginals

This appendix collects the proofs for the locally robust commensurate BART (LRC-BART) model of the main text. We fix notation first. The two control sources are indexed by  $g = 1$  (RCT control) and  $g = 2$  (RWD control), with residual variances  $\sigma_1^2$  and  $\sigma_2^2$ . The control outcomes follow

$$y_i = f(x_i) + D_i g(x_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_{g_i}^2), \quad D_i = \mathbf{1}\{i \text{ is an RCT control}\}, \quad (\text{A11})$$

where  $f$  is a standard BART ensemble of  $H_f$  trees with leaf prior  $\mathcal{N}(0, \lambda_f^2)$  and  $g$  is an ensemble of  $H_g$  trees whose leaves carry the two-component prior

$$\theta_{h\ell} \mid z_{h\ell}, \tau_0^2, \tau_1^2 \sim \mathcal{N}(0, \tau_{1-z_{h\ell}}^2), \quad z_{h\ell} \mid w \stackrel{\text{iid}}{\sim} \text{Bernoulli}(w), \quad (\text{A12})$$

with  $z = 1$  the spike (commensurate component) and  $z = 0$  the slab,  $w \sim \text{Beta}(a_w, b_w)$ ,  $\tau_1^2$  fixed by the range rule, and

$$\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2) \text{ truncated to } (0, \tau_1^2), \quad (\text{A13})$$

which is the prior the sampler implements (its conditional for  $\tau_0^2$  is the scaled-Inv- $\chi^2(\nu_0 + K_0, \cdot)$  law truncated to  $(0, \tau_1^2)$ , which is conjugate for (A13) and not for the untruncated prior). Throughout we impose

**Assumption A1** ((A1)).  $0 < \tau_0^2 < \tau_1^2 < \infty$ .

Under (A13), Assumption A1 holds almost surely under the prior and along the chain. The variance-based calibration in Web Appendix D uses the untruncated scaled-Inv- $\chi^2(\nu_0, s_0^2)$  law. The mass removed by the fitting prior is  $P(\chi_{\nu_0}^2 < \nu_0 s_0^2 / \tau_1^2)$  and depends on the fitted calibration scale; the two laws are distinguished in that appendix.

Within a leaf,  $n_g$  and  $\bar{y}_g$  denote the number and the mean of the source- $g$  partial residuals routed to the leaf,  $v_g = \sigma_g^2 / n_g$ , and  $\phi(\cdot; m, V)$  is the  $\mathcal{N}(m, V)$  density;  $\Phi$  is the standard normal distribution function and  $\bar{\Phi} = 1 - \Phi$ . For a  $g$  leaf only RCT controls enter, and we write  $\bar{y} = \bar{y}_1$ ,  $v = v_1$ ,  $n = n_1$  when no confusion arises. The identity we use repeatedly is the Gaussian product

rule

$$\phi(\theta; 0, \tau^2) \phi(\bar{y}; \theta, v) = \phi(\bar{y}; 0, \tau^2 + v) \phi\left(\theta; \frac{\tau^2}{\tau^2 + v} \bar{y}, \frac{\tau^2 v}{\tau^2 + v}\right), \quad (\text{A14})$$

which follows by completing the square in  $\theta$ .

### A.1 Exact leaf marginals

**Lemma A1** (Exact leaf marginals). *Fix all trees other than tree  $h$ , their leaf parameters, the residual variances,  $\lambda_f^2$ , and  $(\tau_0^2, \tau_1^2, w)$ .*

(a) (*f* leaf.) *Let  $r_i = y_i - f_{-h}(x_i) - D_i g(x_i)$  be the partial residuals of the control observations routed to a leaf of  $f$ -tree  $h$ , so that  $r_i | \theta \stackrel{\text{indep}}{\sim} \mathcal{N}(\theta, \sigma_{g_i}^2)$  and  $\theta \sim \mathcal{N}(0, \lambda_f^2)$ . With  $A = n_1/\sigma_1^2 + n_2/\sigma_2^2$  and  $B = S_1/\sigma_1^2 + S_2/\sigma_2^2$ ,  $S_g = \sum_{i:g_i=g} r_i$ ,*

$$\int \prod_i \phi(r_i; \theta, \sigma_{g_i}^2) \phi(\theta; 0, \lambda_f^2) d\theta = \left[ \prod_i \phi(r_i; 0, \sigma_{g_i}^2) \right] M_f, \quad M_f = \frac{\exp\{\lambda_f^2 B^2 / \{2(1 + \lambda_f^2 A)\}\}}{(1 + \lambda_f^2 A)^{1/2}}. \quad (\text{A15})$$

(b) (*g* leaf.) *Let  $r_i = y_i - f(x_i) - g_{-h}(x_i)$  be the partial residuals of the  $n_1$  RCT controls routed to a leaf of  $g$ -tree  $h$ , with mean  $\bar{y}$ . Under the prior (A12),*

$$\sum_{z \in \{0,1\}} P(z) \int \prod_i \phi(r_i; \theta, \sigma_1^2) \phi(\theta; 0, \tau_{1-z}^2) d\theta = \left[ \prod_i \phi(r_i; 0, \sigma_1^2) \right] M_g, \quad (\text{A16})$$

$$M_g = \frac{w \phi(\bar{y}; 0, \tau_0^2 + v) + (1 - w) \phi(\bar{y}; 0, \tau_1^2 + v)}{\phi(\bar{y}; 0, v)}, \quad (\text{A17})$$

where the index  $1 - z$  selects  $\tau_0^2$  for  $z = 1$  and  $\tau_1^2$  for  $z = 0$ . A leaf with  $n_1 = 0$  has  $M_g = 1$ .

(c) (Leaf conditionals.) *In a  $g$  leaf,*

$$P(z = 1 | \bar{y}) = \frac{w \phi(\bar{y}; 0, \tau_0^2 + v)}{w \phi(\bar{y}; 0, \tau_0^2 + v) + (1 - w) \phi(\bar{y}; 0, \tau_1^2 + v)}, \quad (\text{A18})$$

$$\theta | z, \bar{y} \sim \mathcal{N}\left(\frac{\tau_{1-z}^2}{\tau_{1-z}^2 + v} \bar{y}, \frac{\tau_{1-z}^2 v}{\tau_{1-z}^2 + v}\right).$$

and in an  $f$  leaf  $\theta | \cdot \sim \mathcal{N}(\lambda_f^2 B / (1 + \lambda_f^2 A), \lambda_f^2 / (1 + \lambda_f^2 A))$ .

*Proof.* (a) Expanding  $-(r_i - \theta)^2 / (2\sigma_{g_i}^2) = -r_i^2 / (2\sigma_{g_i}^2) + r_i \theta / \sigma_{g_i}^2 - \theta^2 / (2\sigma_{g_i}^2)$  and summing over  $i$  gives

$\prod_i \phi(r_i; \theta, \sigma_{g_i}^2) = \prod_i \phi(r_i; 0, \sigma_{g_i}^2) \exp(\theta B - \theta^2 A/2)$ . Multiplying by  $\phi(\theta; 0, \lambda_f^2)$  and integrating, the exponent is  $-\frac{1}{2}(A + \lambda_f^{-2})\theta^2 + B\theta$ , a Gaussian integral equal to  $(2\pi)^{1/2}(A + \lambda_f^{-2})^{-1/2} \exp\{B^2/(2(A + \lambda_f^{-2}))\}$ ; dividing by the prior normalizing constant  $(2\pi\lambda_f^2)^{1/2}$  gives  $M_f$ . The conditional of  $\theta$  is read off the same quadratic form.

(b) With a single source, (a) gives  $\prod_i \phi(r_i; \theta, \sigma_1^2) = \prod_i \phi(r_i; 0, \sigma_1^2) \exp\{\theta n \bar{y}/\sigma_1^2 - \theta^2 n/(2\sigma_1^2)\}$ , and the exponential equals  $\phi(\bar{y}; \theta, v)/\phi(\bar{y}; 0, v)$ , since  $-\{(\bar{y} - \theta)^2 - \bar{y}^2\}/(2v) = (2\bar{y}\theta - \theta^2)n/(2\sigma_1^2)$ . Hence the likelihood factors as  $\prod_i \phi(r_i; 0, \sigma_1^2) \times \phi(\bar{y}; \theta, v)/\phi(\bar{y}; 0, v)$ , the sufficiency of  $\bar{y}$ . Integrating  $\phi(\bar{y}; \theta, v)$  against  $\phi(\theta; 0, \tau_{1-z}^2)$  by (A14) gives  $\phi(\bar{y}; 0, \tau_{1-z}^2 + v)$ , and the sum over  $z$  with weights  $w$  and  $1 - w$  gives  $M_g$ . When  $n_1 = 0$  the leaf has no likelihood factor and the integral of the prior is one.

(c) By (A14), the joint density of  $(z, \theta)$  given  $\bar{y}$  is proportional to

$$P(z) \phi(\bar{y}; 0, \tau_{1-z}^2 + v) \phi\left(\theta; \frac{\tau_{1-z}^2}{\tau_{1-z}^2 + v} \bar{y}, \frac{\tau_{1-z}^2 v}{\tau_{1-z}^2 + v}\right).$$

Integrating out  $\theta$  gives the stated Bernoulli weight; the remaining factor is the conditional density of  $\theta$ .  $\square$

**Proposition A1** (The tree moves target the exact posterior). *Condition on  $(\sigma_1^2, \sigma_2^2, \lambda_f^2, \tau_0^2, \tau_1^2, w)$  and on all trees other than  $h$  together with their leaf parameters. Let the update of tree  $h$  consist of a Metropolis–Hastings step on the partition  $T_h$  with grow, prune, and change proposals, acceptance ratio equal to the tree-prior ratio times the proposal ratio times the ratio of products of  $M_f$  (for  $f$  trees) or  $M_g$  (for  $g$  trees) over the affected leaves, followed by a draw of the leaf parameters of  $T_h$  (and, for a  $g$  tree, the indicators) from (A18). Then the composite update leaves the conditional posterior of  $(T_h, \theta_h)$  (resp.  $(T_h, \theta_h, z_h)$ ) invariant. Consequently the full sweep, with the conjugate updates of  $(\sigma_g^2, \tau_0^2, w)$ , the update of  $\tau_0^2$  being the truncated conditional under the prior (A13), is a partially collapsed Gibbs sampler whose invariant law is the exact joint posterior under (A12) and (A13).*

*Proof.* Given the conditioning, the marginal conditional of  $T_h$  with the leaf parameters and indicators integrated out is  $\pi(T_h) \prod_i \phi(r_i; 0, \sigma_{g_i}^2) \prod_{\ell \in T_h} M_\ell$  by Lemma A1, where  $M_\ell$  is  $M_f$  or  $M_g$ . The partial residuals  $r_i$  do not depend on  $T_h$ , and the observation-level product is over all observations of the tree,

so it is the same for the current and the proposed partition and cancels from the Metropolis–Hastings ratio. The stated acceptance ratio is therefore the exact ratio for the collapsed target; for a grow move that splits leaf  $P$  into  $L$  and  $R$ , the leaf factor ratio is  $M_L M_R / M_P$ , and the prune and change ratios follow in the same way. A Metropolis–Hastings kernel with the exact ratio leaves the collapsed conditional invariant. Following that step by a draw of the leaf parameters (and indicators) from their exact conditional given  $T_h$  gives a kernel that preserves the joint conditional of  $(T_h, \theta_h, z_h)$ , which is the marginal-then-conditional construction of Chipman et al. [5]. The conjugate updates of  $\sigma_g^2$ ,  $\tau_0^2$ , and  $w$  are ordinary Gibbs steps, so the sweep is a partially collapsed Gibbs sampler with the joint posterior as invariant law.  $\square$

**Remark A1** (No surrogate is needed). Every factor entering a tree move is closed form because, conditionally on  $(\tau_0^2, \tau_1^2, w)$ , the leaf prior (A12) is a finite Gaussian mixture, and the Gaussian likelihood integrates against each component in closed form. If a leaf-specific discrepancy variance instead has a scaled-Inv- $\chi^2$  prior and is integrated inside the leaf marginal, the resulting additive scale mixture is  $\int \phi(\bar{y}_1; 0, v_1 + \tau^2) \phi(\bar{y}_2; 0, v_2 + \tau^2) p(\tau^2) d\tau^2$ , with no elementary form. In LRC-BART the scale  $\tau_0^2$  is a global parameter updated by its own conjugate step, and the local adaptivity is carried by the per-leaf indicator  $z$ , so the leaf marginal is exact and the tree kernels target the exact posterior (Proposition A1). Posterior invariance therefore follows from the exact update construction without a surrogate-likelihood perturbation bound; Web Appendix F states its scope.

## Web Appendix B: Bounded local influence (T1)

### B.1 A single $g$ leaf

We fix a  $g$  leaf and condition on  $f$ , on the other  $g$  trees, on the partition, and on  $(\sigma_1^2, \tau_0^2, \tau_1^2, w)$ , so that  $\bar{y} | \theta \sim \mathcal{N}(\theta, v)$  with  $v = \sigma_1^2/n$ . Write

$$\rho_j = \frac{\tau_j^2}{\tau_j^2 + v}, \quad \hat{\theta}_j = \rho_j \bar{y}, \quad V_j = \rho_j v = \frac{\tau_j^2 v}{\tau_j^2 + v}, \quad j \in \{0, 1\}, \quad (\text{A19})$$

so that  $\hat{\theta}_1$  is the slab estimate (close to the RCT-only residual mean  $\bar{y}$  when  $\tau_1^2 \gg v$ ) and  $\hat{\theta}_0$  the spike-shrunk estimate (close to zero, that is, to the pooled surface  $f$ , when  $\tau_0^2 \ll v$ ). The subscript  $j$  of  $\rho_j$ ,  $\hat{\theta}_j$ ,  $V_j$  indexes the scale  $\tau_j^2$ , so  $j = 1$  is the slab, whereas the indicator convention of (A12) has

$z = 1$  for the spike. Under Assumption A1,  $0 < \rho_0 < \rho_1 < 1$ . Let  $p = P(z = 0 | \bar{y})$  be the posterior slab probability and define

$$\kappa = \frac{1}{2} \left( \frac{1}{\tau_0^2 + v} - \frac{1}{\tau_1^2 + v} \right), \quad a = \log \frac{1-w}{w} + \frac{1}{2} \log \frac{\tau_0^2 + v}{\tau_1^2 + v}, \quad C = e^{-a} = \frac{w}{1-w} \sqrt{\frac{\tau_1^2 + v}{\tau_0^2 + v}}. \quad (\text{A20})$$

Under Assumption A1,  $\kappa > 0$ , and a direct computation gives

$$\rho_1 - \rho_0 = \frac{v(\tau_1^2 - \tau_0^2)}{(\tau_1^2 + v)(\tau_0^2 + v)} = 2v\kappa. \quad (\text{A21})$$

**Theorem A1** (Bounded local influence). *Under Assumption A1 and the conditioning above, the following hold for every  $\bar{y} \in \mathbb{R}$ ; parts (b) to (d) assume  $0 < w < 1$ .*

(a) (Posterior.)  $\theta | \bar{y} \sim p\mathcal{N}(\hat{\theta}_1, V_1) + (1-p)\mathcal{N}(\hat{\theta}_0, V_0)$ , and hence

$$\mathbb{E}[\theta | \bar{y}] = p \frac{\tau_1^2}{\tau_1^2 + v} \bar{y} + (1-p) \frac{\tau_0^2}{\tau_0^2 + v} \bar{y}. \quad (\text{A22})$$

(b) (Slab probability.)  $\text{logit} p = \log \frac{p}{1-p} = a + \kappa \bar{y}^2$ , with  $a$  and  $\kappa$  as in (A20).

(c) (Bounded spike contribution and bounded RWD-induced shift.) *The spike-shrunk part of (A22) satisfies  $|(1-p)\hat{\theta}_0| \leq |\bar{y}| \tau_0^2 / (\tau_0^2 + v)$ . The shift of the posterior mean relative to the slab estimate,*

$$I(\bar{y}) = \mathbb{E}[\theta | \bar{y}] - \hat{\theta}_1 = -(1-p)(\rho_1 - \rho_0) \bar{y}, \quad (\text{A23})$$

*satisfies  $|I(\bar{y})| \leq (\rho_1 - \rho_0) |\bar{y}| \min\{1, Ce^{-\kappa \bar{y}^2}\}$  and*

$$\sup_{\bar{y} \in \mathbb{R}} |I(\bar{y})| \leq 2v \sqrt{\kappa \max\{\log C, \frac{1}{2}\}} \leq \sqrt{2v \max\{\log C, \frac{1}{2}\}}. \quad (\text{A24})$$

(d) (Gaussian rate.)  $1-p \leq Ce^{-\kappa \bar{y}^2}$ , so  $p \rightarrow 1$  and  $I(\bar{y}) \rightarrow 0$  as  $|\bar{y}| \rightarrow \infty$ , at a Gaussian rate.

(e) (Nested cases.) *With  $w = 1$  (C-BART),  $p \equiv 0$  and  $I(\bar{y}) = -(\rho_1 - \rho_0)\bar{y}$ , linear and unbounded; if in addition  $\tau_0^2 \rightarrow 0$  (BART-CP),  $\mathbb{E}[\theta | \bar{y}] \rightarrow 0$  and  $I(\bar{y}) \rightarrow -\rho_1 \bar{y}$ . With  $w = 0$ ,  $p \equiv 1$  and  $I \equiv 0$ . If Assumption A1 fails with  $\tau_0^2 > \tau_1^2$ , then  $\kappa < 0$ , the roles of the two components are exchanged, and (c), (d) hold with the wide component in the role of the slab; if  $\tau_0^2 = \tau_1^2$ ,*

$p \equiv 1 - w$  and  $I \equiv 0$ .

*Proof.* (a) The posterior density is proportional to  $\{w\phi(\theta; 0, \tau_0^2) + (1 - w)\phi(\theta; 0, \tau_1^2)\}\phi(\bar{y}; \theta, v)$ . Applying (A14) to each term gives  $w\phi(\bar{y}; 0, \tau_0^2 + v)\phi(\theta; \hat{\theta}_0, V_0) + (1 - w)\phi(\bar{y}; 0, \tau_1^2 + v)\phi(\theta; \hat{\theta}_1, V_1)$ , a two-component Gaussian mixture with weights proportional to  $w\phi(\bar{y}; 0, \tau_0^2 + v)$  and  $(1 - w)\phi(\bar{y}; 0, \tau_1^2 + v)$ ; normalizing gives  $1 - p$  and  $p$  as in (A18), and (A22) is the mean of the mixture.

(b) The log odds of the two weights are

$$\log \frac{(1 - w)\phi(\bar{y}; 0, \tau_1^2 + v)}{w\phi(\bar{y}; 0, \tau_0^2 + v)} = \log \frac{1 - w}{w} - \frac{1}{2} \log \frac{\tau_1^2 + v}{\tau_0^2 + v} + \frac{\bar{y}^2}{2} \left( \frac{1}{\tau_0^2 + v} - \frac{1}{\tau_1^2 + v} \right) = a + \kappa \bar{y}^2.$$

(c) The first bound is immediate from  $0 \leq 1 - p \leq 1$ . For  $I$ , subtract  $\hat{\theta}_1$  from (A22). Since  $1 - p = (1 + e^{a + \kappa \bar{y}^2})^{-1} \leq \min\{1, e^{-a - \kappa \bar{y}^2}\} = \min\{1, Ce^{-\kappa \bar{y}^2}\}$ , the pointwise bound follows. For the supremum we show that for  $\kappa > 0$  and  $C > 0$ ,

$$\sup_{u \in \mathbb{R}} |u| \min\{1, Ce^{-\kappa u^2}\} \leq \sqrt{\max\{\log C, \frac{1}{2}\}} / \kappa. \quad (\text{A25})$$

Put  $t = \kappa u^2$ . On  $\{t \leq \log C\}$  (empty if  $C \leq 1$ ) the left side is at most  $|u| \leq \sqrt{\log C / \kappa}$ . On  $\{t > \max(\log C, 0)\}$  the left side equals  $\sqrt{t / \kappa} Ce^{-t}$ . If  $\log C \geq \frac{1}{2}$ , the function  $\sqrt{t} e^{-t}$  is decreasing on  $[\frac{1}{2}, \infty)$ , so the value is at most  $\sqrt{\log C / \kappa} Ce^{-\log C} = \sqrt{\log C / \kappa}$ . If  $\log C < \frac{1}{2}$ , then  $\sup_{t > 0} \sqrt{t} e^{-t} = (2e)^{-1/2}$  gives at most  $C(2e\kappa)^{-1/2} < \sqrt{1/(2\kappa)}$  because  $C < e^{1/2}$ . This proves (A25). Combining with (A21),  $\sup |I| \leq 2v\kappa \sqrt{\max\{\log C, \frac{1}{2}\}} / \kappa = 2v \sqrt{\kappa \max\{\log C, \frac{1}{2}\}}$ , and  $\kappa \leq 1/(2(\tau_0^2 + v)) \leq 1/(2v)$  gives the second inequality in (A24).

(d) is the pointwise bound in (c) together with  $\kappa > 0$ . (e) With  $w = 1$  the slab weight is zero for every  $\bar{y}$ ; with  $w = 0$  the spike weight is zero; the remaining statements are read off (A20) and (A23).  $\square$

**Remark A2** (What is and is not bounded). Theorem A1 bounds the influence of the commensurate component relative to the slab estimate  $\hat{\theta}_1 = \rho_1 \bar{y}$ . The slab estimate itself is shrunk toward zero, that is toward the pooled surface  $f$ , by the factor  $\rho_1$ :  $\mathbb{E}[\theta | \bar{y}] - \bar{y} = -(1 - \rho_1)\bar{y} + I(\bar{y})$  with  $1 - \rho_1 = v/(\tau_1^2 + v)$ . This linear term is the range-rule regularization that any BART leaf with prior variance  $\tau_1^2$  applies, and it is of the same form as the shrinkage of a BART-NP leaf toward its ensemble center; at  $n = 50$ ,  $\sigma_1 = 1.5$  and  $\tau_1^2 = 1.25$  its slope is 0.035. The bounded term  $I$  is the

part specific to borrowing.

**Remark A3** (Numerical illustration). With  $\sigma_1 = 1.5$ ,  $n = 50$ ,  $\tau_0^2 = 0.001$  (the scale calibrated for  $N_{\text{target}} = 100$  in the Gaussian study),  $\tau_1^2 = 1.25$  (the range rule at  $H_g = 5$  for an outcome range of 10) and  $w = 0.8$  fixed for illustration,  $v = 0.045$ ,  $\kappa = 10.5$ ,  $C = 21.2$ ,  $\log C = 3.06$ , and  $\rho_1 - \rho_0 = 0.94$ . The bound (A24) gives  $\sup |I| \leq 0.51$ ; the exact supremum of (A23), a one-dimensional maximization, is 0.31, attained at  $\bar{y} = 0.43$ . By comparison C-BART's shift is  $0.94 |\bar{y}|$ , which exceeds 0.51 once  $|\bar{y}| > 0.54$ .

## B.2 The estimand level: fixed partitions of the $g$ ensemble

The estimand is the RCT-standardized control mean  $\mu = N^{-1} \sum_{i \leq N} f_1(x_i)$  over the  $N$  RCT covariate profiles, and  $f_1 = f + g$ , so  $\mu = \mu_f + \mu_g$  with  $\mu_g = N^{-1} \sum_{i \leq N} g(x_i)$ . We condition on  $f$ , on the partitions  $\mathcal{T} = (T_1, \dots, T_{H_g})$  of the  $g$  trees, and on  $(\sigma_1^2, \tau_0^2, \tau_1^2, w)$ ; the RWD then enters  $\mu_g$  only through the shrinkage of  $g$  toward zero. Index the  $L_g$  leaves of all  $g$  trees by  $j$ , let  $\theta \in \mathbb{R}^{L_g}$  collect the leaf parameters and  $z \in \{0, 1\}^{L_g}$  the indicators, and let  $r \in \mathbb{R}^{n_1}$  be the residuals  $y_i - f(x_i)$  of the RCT controls. Let  $X \in \{0, 1\}^{n_1 \times L_g}$  be the incidence matrix ( $X_{ij} = 1$  if control  $i$  falls in leaf  $j$ ), so each row of  $X$  has exactly  $H_g$  ones and  $g(x_i) = (X\theta)_i$ . Let  $a \in \mathbb{R}^{L_g}$  with  $a_j = N_j/N$ ,  $N_j$  the number of estimand profiles in leaf  $j$ , so  $\mu_g = a^\top \theta$ ,  $\|a\|_1 = H_g$ , and  $\|a\|_2^2 \leq \max_j a_j \sum_j a_j \leq H_g$ , so  $\|a\|_2 \leq \sqrt{H_g}$ . Then

$$r | \theta \sim \mathcal{N}(X\theta, \sigma_1^2 I), \quad \theta | z \sim \mathcal{N}(0, D_z), \quad D_z = \text{diag}(\tau_{1-z_j}^2), \quad \pi(z) = \prod_j w^{z_j} (1-w)^{1-z_j}. \quad (\text{A26})$$

Standard Gaussian algebra gives, with  $V_z = XD_zX^\top + \sigma_1^2 I$  and  $\Sigma_z = (X^\top X/\sigma_1^2 + D_z^{-1})^{-1}$ ,

$$P(z | r) \propto \pi(z) \phi_{n_1}(r; 0, V_z), \quad \theta | z, r \sim \mathcal{N}(\hat{\theta}(z), \Sigma_z), \quad \hat{\theta}(z) = D_z X^\top V_z^{-1} r, \quad (\text{A27})$$

where the form of  $\hat{\theta}(z)$  follows from  $(X^\top X/\sigma^2 + D^{-1})^{-1} X^\top/\sigma^2 = DX^\top (XD X^\top + \sigma^2 I)^{-1}$ . We write  $\mathbf{0}$  for the all-slab configuration; the  $w = 0$  model (the tree analogue of BART-PP, a free source offset regularized by the range rule) has  $\mathbb{E}_{w=0}[\mu_g | r] = a^\top \hat{\theta}(\mathbf{0})$ , and we call

$$\Delta(r) = \mathbb{E}[\mu_g | r] - a^\top \hat{\theta}(\mathbf{0}) = \sum_{z \neq \mathbf{0}} P(z | r) a^\top (\hat{\theta}(z) - \hat{\theta}(\mathbf{0})) \quad (\text{A28})$$

the RWD-induced shift of the estimand at the given partitions.

**Corollary A1** (Bounded shift of the standardized control mean). *Under Assumption A1,  $0 < w < 1$ , and the conditioning of this subsection, let  $\delta = \tau_1^2 - \tau_0^2$  and  $b = w\tau_1/\{(1-w)\tau_0\}$ .*

$$(a) \text{ (General partitions.) } \sup_{r \in \mathbb{R}^{n_1}} |\Delta(r)| \leq \sqrt{H_g} \left(1 + \frac{\tau_1}{\tau_0}\right) (2^{L_g} - 1) \sqrt{2\delta \max\{L_g \log b, \frac{1}{2}\}} < \infty.$$

(b) (Single tree,  $H_g = 1$ .) *With leaves  $\ell$ , RCT counts  $n_{1\ell}$ ,  $v_\ell = \sigma_1^2/n_{1\ell}$ , leaf means  $\bar{y}_\ell$ , and  $\kappa_\ell, C_\ell$  as in (A20) with  $v = v_\ell$ ,*

$$\Delta(r) = \sum_{\ell: n_{1\ell} > 0} a_\ell I_\ell(\bar{y}_\ell), \quad \sup_r |\Delta(r)| \leq \sum_\ell a_\ell \sqrt{2v_\ell \max\{\log C_\ell, \frac{1}{2}\}},$$

*and the right side is at most  $\max_\ell \sqrt{2v_\ell \max\{\log C_\ell, \frac{1}{2}\}}$ .*

(c) ( $H_g$  trees with a common partition.) *If all  $g$  trees share one partition with leaves  $\ell$ , write  $\vartheta_\ell = \sum_h \theta_{h\ell}$ ,  $T_k = k\tau_0^2 + (H_g - k)\tau_1^2$ ,  $\rho_\ell^{(k)} = T_k/(T_k + v_\ell)$ ,*

$$\kappa_{k\ell} = \frac{1}{2} \left( \frac{1}{T_k + v_\ell} - \frac{1}{T_0 + v_\ell} \right), \quad C_{k\ell} = \binom{H_g}{k} \left( \frac{w}{1-w} \right)^k \sqrt{\frac{T_0 + v_\ell}{T_k + v_\ell}}.$$

*Then  $P(k \text{ spikes in leaf } \ell \mid \bar{y}_\ell) \propto \binom{H_g}{k} w^k (1-w)^{H_g-k} \phi(\bar{y}_\ell; 0, T_k + v_\ell)$ , the leaves are independent a posteriori,*

$$\Delta(r) = - \sum_\ell a_\ell \bar{y}_\ell \sum_{k=1}^{H_g} P(k \mid \bar{y}_\ell) (\rho_\ell^{(0)} - \rho_\ell^{(k)}), \tag{A29}$$

$$\sup_r |\Delta(r)| \leq \max_\ell 2v_\ell \sum_{k=1}^{H_g} \sqrt{\kappa_{k\ell} \max\{\log C_{k\ell}, \frac{1}{2}\}} \leq \max_\ell H_g v_\ell \sqrt{\frac{2 \max_k \max\{\log C_{k\ell}, \frac{1}{2}\}}{H_g \tau_0^2 + v_\ell}}. \tag{A30}$$

(d) (Contrasts.) *Under complete pooling of the control surface ( $w = 1$ ,  $\tau_0^2 \rightarrow 0$ ) the shift is  $-a^\top \hat{\theta}(\mathbf{0})$ , and under C-BART ( $w = 1$ ) it is  $a^\top (\hat{\theta}(\mathbf{1}) - \hat{\theta}(\mathbf{0}))$ ; both are linear in  $r$ , and each is unbounded on  $\mathbb{R}^{n_1}$  unless the corresponding linear functional vanishes identically. For a single tree the C-BART functional is  $-\sum_\ell a_\ell (\rho_{1\ell} - \rho_{0\ell}) \bar{y}_\ell$ , which is nonzero whenever some leaf has  $a_\ell > 0$  and  $n_{1\ell} > 0$ .*

*Proof.* (a) Fix  $z \neq \mathbf{0}$ , let  $S = \{j : z_j = 1\}$  be its spike leaves,  $E_S$  the diagonal indicator of  $S$ ,  $X_S$  the corresponding columns of  $X$ , and  $u = X_S^\top V_0^{-1} r \in \mathbb{R}^{|S|}$ . Since  $D_0 - D_z = \delta E_S$ ,

$$V_z^{-1} - V_0^{-1} = V_z^{-1}(V_0 - V_z)V_0^{-1} = \delta V_z^{-1} X_S X_S^\top V_0^{-1}, \quad \text{so} \quad V_z^{-1} r = V_0^{-1} r + \delta V_z^{-1} X_S u.$$

Three consequences follow. First, the quadratic form in the likelihood ratio is

$$r^\top (V_z^{-1} - V_0^{-1}) r = \delta (V_z^{-1} r)^\top X_S u = \delta (u + \delta X_S^\top V_z^{-1} X_S u)^\top u = \delta \|u\|^2 + \delta^2 u^\top X_S^\top V_z^{-1} X_S u \geq \delta \|u\|^2.$$

Second, the difference of posterior means is

$$\hat{\theta}(z) - \hat{\theta}(\mathbf{0}) = (D_z - D_0) X^\top V_0^{-1} r + D_z X^\top (V_z^{-1} - V_0^{-1}) r = -\delta \iota_S u + \delta D_z X^\top V_z^{-1} X_S u,$$

where  $\iota_S$  embeds  $\mathbb{R}^{|S|}$  into the  $S$  coordinates of  $\mathbb{R}^{L_g}$ . The matrix  $D_z X^\top V_z^{-1} X = D_z^{1/2} \{M^\top (M M^\top + \sigma_1^2 I)^{-1} M\} D_z^{-1/2}$  with  $M = X D_z^{1/2}$ , and the bracket is symmetric with eigenvalues  $\lambda_i / (\lambda_i + \sigma_1^2) \in [0, 1)$ , so  $\|D_z X^\top V_z^{-1} X_S\| \leq \|D_z X^\top V_z^{-1} X\| \leq \|D_z^{1/2}\| \|D_z^{-1/2}\| \leq \tau_1 / \tau_0$ , and  $\|\hat{\theta}(z) - \hat{\theta}(\mathbf{0})\| \leq \delta(1 + \tau_1 / \tau_0) \|u\|$ . Third,  $|V_z| = \sigma_1^{2n_1} |D_z| |D_z^{-1} + X^\top X / \sigma_1^2|$  and  $D_z^{-1} \succeq D_0^{-1}$  give  $|V_0| / |V_z| \leq |D_0| / |D_z| = (\tau_1^2 / \tau_0^2)^{|S|}$ , so

$$\begin{aligned} P(z | r) &\leq \frac{P(z | r)}{P(\mathbf{0} | r)} = \frac{\pi(z)}{\pi(\mathbf{0})} \sqrt{\frac{|V_0|}{|V_z|}} \exp\left\{-\frac{1}{2} r^\top (V_z^{-1} - V_0^{-1}) r\right\} \\ &\leq c_z e^{-\delta \|u\|^2 / 2}, \quad c_z = b^{|S|}. \end{aligned}$$

Combining, and using (A25) in the variable  $\|u\|$  with  $\kappa = \delta/2$ ,

$$\begin{aligned} |P(z | r) a^\top (\hat{\theta}(z) - \hat{\theta}(\mathbf{0}))| &\leq \|a\| \delta \left(1 + \frac{\tau_1}{\tau_0}\right) \|u\| \min\{1, c_z e^{-\delta \|u\|^2 / 2}\} \\ &\leq \sqrt{H_g} \left(1 + \frac{\tau_1}{\tau_0}\right) \sqrt{2\delta \max\{|S| \log b, \frac{1}{2}\}}. \end{aligned}$$

Summing over the  $2^{L_g} - 1$  configurations  $z \neq \mathbf{0}$  and using  $|S| \leq L_g$  gives (a).

(b) For a single tree  $X^\top X$  is diagonal, the residual vector splits by leaf, and (A26) factorizes over leaves, so the leaves are independent a posteriori and each is the single-leaf model of Section B.1 with  $v = v_\ell$ ; leaves with  $n_{1\ell} = 0$  have posterior equal to prior under every  $w$  and contribute zero to

$\Delta$ . Theorem A1(c) bounds each  $|I_\ell|$ , and  $\sum_\ell a_\ell = 1$  gives the last inequality.

(c) With a common partition, the residual in leaf  $\ell$  is  $r_i = \vartheta_\ell + \varepsilon_i$  and, given the indicators,  $\vartheta_\ell \sim \mathcal{N}(0, T_{k_\ell})$  with  $k_\ell$  the number of spikes among the  $H_g$  leaves stacked on  $\ell$ ; the leaves are independent a posteriori and  $\mu_g = \sum_\ell a_\ell \vartheta_\ell$ . Summing  $\pi(z)\phi(\bar{y}_\ell; 0, T_k + v_\ell)$  over the  $\binom{H_g}{k}$  configurations with  $k$  spikes gives the stated posterior of  $k$ , and (A14) gives  $\mathbb{E}[\vartheta_\ell | k, \bar{y}_\ell] = \rho_\ell^{(k)} \bar{y}_\ell$ . Under  $w = 0$ ,  $k = 0$  almost surely, which gives the expression for  $\Delta$ . For  $k \geq 1$ ,  $P(k | \bar{y}_\ell) \leq P(k | \bar{y}_\ell)/P(0 | \bar{y}_\ell) = C_{k\ell} e^{-\kappa_{k\ell} \bar{y}_\ell^2}$ , and  $\rho_\ell^{(0)} - \rho_\ell^{(k)} = v_\ell(T_0 - T_k)/\{(T_0 + v_\ell)(T_k + v_\ell)\} = 2v_\ell \kappa_{k\ell}$ ; the  $k$ -th term is therefore at most  $2v_\ell \kappa_{k\ell} |\bar{y}_\ell| \min\{1, C_{k\ell} e^{-\kappa_{k\ell} \bar{y}_\ell^2}\} \leq 2v_\ell \sqrt{\kappa_{k\ell} \max\{\log C_{k\ell}, \frac{1}{2}\}}$  by (A25). The final form uses  $\kappa_{k\ell} \leq \kappa_{H_g \ell} \leq 1/\{2(H_g \tau_0^2 + v_\ell)\}$  and  $\sum_\ell a_\ell = 1$ .

(d) With  $w = 1$  the posterior over  $z$  is degenerate at  $\mathbf{1}$ , so  $\mathbb{E}[\mu_g | r] = a^\top \hat{\theta}(\mathbf{1})$ , linear in  $r$ ; as  $\tau_0^2 \rightarrow 0$ ,  $\hat{\theta}(\mathbf{1}) \rightarrow 0$ . A nonzero linear functional on  $\mathbb{R}^{n_1}$  is unbounded. For a single tree,  $\hat{\theta}_\ell(\mathbf{1}) - \hat{\theta}_\ell(\mathbf{0}) = (\rho_{0\ell} - \rho_{1\ell}) \bar{y}_\ell$  by (b).  $\square$

**Remark A4** (On the constants). The constant in (a) is finite for every design and every partition, which is the content of the statement, but it is far from sharp: it treats each of the  $2^{L_g} - 1$  non-slab configurations separately and bounds the posterior probability of each by its odds against the all-slab configuration. Parts (b) and (c) are the cases in which the sum collapses, and they display the dependence on  $H_g$ , on  $\tau_0$  through  $\kappa$ , and on the leaf shrinkage  $\rho_\ell^{(H_g)} = H_g \tau_0^2 / (H_g \tau_0^2 + v_\ell)$  of the all-spike state. For the numbers of Remark A3 with  $\tau_0^2 = 0.0012$  and the study's configuration  $H_g = 5$ , the common-partition bound in (c) is 1.15 per leaf, while the exact supremum of (A29) is 0.148, attained at  $\bar{y}_\ell = 0.33$  (at  $H_g = 10$  the corresponding values are 1.80 and 0.067 at 0.30); the ensemble abstains sooner than a single leaf because there are  $H_g$  ways to place one slab, and one slab leaf absorbs almost all of the discrepancy. The exact supremum is a one-dimensional maximization and is what we recommend reporting.

## Web Appendix C: Detection and the borrowing map (T2)

### C.1 A single $g$ leaf under a true local shift

We keep the conditioning of Section B.1 and add a data-generating assumption: the RCT residuals in the leaf are  $r_i = d + \epsilon_i$  with  $\epsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_1^2)$ , where  $d$  is the true local discrepancy between the RCT

control surface and the pooled surface  $f$  (taken as fixed; see the scope discussion in Section C.3 for the estimated  $f$ ). Then  $\bar{y} \sim \mathcal{N}(d, v)$ ,  $v = \sigma_1^2/n$ . Let  $P_{\text{spike}}(\bar{y}) = 1 - p = P(z = 1 \mid \bar{y})$ .

**Theorem A2** (Leaf-level detection). *Under Assumption A1 and the conditioning above:*

- (a) (Spike probability.)  $P_{\text{spike}}(\bar{y}) = \{1 + \exp(a + \kappa \bar{y}^2)\}^{-1}$  with  $a, \kappa$  as in (A20); it depends on  $(\bar{y}, n)$  only through  $\bar{y}^2$  and  $v = \sigma_1^2/n$ , is decreasing in  $|\bar{y}|$ , and satisfies  $P_{\text{spike}}(0) = \{1 + e^a\}^{-1}$ .
- (b) (Half-detection discrepancy.) If  $\log \frac{w}{1-w} + \frac{1}{2} \log \frac{\tau_1^2 + v}{\tau_0^2 + v} > 0$ , the equation  $P_{\text{spike}}(\bar{y}) = \frac{1}{2}$  has the unique positive solution

$$d_{1/2} = \sqrt{\frac{2(\tau_0^2 + v)(\tau_1^2 + v)}{\tau_1^2 - \tau_0^2} \left\{ \log \frac{w}{1-w} + \frac{1}{2} \log \frac{\tau_1^2 + v}{\tau_0^2 + v} \right\}}; \quad (\text{A31})$$

otherwise  $P_{\text{spike}} \leq \frac{1}{2}$  for every  $\bar{y}$  (with equality only at  $\bar{y} = 0$  when the bracket vanishes) and we set  $d_{1/2} = 0$ . When  $\tau_1^2 \gg \tau_0^2 + v$ ,  $d_{1/2}^2 \approx 2(\tau_0^2 + v) \left\{ \log \frac{w}{1-w} + \frac{1}{2} \log \frac{\tau_1^2}{\tau_0^2 + v} \right\}$ , a few multiples of  $\tau_0^2 + \sigma_1^2/n$ . The probability that the leaf's slab probability exceeds one half, under the true shift  $d$ , is

$$P_d(p(\bar{y}) > \frac{1}{2}) = P_d(|\bar{y}| > d_{1/2}) = \bar{\Phi}\left(\frac{d_{1/2} - d}{\sqrt{v}}\right) + \bar{\Phi}\left(\frac{d_{1/2} + d}{\sqrt{v}}\right). \quad (\text{A32})$$

- (c) (Large  $n$ ,  $\tau_0^2$  fixed.) As  $n \rightarrow \infty$ , almost surely

$$P_{\text{spike}}(\bar{y}) \rightarrow P_{\text{spike}}^\infty(d) = \left\{ 1 + \frac{(1-w)\tau_0}{w\tau_1} \exp\left(\frac{d^2}{2} \left(\frac{1}{\tau_0^2} - \frac{1}{\tau_1^2}\right)\right) \right\}^{-1} \in (0, 1). \quad (\text{A33})$$

In particular, when  $d = 0$  (borrowing is warranted),  $P_{\text{spike}} \rightarrow w\tau_1 / \{w\tau_1 + (1-w)\tau_0\} > w$ ; when  $d \neq 0$ ,  $P_{\text{spike}}^\infty(d) \leq \frac{w\tau_1}{(1-w)\tau_0} \exp\{-\frac{d^2}{2}(\tau_0^{-2} - \tau_1^{-2})\}$ , which tends to zero as  $|d|/\tau_0 \rightarrow \infty$  but is strictly positive for every fixed  $d$ .

- (d) (Large  $n$  with a vanishing spike.) If  $\tau_0^2 = \tau_{0,n}^2 \rightarrow 0$  as  $n \rightarrow \infty$  (with  $\tau_1^2$ ,  $w$  fixed), then for every  $d \neq 0$ ,  $P_{\text{spike}}(\bar{y}) \rightarrow 0$  almost surely.

*Proof.* (a) is Theorem A1(b) with  $\kappa > 0$ . (b) Setting  $a + \kappa \bar{y}^2 = 0$  gives  $\bar{y}^2 = -a/\kappa$ , which is (A31) after substituting (A20); the condition is  $-a > 0$ . The approximation drops  $v$  and  $\tau_0^2$  against  $\tau_1^2$  in  $(\tau_1^2 + v)/(\tau_1^2 - \tau_0^2)$ . Since  $p > \frac{1}{2}$  iff  $|\bar{y}| > d_{1/2}$  and  $\bar{y} \sim \mathcal{N}(d, v)$ , (A32) follows.

(c) By the strong law,  $\bar{y} \rightarrow d$  almost surely;  $v \rightarrow 0$  gives  $a \rightarrow a_\infty = \log \frac{1-w}{w} + \frac{1}{2} \log \frac{\tau_0^2}{\tau_1^2}$  and  $\kappa \rightarrow \kappa_\infty = \frac{1}{2}(\tau_0^{-2} - \tau_1^{-2})$ , both finite. Hence  $a + \kappa \bar{y}^2 \rightarrow a_\infty + \kappa_\infty d^2$  almost surely, and (A33) follows by continuity of  $t \mapsto (1 + e^t)^{-1}$ ; the limit is strictly between 0 and 1 because the exponent is finite. The bound on  $P_{\text{spike}}^\infty(d)$  is  $(1 + e^t)^{-1} \leq e^{-t}$ .

(d) Write  $\epsilon_n = \tau_{0,n}^2 + v_n \rightarrow 0$ . Then  $a_n + \kappa_n \bar{y}_n^2 \geq \log \frac{1-w}{w} + \frac{1}{2} \log \frac{\epsilon_n}{\tau_1^2 + v_n} + \bar{y}_n^2 \left( \frac{1}{2\epsilon_n} - \frac{1}{2\tau_1^2} \right)$ . Almost surely  $\bar{y}_n^2 \rightarrow d^2 > 0$ , and  $\frac{d^2}{2\epsilon_n} + \frac{1}{2} \log \epsilon_n \rightarrow \infty$ , so the log odds of the slab diverge and  $P_{\text{spike}} \rightarrow 0$ .  $\square$

**Fixed and vanishing spike scales.** Theorem A2(c) distinguishes increasing local sample size from decreasing spike scale: because the spike is a  $\mathcal{N}(0, \tau_0^2)$  component and not a point mass, a discrepancy of the order of  $\tau_0$  is consistent with it, and the limiting spike probability (A33) is positive for every  $d$ . What is true is (i) the limit is exponentially small in  $d^2/(2\tau_0^2)$ , so abstention is consistent in the regime  $|d|/\tau_0 \rightarrow \infty$ , and (ii) abstention is consistent for every fixed  $d \neq 0$  when the spike scale shrinks with  $n$  (part (d)). For a fixed-scale illustration, at  $\tau_0^2 = 0.001$  a shift of  $d = 0.5$  has  $d^2/(2\tau_0^2) = 125$ . The fitted two-arm sampler learns  $\tau_0^2$  from the spike leaves under a prior calibrated to the order  $10^{-3}$ ; the fixed-scale limit therefore does not establish its finite-sample detection behavior, which is assessed in Section 3. The consistency statement that does hold for every fixed  $d \neq 0$ , without a vanishing spike, is the one for the discrepancy map in Theorem A3 below.

**Remark A5** (Numerical illustration). With the constants of Remark A3, (A31) gives  $d_{1/2} = 0.54$ , and (A32) gives leaf abstention probabilities 0.011, 0.43, 0.98 and 1.00 at true shifts  $d = 0, 0.5, 1$  and 2. The limit (A33) at  $d = 0$  is 0.993.

## C.2 The borrowing map $P(|g(x)| < c \mid \text{data})$

The map we display is the posterior probability that the local discrepancy is smaller than a pre-specified outcome-scale threshold  $c > 0$ ,  $\mathcal{B}_c(x) = P(|g(x)| < c \mid \text{data})$ , together with the all-spike map  $B(x) = P(\text{all } H_g \text{ leaves covering } x \text{ are spike} \mid \text{data})$  as a diagnostic. For a single tree and a fixed partition,  $g(x) = \theta_\ell$  for  $x$  in leaf  $\ell$ , and the two are related by the leaf mixture of Theorem A1(a).

**Theorem A3** (The discrepancy map for a single tree). *Under Assumption A1, for a single  $g$  tree with a fixed partition,  $f$  and  $(\sigma_1^2, \tau_0^2, \tau_1^2, w)$  fixed, and  $x$  in a leaf with statistics  $(\bar{y}, n)$ :*

(a) (Exact form.)  $\mathcal{B}_c(x) = p \Psi_1(c) + (1 - p) \Psi_0(c)$  with

$$\Psi_j(c) = \Phi\left(\frac{c - \rho_j \bar{y}}{\sqrt{\rho_j v}}\right) - \Phi\left(\frac{-c - \rho_j \bar{y}}{\sqrt{\rho_j v}}\right), \quad j \in \{0, 1\}.$$

(b) (Relation to the spike probability.)  $\mathcal{B}_c(x) \leq P_{\text{spike}}(\bar{y}) + p \Psi_1(c)$ , and on  $\{\rho_0 |\bar{y}| < c\}$ ,

$$\mathcal{B}_c(x) \geq P_{\text{spike}}(\bar{y}) \left\{ 1 - 2\bar{\Phi}\left(\frac{c - \rho_0 |\bar{y}|}{\tau_0}\right) \right\}.$$

(c) (Consistency.) If the RCT residuals in the leaf are  $d + \epsilon_i$  as in Section C.1 and  $|d| \neq c$ , then as  $n \rightarrow \infty$ , almost surely  $\mathcal{B}_c(x) \rightarrow \mathbf{1}\{|d| < c\}$ , whatever the limit of  $P_{\text{spike}}$ .

*Proof.* (a) By Theorem A1(a), the posterior of  $g(x) = \theta$  is the two-component mixture with components  $\mathcal{N}(\rho_j \bar{y}, \rho_j v)$ , and  $\Psi_j(c)$  is the probability that component  $j$  lies in  $(-c, c)$ .

(b) The upper bound uses  $\Psi_0 \leq 1$ . For the lower bound,  $\mathcal{B}_c \geq (1 - p)\Psi_0(c)$  and, for a  $\mathcal{N}(m, V)$  variable with  $|m| < c$ ,  $P(|N| \geq c) \leq P(|N - m| \geq c - |m|) = 2\bar{\Phi}((c - |m|)/\sqrt{V})$ ; here  $|m| = \rho_0 |\bar{y}|$  and  $V = \rho_0 v = \tau_0^2 v / (\tau_0^2 + v) \leq \tau_0^2$ , and  $\bar{\Phi}$  is decreasing.

(c) Almost surely  $\bar{y} \rightarrow d$ , and  $\rho_j \rightarrow 1$ ,  $\rho_j v \rightarrow 0$  for both  $j$ , so  $\rho_j \bar{y} \rightarrow d$  and each component concentrates at  $d$ :  $\Psi_j(c) \rightarrow \mathbf{1}\{|d| < c\}$  when  $|d| \neq c$ . A convex combination of two sequences with the same limit has that limit, regardless of the weights.  $\square$

**Remark A6.** Parts (b) and (c) are the theoretical counterpart of the empirical finding that  $\mathcal{B}_c$  separates shifted from compatible regions while  $B(x)$  does not. The spike indicator answers whether the leaf discrepancy is of the order of  $\tau_0$ , a question whose answer stays uncertain in the limit (Theorem A2(c));  $\mathcal{B}_c$  answers whether it is smaller than  $c$  on the outcome scale, a question that the RCT controls settle as  $n$  grows. The threshold  $c$  must be pre-specified on the outcome scale; the study uses one third of  $\sigma_{\text{RCT}}$ .

### C.3 The ensemble with fixed partitions, and what is not proved

We return to the fixed-partition linear form (A26). For a profile  $x$ , let  $e_x \in \{0, 1\}^{L_g}$  select the  $H_g$  leaves covering  $x$ , so  $g(x) = e_x^\top \theta$ . The atoms of the common refinement of the  $H_g$  partitions are the cells  $A_1, \dots, A_m$ ;  $e_x$  is constant on each cell, and if  $x$  and RCT control  $i$  lie in the same cell then  $e_x$

is row  $i$  of  $X$ . Let  $n_{1k}$  be the number of RCT controls in cell  $A_k$ .

**Proposition A2** (The discrepancy map for  $H_g$  trees). *Under Assumption A1 and the conditioning of Section B.2:*

- (a) (Exact form.)  $\mathcal{B}_c(x) = \sum_z P(z|r) \Psi_z(c)$  with  $\Psi_z(c) = P\{|\mathcal{N}(e_x^\top \hat{\theta}(z), e_x^\top \Sigma_z e_x)| < c\}$ , and  $B(x) = \sum_{z \in \mathcal{Z}_x} P(z|r)$  where  $\mathcal{Z}_x = \{z : z_j = 1 \text{ for all } j \text{ with } (e_x)_j = 1\}$ .
- (b) (Union bound.)  $1 - B(x) \leq \sum_{h=1}^{H_g} P(z_{h, \ell_h(x)} = 0 | r)$  and  $B(x) \leq \min_h P(z_{h, \ell_h(x)} = 1 | r)$ , where  $\ell_h(x)$  is the leaf of tree  $h$  containing  $x$ . For  $z \in \mathcal{Z}_x$ ,  $e_x^\top \Sigma_z e_x \leq e_x^\top D_z e_x = H_g \tau_0^2$ , so on  $\{|e_x^\top \hat{\theta}(z)| < c\}$ ,  $\Psi_z(c) \geq 1 - 2\bar{\Phi}\{(c - |e_x^\top \hat{\theta}(z)|)/(\sqrt{H_g} \tau_0)\}$ .
- (c) (Consistency at fixed partitions.) Suppose the true residual means are representable on the partitions,  $\mathbb{E}[r|x] = X\theta^*$  for some  $\theta^* \in \mathbb{R}^{L_g}$ , and let  $x \in A_k$  with  $m_k^* = e_x^\top \theta^*$  and  $|m_k^*| \neq c$ . If  $n_{1k} \rightarrow \infty$  (the other cell counts arbitrary), then  $\mathcal{B}_c(x) \rightarrow \mathbf{1}\{|m_k^*| < c\}$  in probability.

*Proof.* (a) is (A27) and the definition of the two maps. (b) The first two inequalities are the union bound and monotonicity for the events  $\{z_{h, \ell_h(x)} = 0\}$ ,  $h = 1, \dots, H_g$ . For Gaussians the posterior covariance is dominated by the prior covariance,  $\Sigma_z = (X^\top X / \sigma_1^2 + D_z^{-1})^{-1} \preceq D_z$ , and for  $z \in \mathcal{Z}_x$ ,  $e_x^\top D_z e_x = H_g \tau_0^2$ ; the tail bound is as in Theorem A3(b).

(c) Fix  $z$ ; write  $e = e_x$ ,  $G = X^\top X / \sigma_1^2$ ,  $m = e^\top \theta$ . *Variance.* The data enter the Gaussian model (A26) through the cell means  $\bar{r}_{k'} | \theta \stackrel{\text{indep}}{\sim} \mathcal{N}(e^{(k')\top} \theta, \sigma_1^2 / n_{1k'})$ , where  $e^{(k')}$  is the common row of  $X$  in cell  $k'$ . In the joint Gaussian law of  $(m, \bar{r}_1, \dots, \bar{r}_m)$  given  $z$ , conditional variances are constants and the law of total variance gives  $\text{Var}(m | \bar{r}_k) = \mathbb{E}[\text{Var}(m | \bar{r}) | \bar{r}_k] + \text{Var}(\mathbb{E}[m | \bar{r}] | \bar{r}_k) \geq \text{Var}(m | \bar{r})$ ; and  $\text{Var}(m | \bar{r}_k) = \omega^2 (\sigma_1^2 / n_{1k}) / (\omega^2 + \sigma_1^2 / n_{1k}) \leq \sigma_1^2 / n_{1k}$  with  $\omega^2 = e^\top D_z e$ . Hence  $e^\top \Sigma_z e \leq \sigma_1^2 / n_{1k} \rightarrow 0$ . *Mean.* With  $r = X\theta^* + \epsilon$ , (A27) gives  $\mathbb{E}[\theta | z, r] = \Sigma_z G \theta^* + \Sigma_z X^\top \epsilon / \sigma_1^2$ , and  $\Sigma_z G = I - \Sigma_z D_z^{-1}$ , so

$$e^\top \mathbb{E}[\theta | z, r] - m_k^* = -e^\top \Sigma_z D_z^{-1} \theta^* + e^\top \Sigma_z X^\top \epsilon / \sigma_1^2.$$

By the Cauchy–Schwarz inequality in the  $\Sigma_z$  inner product,

$$|e^\top \Sigma_z D_z^{-1} \theta^*| \leq \sqrt{e^\top \Sigma_z e} \sqrt{(D_z^{-1} \theta^*)^\top \Sigma_z (D_z^{-1} \theta^*)} \leq \sqrt{\sigma_1^2 / n_{1k}} \tau_1 \|\theta^*\| / \tau_0^2,$$

using  $\Sigma_z \preceq D_z \preceq \tau_1^2 I$  and  $\|D_z^{-1}\| \leq \tau_0^{-2}$ . The noise term has mean zero and variance  $e^\top \Sigma_z G \Sigma_z e \leq$

$e^\top \Sigma_z e \leq \sigma_1^2/n_{1k}$ , because  $G \preceq G + D_z^{-1} = \Sigma_z^{-1}$ . Both terms vanish in probability. *Conclusion.* For each of the finitely many  $z$ ,  $\Psi_z(c) = P\{|\mathcal{N}(\hat{m}_z, \hat{V}_z)| < c\}$  with  $\hat{m}_z \rightarrow m_k^*$  in probability and  $\hat{V}_z \rightarrow 0$ , so  $\Psi_z(c) \rightarrow \mathbf{1}\{|m_k^*| < c\}$  in probability when  $|m_k^*| \neq c$ , uniformly over  $z$ ; the mixture  $\sum_z P(z|r)\Psi_z(c)$  then has the same limit whatever the weights.  $\square$

**Scope under random partitions.** Everything in this appendix about the map is conditional on the partitions of the  $g$  trees. Three things are not proved. First, the representability assumption in Proposition A2(c) is a real restriction: with two trees splitting on different variables the incidence matrix has rank three on four cells, so a discrepancy with an interaction on those cells is not representable, and the fitted cell means then converge to a weighted projection of the truth onto the additive span of the partitions rather than to the truth. Second, we have no statement about the posterior over the partitions themselves, hence none about  $\mathcal{B}_c(x)$  or  $B(x)$  under the full posterior; posterior concentration for a single standard BART ensemble is known [20], but the two-ensemble decomposition with a spike-and-slab discrepancy prior is outside the available theory, and the empirical separation of regions by  $\mathcal{B}_c$  (Section 3) is the evidence we have. Third, the leaf-level results treat  $f$  as fixed; with  $f$  estimated, the residual in a  $g$  leaf carries the error of  $f$  at the RCT profiles, which is  $O_p$  of the RWD posterior standard deviation of  $f$  there, and enters  $d$  as an additive perturbation. On the estimand, this perturbation is what  $V_\mu^f$  of Web Appendix D measures.

## Web Appendix D: Variance-based ESS and calibration (T3)

We use a normal location experiment with known variance  $\sigma_1^2$  as the reference information unit. For a smooth scalar prior density  $p(\mu)$ , the ELIR definition of Neuenschwander et al. [18] is

$$\text{ESS}_{\text{ELIR}} = \sigma_1^2 \mathbb{E}_p[-\partial_\mu^2 \log p(\mu)].$$

For a normal prior of variance  $V$ , this equals  $\sigma_1^2/V$ . For a mixture, the curvature is that of the *log marginal* density, not the average of the component curvatures. Accordingly, our calibration is an average of working precision ratios, with  $\sigma_1^2$  providing normal-reference units. It is not an exact marginal ELIR calculation and does not inherit ELIR's predictive-consistency property. In the survival analysis these units refer to uncensored log times.

## D.1 Leaf aggregation and the calibration reference

At fixed discrepancy partitions  $\mathcal{T}$ , define  $a_{h\ell} = N_{h\ell}/N$  from the profiles used to standardize the estimand. With  $\mu_g = \sum_{h,\ell} a_{h\ell} \theta_{h\ell}$ , conditional independence of the leaf priors gives

$$V_g(\mathcal{T}, z, t) = \text{Var}(\mu_g | \mathcal{T}, z, \tau_0^2 = t) = \sum_{h,\ell} a_{h\ell}^2 \{z_{h\ell} t + (1 - z_{h\ell}) \tau_1^2\}. \quad (\text{A34})$$

Leaves with  $a_{h\ell} = 0$  contribute nothing. Let  $L_+$  count the remaining leaves. For fixed  $w$  and a specified scale law  $P_s$ , averaging the corresponding working precision over independent leaf states gives

$$\text{ESS}_{\mathcal{T}}(w, s) = \sigma_1^2 \sum_{z \in \{0,1\}^{L_+}} w^{|z|} (1-w)^{L_+ - |z|} \int \frac{P_s(dt)}{V_\mu^f + V_g(\mathcal{T}, z, t)}.$$

An additional expectation over  $\mathcal{T}$  averages over the partition prior. When one leaf per tree has positive weight, its weight is one,  $L_+ = H_g$ , and the sum reduces to a binomial average with variance  $kt + (H_g - k)\tau_1^2$ . This includes a pointwise profile and stump-only ensembles. With several occupied leaves per tree, their squared proportions and independent indicators must be retained. Merely counting the total number of spikes is insufficient when their weights differ.

For the informative calibration reference we set every discrepancy leaf to the spike. Then  $V_g = tC(\mathcal{T})$ , where  $C(\mathcal{T}) = \sum_{h,\ell} a_{h\ell}^2$  and  $\mathbb{E}C(\mathcal{T}) = c_g H_g$ . The implemented convention replaces  $C(\mathcal{T})$  by this mean and uses an untruncated scale law:

$$\text{ESS}_0(s_0^2) = \sigma_1^2 \mathbb{E}_{\tau_0^2 \sim \text{scaled-Inv-}\chi^2(\nu_0, s_0^2)} \left[ \frac{1}{V_\mu^f + c_g H_g \tau_0^2} \right]. \quad (\text{A35})$$

Here  $V_\mu^f > 0$  is the variance of the standardized RWD-only BART surface. Its use represents that marginal distribution by its variance. By convexity, substituting  $\mathbb{E}C$  before inversion gives no larger a value than averaging  $1/(V_\mu^f + tC)$  over partitions at fixed  $t$ . Averaging these conditional precisions over  $t$  also differs from taking the inverse of the variance after mixing over  $t$ . These are distinct operations.

For block variances  $V^{(b)}$ , the grid rule is

$$\hat{s}_{0*}^2 = \inf\{s_0^2 \in \mathcal{S} : \hat{G}(s_0^2) \geq q\}, \quad \hat{G}(s_0^2) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\text{ESS}_0(V^{(b)}; s_0^2) \leq N_{\text{target}}\}. \quad (\text{A36})$$

Write  $G(s) = P\{\text{ESS}_0(V; s) \leq N_{\text{target}}\}$  for a specified positive block-variance law. The theorem concerns exact evaluation of (A35), a fixed coefficient  $c_g$ , and the rule (A36) before implementation overrides.

**Theorem A4** (Monotone, well-posed calibration). *Fix  $\sigma_1^2 > 0$ ,  $V > 0$ ,  $c = c_g H_g > 0$ ,  $\nu_0 > 0$ ,  $N_{\text{target}} > 0$  and  $q \in (0, 1)$ , and write  $s = s_0^2$ .*

(a) (Monotonicity and limits.) *Under the untruncated law,  $\text{ESS}_0(V; s)$  is continuous, differentiable, convex and strictly decreasing, with limits  $\sigma_1^2/V$  and 0 as  $s \downarrow 0$  and  $s \uparrow \infty$ . Under truncation to  $(0, \tau_1^2)$  it remains continuous and strictly decreasing, with limits  $\sigma_1^2/V$  and  $\sigma_1^2/(V + c\tau_1^2) > 0$ . Convexity is asserted only for the untruncated law.*

(b) (Population selection.) *Under the untruncated law, extend  $\text{ESS}_0(V; 0) = \sigma_1^2/V$ . Then  $\{s \geq 0 : \text{ESS}_0(V; s) \leq N_{\text{target}}\} = [s^\dagger(V), \infty)$ , where  $s^\dagger$  is the unique positive root when  $N_{\text{target}} < \sigma_1^2/V$  and zero otherwise. The function  $s^\dagger$  is continuous and strictly decreasing in  $V$  where positive. Thus  $G(s) = P\{s^\dagger(V) \leq s\}$  is a distribution function, and its  $q$ -quantile  $s_\star$  is the unique attained infimum of  $\{G \geq q\}$ . Moreover,*

$$s_\star > 0 \iff P\{N_{\text{target}} < \sigma_1^2/V\} > 1 - q. \quad (\text{A37})$$

*Under truncation, a positive finite root requires the target to lie strictly between the lower limit in (a) and the ceiling. At or below the lower limit, no finite scale satisfies the target.*

(c) (Monte Carlo limits.) *Let the full Stage-1 state process be a stationary, aperiodic Harris-ergodic Markov chain. Suppose the standardized surface is a square-integrable measurable function of that state, with variance  $V_\star > 0$ . Let  $U_b$  be the sample variance in its  $b$ th block of length  $m \geq 2$ , and set  $V^{(b)} = \gamma_B U_b$  with  $\gamma_B = \hat{V}_{\text{whole}}/(B^{-1} \sum_b U_b)$ . For fixed  $B$  and  $m \rightarrow \infty$ , every  $V^{(b)} \rightarrow V_\star$  almost surely. On a finite grid avoiding equality  $\text{ESS}_0(V_\star; s) = N_{\text{target}}$  and containing an admissible point, selection by (A36) eventually equals the smallest admissible grid value. For fixed  $m$  and  $B \rightarrow \infty$ , suppose  $U_b > 0$  almost surely and  $u_m = \mathbb{E}U_b > 0$ . Put  $\gamma = V_\star/u_m$  and*

$$G_m(s) = P\{\text{ESS}_0(\gamma U_b; s) \leq N_{\text{target}}\}.$$

At any  $s$  for which  $P\{\text{ESS}_0(\gamma U_b; s) = N_{\text{target}}\} = 0$ ,  $\widehat{G}(s) \rightarrow G_m(s)$  almost surely. On a finite grid satisfying this condition, with  $G_m(s) \neq q$  at every point and at least one  $G_m(s) > q$ , the empirical and population grid selections eventually agree.

(d) (Single-tree reference.) Condition on one fixed  $f$  tree and  $\sigma_2^2$ . Let  $a_\ell$  and  $\pi_\ell = n_{2\ell}/n_2$  be target and RWD leaf proportions, with  $\pi_\ell > 0$  whenever  $a_\ell > 0$ . Terms with  $a_\ell = \pi_\ell = 0$  contribute zero to the sums below. Then

$$V_\mu^f = \sum_\ell a_\ell^2 (\lambda_f^{-2} + n_{2\ell}/\sigma_2^2)^{-1} \leq \frac{\sigma_2^2}{n_2} \sum_\ell \frac{a_\ell^2}{\pi_\ell}, \quad \frac{\sigma_1^2}{V_\mu^f} \geq \frac{n_2 \sigma_1^2 / \sigma_2^2}{\sum_\ell a_\ell^2 / \pi_\ell}.$$

Equality is attained in the flat-prior limit. In that limit the denominator is at least one, with equality exactly when  $a = \pi$ . At finite  $\lambda_f^2$  the prior contributes information, and covariate shift need not lower the ceiling relative to its value at  $a = \pi$ .

*Proof.* (a) Write  $\tau_0^2 = \nu_0 s / Q$ ,  $Q \sim \chi_{\nu_0}^2$ . The integrand  $(V + c\nu_0 s / Q)^{-1}$  is continuous, strictly decreasing and convex in  $s$ , bounded by  $1/V$ . Dominated convergence gives continuity and the limits; integration preserves monotonicity and convexity. Its derivative is bounded in absolute value by  $1/(4Vs)$ , which permits differentiation on compact subsets of  $(0, \infty)$ . Under truncation, the density ratio at  $s' > s$  is proportional to  $\exp\{-\nu_0(s' - s)/(2t)\}$ , strictly increasing in  $t$ . The truncated family is therefore stochastically increasing, giving strict decrease of the expected reciprocal. Continuity follows from integration on a fixed interval. As  $s \downarrow 0$ , the untruncated law concentrates at zero and its truncation probability tends to one. As  $s \uparrow \infty$ , compare any interval below  $\tau_1^2 - \epsilon$  with one inside  $(\tau_1^2 - \epsilon/2, \tau_1^2)$ : the exponential density factor makes the former probability tend to zero. The truncated law concentrates at  $\tau_1^2$ , giving the stated lower limit.

(b) Continuity, strict decrease and the limits in (a) give the root and sublevel set. Joint continuity and strict decrease in  $V$  give continuity and strict decrease of a positive root. Since  $s^\dagger(V)$  is finite almost surely,  $G$  is a distribution function with an attained finite  $q$ -quantile. Right-continuity gives  $s_\star > 0$  exactly when  $G(0) < q$ , equivalent to (A37). The truncated range in (a) gives its feasibility restriction.

(c) At fixed  $B$ , ergodic laws for the draw and its square imply convergence of every growing block variance to  $V_\star$ . A block sum is the difference of two partial sums, so the same conclusion holds

for later blocks whose starting index also increases. The whole-chain and mean block variances have that limit, giving  $\gamma_B \rightarrow 1$ . Continuity and strict grid separation yield the selection result. At fixed  $m$ , the nonoverlapping block sequence is stationary and ergodic. Thus  $B^{-1} \sum_b U_b \rightarrow u_m$  and  $\gamma_B \rightarrow \gamma$  almost surely. For each small  $\epsilon > 0$ , eventually  $(\gamma - \epsilon)U_b \leq \gamma_B U_b \leq (\gamma + \epsilon)U_b$  for every block. Monotonicity in  $V$  bounds the empirical indicators between those at the two fixed scale factors. Apply the ergodic theorem to each bound and let  $\epsilon \downarrow 0$ ; absence of mass at the threshold makes both limits  $G_m(s)$ . A finite set of grid points with  $G_m(s) \neq q$  eventually has the correct sign relative to  $q$ , proving selection. No central limit theorem or convergence rate is asserted.

(d) The fixed-tree posterior leaf variances are  $(\lambda_f^{-2} + n_{2\ell}/\sigma_2^2)^{-1}$  and the leaves are independent, giving the expression for  $V_\mu^f$ . Drop  $\lambda_f^{-2}$  for the upper bound. Cauchy–Schwarz gives  $1 = (\sum_\ell a_\ell)^2 \leq \sum_\ell a_\ell^2/\pi_\ell$ , with equality exactly at  $a = \pi$ . If a target leaf has no RWD observations, its finite-prior variance remains  $\lambda_f^2$ ; the flat-prior reference for that target is unavailable.  $\square$

**Remark A7** (Units and regularization). The fitted ceiling  $\sigma_1^2/V_\mu^f$  is a property of the working precision summary. Both external observations and the BART prior determine  $V_\mu^f$ . The single-tree reference in part (d) does not justify subtracting a universal prior contribution from a fitted ensemble. Under the range rule,  $H_f \lambda_f^2 = (\max y - \min y)^2/(4k^2)$ , so increasing the number of trees alone does not remove prior information.

**Remark 4** (Calibration implementation). Stage 1 fits BART to the RWD and evaluates the standardized surface at the target profiles. In two-arm analyses,  $\sigma_1^2$  is the posterior mean residual variance from a separate BART fitted to the RCT controls; a design value or the RWD estimate is used when controls are unavailable. The code fixes  $B = 20$ , estimates  $c_g$  with 2,000 prior trees, and evaluates the untruncated scale expectation with 4,000 common chi-squared draws across the grid. These finite approximations are separate from the exact-integral theorem. For a fixed positive variance and finite grid, their errors vanish as the inner sample sizes grow; conclusions about selection require strict separation at the limiting grid values.

A requested target at or above the whole-chain ceiling triggers an override: the smallest grid scale is returned, without requiring the block rule to pass. Otherwise the first grid value satisfying the rule is selected. If none passes, the largest value is returned with an upper-boundary flag; it does not thereby meet the target. A small scale approximates complete pooling of  $f_1$  with  $f$  only

when  $w = 1$ . With  $w < 1$ , slab discrepancies remain possible. Equation (A35) uses an untruncated calibration law, while posterior fitting uses (A13); replacing either law or the boundary rule would change the fitted analyses.

The realized diagnostic averages  $\sigma_1^2/\{V_\mu^f + V_g(\mathcal{T}, z, \tau_0^2)\}$  over posterior states, using (A34). This evaluates prior variance at learned states; it is not the posterior variance of the discrepancy or a subtraction of posterior precisions. The application map uses  $\hat{\sigma}_1^2/\{V_f(x) + H_g\bar{\tau}_0^2\}$ , with the fitted mean scale  $\bar{\tau}_0^2$ . In single-arm mode this estimates the mean under the calibrated truncated prior. The supplementary pointwise prior summaries instead average  $\hat{\sigma}_1^2/\{V_f(x) + H_g\tau_0^2\}$  over the untruncated calibration prior. The inversion and expectation occur in different orders. All regional summaries use their own standardization profiles and are not additive.

## Web Appendix E: Identifiability of $f$ and $g$

**Proposition A3** (Population identification). *Let  $\mathcal{X}_1$  and  $\mathcal{X}_2$  be the supports of the covariate distributions of the RCT controls and of the RWD, and suppose the regression functions  $x \mapsto \mathbb{E}[y | x, D = 0]$  on  $\mathcal{X}_2$  and  $x \mapsto \mathbb{E}[y | x, D = 1]$  on  $\mathcal{X}_1$  are known (the infinite-data limit). Under model (A11):*

- (a)  *$f$  is identified on  $\mathcal{X}_2$  and  $f + g$  is identified on  $\mathcal{X}_1$ .*
- (b)  *$g$  is identified on  $\mathcal{X}_1 \cap \mathcal{X}_2$ , as  $g = (f + g) - f$ .*
- (c) *On  $\mathcal{X}_1 \setminus \mathcal{X}_2$  only the sum  $f + g$  is identified; the split between  $f$  and  $g$  is determined by the prior, with  $f$  the BART extrapolation of the RWD surface and  $g$  the remainder.*
- (d) *On  $\mathcal{X}_2 \setminus \mathcal{X}_1$ ,  $f$  is identified but  $g$  is not identified from the source-specific outcome distributions. Its posterior predictions depend on extrapolation through tree partitions and on the prior. This population statement does not imply that its fitted posterior law at those profiles equals the prior.*

*These statements concern unrestricted population regression functions. Fixed tree partitions impose additional restrictions that can transmit information across these regions. In the single-arm case  $\mathcal{X}_1 = \emptyset$ , the likelihood does not involve  $g$  anywhere. Its entire posterior law equals its calibrated prior, and, conditional on the scales and  $w$ , its pointwise variance is  $H_g\{w\tau_0^2 + (1 - w)\tau_1^2\}$ .*

*Proof.* (a) is the model:  $\mathbb{E}[y | x, D = 0] = f(x)$  on  $\mathcal{X}_2$  and  $\mathbb{E}[y | x, D = 1] = f(x) + g(x)$  on  $\mathcal{X}_1$ . (b), (c), (d) are immediate from (a) and from the fact that no other moment of the data involves  $f$  or  $g$ . For the single-arm variance, condition on the scales and  $w$ . Then  $g(x) = \sum_h \theta_{h, \ell_h(x)}$  is a sum of  $H_g$  independent leaf contributions, each with variance  $w\tau_0^2 + (1 - w)\tau_1^2$ .  $\square$

**Remark A8** (Finite samples). In finite samples the statement is leaf-wise. A  $g$  leaf with no RCT controls has  $M_g = 1$  (Lemma A1(b)) and its  $(z, \theta)$  is drawn from the prior at the current  $(\tau_0^2, w)$ ; a  $g$  leaf that straddles the RCT support extends the RCT-informed discrepancy as a constant into the region without RCT controls, which is a piecewise-constant extrapolation and not a prior draw. On  $\mathcal{X}_1 \cap \mathcal{X}_2$  the separation of  $f$  and  $g$  rests on the two priors:  $f$  is pinned by the RWD, whose observations do not involve  $g$ , and the spike shrinks  $g$  toward zero, so a shift common to both sources is attributed to  $f$  and a shift of the RCT controls alone to  $g$ . The mixture (A12) is identifiable as a two-component scale mixture as long as  $\tau_0^2 \neq \tau_1^2$ , and Assumption A1 fixes which component is the spike, so the label of  $z$  is well defined.

## Web Appendix F: Scope of the posterior-invariance result

The leaf marginals in Lemma A1 and the marginal-then-conditional update in Proposition A1 establish invariance for the specified posterior kernel. This construction uses the global discrepancy scale as a conditioned-on parameter during a tree move and updates it separately from its truncated conditional. Thus no moment-matched likelihood or approximation-error bound enters the tree acceptance ratio. Invariance does not imply independent posterior draws, rapid mixing, or adequate exploration by a finite run. Those properties require separate analysis and diagnostics. Likewise, the conditional influence and map results in Appendices B and C and the working calibration result in Appendix D have their own assumptions; the posterior-invariance proposition does not establish them for the full fitted ensemble.

## Web Appendix G: Related work, methodological remarks, and simulation and application details

### G.0 Related work: covariate-adaptive borrowing

In addition to the study-level priors and tree methods discussed in Section 1.1, several approaches allow borrowing to depend on covariates. The propensity-score-integrated power prior [25] and composite likelihood [4] stratify the pooled patients on the estimated probability of belonging to the trial and give each stratum its own discount, so borrowing is constant within strata fixed by a one-dimensional summary before the outcomes are seen. The covariate-adaptive historical control borrowing method of Jin et al. [13] places a commensurate prior on the current control mean, centered on the historical mean function, with a covariate-dependent precision fitted by kernel local regression and defines a covariate-conditional effective sample size; it is the closest kernel-local relative of our borrowing map, with locality from a kernel bandwidth and a single covariate-adjusted mean model. Multisource exchangeability models [14] average over the subsets of sources deemed exchangeable with the current study, and Kotalik et al. [15] extend the averaging to treatment effects that vary with covariates. The latent exchangeability prior [1] attaches to each historical patient a latent class, so that a subset of patients is borrowed from, with membership driven by the outcome model. Elastic priors [12] set the prior precision through a global congruence statistic and are not local. SAM-HC [2] and Chandra et al. [3] cluster trial and external patients on the covariates with shared- or common-atoms mixtures and borrow only within clusters both sources occupy; their borrowing regions are determined before fitting the outcome model. In LRC-BART, these regions are learned through the outcome-model partition.

### G.1 Data-generating constants

We choose the intercept  $\alpha$  in (10) to maintain the outcome location after including the bounded bump term:  $\alpha = 0.42$  under Sc1, Sc4 and Sc5 and  $\alpha = -0.08$  under Sc2 and Sc3 for Gaussian outcomes, and  $\alpha = -0.20$  and  $-0.70$  on the log-time scale for survival outcomes. Under Sc2 and Sc3 the source is assigned by  $P(D_i = 1 | \mathbf{W}_i) = \text{logit}^{-1}\{(\ell_i - q_{0.5}(\ell))/0.5\}$  with  $\ell_i = -1.20(W_{i5} + W_{i6})$ , where  $\mathbf{W}_i = (X_{i5}, X_{i6})$  under Sc2 and  $(U_{i1}, U_{i2})$  under Sc3; the median centring and the temperature

0.5 give the two sources equal size on average. Under Sc3 the unmeasured confounders are  $U_j \sim \mathcal{N}(\mu_{j+4}, 1)$  and the observed proxies are  $X_{j+4} = \rho \tilde{U}_j + \sqrt{1 - \rho^2} \epsilon_j$  with  $\tilde{U}_j$  the standardized  $U_j$  and  $\epsilon_j \sim \mathcal{N}(0, 1)$ , so that  $\mathbf{U}$  drives both the assignment and the outcome and  $(X_5, X_6)$  carry information about it only through  $\rho$ . Under Sc4 and Sc5 the RWD mean is  $f_0(\mathbf{X}) + \delta_2 \mathbf{1}\{\mathbf{X} \notin \mathcal{R}\}$  with the RCT mean unchanged, so the true local discrepancy  $g = f_1 - f$  is  $-\delta_2$  outside  $\mathcal{R}$  and 0 inside it; the random-number stream is arranged so that Sc4 with  $\delta_2 = 0$  reproduces Sc1 replicate by replicate. Each replicate draws 300 RCT subjects and 300 RWD controls afresh. We compute the true estimands by standardizing to each replicate’s RCT profiles: the Gaussian ATE is  $\delta = 1$  exactly, and for survival outcomes the population median of the standardized log-normal mixture is obtained by numerical root-finding for each arm and the RMST at  $t^* = 3$  is the average over profiles of the closed-form log-normal RMST, both with  $\sigma_1$  in the control arm.

## G.2 Implementation of the comparators

We implement the comparators in Section 3.1 as follows. LM-NP, LM-CP and LM-PP are Bayesian linear regressions fitted to the RCT controls alone, to the pooled controls, and to the pooled controls with the source indicator as a covariate, and AFT-NP, AFT-CP and AFT-PP are their log-normal AFT counterparts; BART-NP, BART-CP and BART-PP are standard BART and AFT-BART [5, 24] in the same three configurations, BART-PP being the implicit borrowing of Zhou and Ji [27]. PSCL is the propensity-score composite likelihood of Chen et al. [4] with a borrowed size of 100; MAP is the meta-analytic predictive prior for the control mean, without covariate adjustment, with its heterogeneity scale calibrated to the prior ESS target; HierLM and HierAFT give the two sources distinct coefficients drawn from common means with  $\tau^2 \sim \text{Inv-Gamma}(1.5, 0.75)$ . We also include two nested LRC-BART models: C-BART fixes  $w = 1$ , the commensurate model with a single learned scale and no slab, and LRC-BART ( $w = 0$ ) fixes every leaf in the slab, the free-offset model. We implement the analysis component of covariate-adaptive historical control borrowing (CAHB) from Jin et al. [13]. Its kernel-local control mean has a commensurate prior centred on a historical mean function, with covariate-dependent precision. Local precision estimates are thresholded and projected onto an  $L_1$  ball. We retain the fixed allocation of our simulation; the sequential allocation component of CAHB is not used.

Five transfer choices define our primary CAHB analysis. We estimate the historical mean by

Nadaraya–Watson regression on the RWD with the same kernel used for the current control mean, treating that estimate as known. We select four kernel covariates by their variable-inclusion counts in an RWD-only BART fit. We enlarge Scott’s bandwidth until the median local control sample size is 20. We rescale the precision constants to  $\gamma^2 = 3 \times 0.11/\hat{\phi}^2$  and  $\lambda_2 = 300 \log(n) \times 0.11/\hat{\phi}^2$ , where  $\hat{\phi}^2$  is the residual variance of the current-control no-borrowing smoother; the precision threshold uses its 10% quantile. Finally, we preserve the profile-specific normal marginals and induce dependence across posterior draws through the smoother weights before standardizing to the trial profiles. This last choice affects posterior SD, coverage and power, but not the point estimates or fitted surfaces. The treated mean is fitted to trial treated subjects with a flat prior. Precision estimation and projection use all trial profiles.

CAHB-p10 uses all ten covariates with the other primary choices unchanged. CAHB-lit uses historical linear regression, all ten covariates, Scott’s bandwidth without enlargement,  $\gamma^2 = 3$ ,  $\lambda_2 = 300 \log(n)$  and independent profile draws. The primary choices were assessed on diagnostic seeds 2027, 2050 and 2101, which overlap the study replicates. All variants use 1,000 draws,  $R = 200$  and data seed  $2026 + r$  for replicate  $r$ ; code is in `01-code/R/cahb.R` and the run configuration in `02-validation/cahb/run-cahb.R`. The comparison therefore assesses our transfer of the analysis, including its interval construction.

For CAHB, the local summary is the historical precision share  $1 - 1/R(x)$ , where  $R(x)$  is its information ratio, and its effective size is the sum of  $R(x) - 1$  over RCT controls. These are distinct from  $\mathcal{B}_c$  and the estimand-level ESS of LRC-BART. The literal variant has Sc1 posterior SD 12.226 and RMSE 0.950, compared with 0.542 and 0.262 for CAHB-p10 and 0.410 and 0.258 for primary CAHB. The local kernel fits are unstable or oversmoothed in this design; these results support examining tree partitions for locality, subject to the limitations of the transfer.

| Scenario | Transfer | Bias   | SD     | RMSE  | Share, $\mathcal{R}$ | Share, $\mathcal{R}^c$ |
|----------|----------|--------|--------|-------|----------------------|------------------------|
| Sc1      | CAHB     | -0.003 | 0.410  | 0.258 | 0.52                 | 0.51                   |
| Sc1      | CAHB-p10 | -0.018 | 0.542  | 0.262 | 0.50                 | 0.50                   |
| Sc1      | CAHB-lit | -0.087 | 12.226 | 0.950 | 0.01                 | 0.01                   |
| Sc4 (1)  | CAHB     | -0.208 | 0.413  | 0.333 | 0.53                 | 0.48                   |
| Sc4 (1)  | CAHB-p10 | -0.260 | 0.545  | 0.380 | 0.50                 | 0.49                   |
| Sc4 (1)  | CAHB-lit | -0.088 | 12.207 | 0.948 | 0.01                 | 0.01                   |
| Sc4 (2)  | CAHB     | -0.333 | 0.422  | 0.453 | 0.49                 | 0.41                   |
| Sc4 (2)  | CAHB-p10 | -0.265 | 0.585  | 0.533 | 0.31                 | 0.31                   |
| Sc4 (2)  | CAHB-lit | -0.089 | 12.187 | 0.947 | 0.00                 | 0.00                   |
| Sc5 (1)  | CAHB     | -0.437 | 0.419  | 0.546 | —                    | 0.47                   |
| Sc5 (1)  | CAHB-p10 | -0.290 | 0.581  | 0.549 | —                    | 0.32                   |
| Sc5 (1)  | CAHB-lit | -0.089 | 12.198 | 0.948 | —                    | 0.01                   |
| Sc5 (2)  | CAHB     | -0.067 | 0.479  | 0.400 | —                    | 0.03                   |
| Sc5 (2)  | CAHB-p10 | -0.028 | 0.648  | 0.340 | —                    | 0.00                   |
| Sc5 (2)  | CAHB-lit | -0.090 | 12.165 | 0.947 | —                    | 0.00                   |

The linear models use  $\alpha \sim \mathcal{N}(0, 5^2)$  and  $\beta \sim \mathcal{N}(0, 2^2 I)$ , BART the default prior with  $H = 50$  trees, and the residual variance of these non-CAHB methods the BART-style prior with  $\nu = 3$  and  $q = 0.9$ . The implementation details are given below. The non-CAHB samplers run 1,000 burn-in and 1,000 retained iterations. The linear, AFT and hierarchical models are fitted by exact conjugate Gibbs samplers, with right censoring handled by data augmentation of the log survival times; the HierLM sampler was checked against a Hamiltonian Monte Carlo implementation in Stan on the same data. HierLM and HierAFT use one hypermean per coefficient,  $\mu_{\beta j} \sim \mathcal{N}(0, 10^2)$ , and the group residual variances carry the BART-style prior with the scale from a group-specific least-squares or parametric AFT fit. AFT-NP, AFT-CP and AFT-PP use the same coefficient priors as the linear models. The MAP prior is built from the RWD sample mean and, with a fixed Half-Normal(0, 1) heterogeneity scale, gives an expected-local-information-ratio ESS of about 2 with a single external study of 300, so we calibrate the half-normal scale so that the ESS of the mixture-approximated MAP prior equals the target, falling back to the largest attainable ESS where the target is out of reach, with the mixture, ESS and posterior update from the RBeST package.

For PSCL, we follow Chen et al. [4]: a logistic propensity model on  $X_1$  to  $X_{10}$  over both sources, trimming of RWD subjects outside the RCT propensity range, five strata at the RCT propensity quintiles, the total borrowed size of 100 allocated across strata in proportion to the overlap coefficient of the stratum propensity distributions, composite-likelihood stratum means with

a sandwich variance, and standardization by the RCT stratum shares; its interval is Wald. BART uses the BART package with  $H = 50$ ,  $\alpha = 0.95$ ,  $\beta = 2$ ,  $k = 2$ ,  $\nu = 3$  and  $q = 0.9$ ; the treated arm always has its own ensemble on the RCT treated subjects.

The simulation summaries use the main-text definitions. Comparisons with the previous analysis require several distinctions. Previously, power was computed as the fraction of replicates with  $P(\text{ATE} > 0.5 | \text{data}) > 0.95$ , a one-sided test against 0.5, and for survival as  $P(\text{ratio} > 1 | \text{data}) > 0.95$ , although the table caption specified the 95% interval excluding the null; we use that interval-based definition. Under it, every tree model has power 1.00 at  $\delta = 1$  for Gaussian outcomes. The 90%-interval version (`pow90` in the result files) is 1.00 for the tree models and 0.90 to 0.91 for LM-NP, LM-PP and HierLM in Sc1, so neither version discriminates among the tree models.

The previous survival point estimate was the posterior median of the ratio draws; we use the posterior mean throughout, which moves the BART-NP median-ratio bias in Sc1 by about 0.05 and does not account for the remaining difference from the earlier row, which we could not reproduce from the code and the stated design.

The previous single-arm tables used a different mechanism, with a selection slope of  $-0.55$  at temperature 1.0 in place of  $-1.20$  at 0.5, different intercepts and residual scales, a dropout rate of 0.09, entry over  $(0, 0.5)$ , administrative censoring at 5.5 minus entry, an RMST horizon of 5, and a hierarchical hyperprior scale of 0.05 with covariates centred at the trial means; Section 3.4 reports the stated design.

### G.3 Configuration of the discrepancy ensemble

The configuration of  $g$  was chosen on a development run of  $R = 200$  Gaussian replicates paired with the first 200 replicates of the full study. The initial configuration,  $H_g = 10$  with  $w \sim \text{Beta}(4, 1)$ , gave a bias of 0.06 under Sc3 and the same Sc4 map as the final one. A tuning study varied  $H_g$ , the split prior, the slab scale, the leaf minimum, the prior on  $w$ , the number of  $g$  sweeps per iteration, and a warm start of  $g$  from the BART-PP source offset, and ran oracle fits with  $f$  fixed at the truth.

The configuration was informed by two observations. Underestimation of  $g$  at  $\delta_2 = 2$  persisted under every prior, under long chains, and with  $f$  known, and was not resolved by those configurations. The observed tradeoffs involved both  $H_g$  and the slab range rule, which increases  $\tau_1$  when the number of discrepancy trees decreases. These joint changes do not isolate one cause of the moderate-shift

bias. Web Appendix G.6 gives a conditional prior-cost comparison, with its limits stated explicitly.

The final configuration,  $H_g = 5$ ,  $(\alpha_g, \beta_g) = (0.5, 3)$ , the range-rule slab, a leaf minimum of 5, one  $g$  sweep per iteration and  $w \sim \text{Beta}(1, 1)$ , lowered the Sc3 bias to 0.04 and left every other quantity within Monte Carlo error of the initial one; LRC-BART ( $H_g = 10$ ) in Web Table S1 is the initial ensemble size under the final prior on  $w$ .

The development run evaluated 25 pre-specified criteria, of which 18 pass at  $R = 500$ ; the seven that fail are the Sc1 power criterion, which is uninformative at  $\delta = 1$ , the Sc4  $\delta_2 = 1$  RMSE criterion against BART-PP, and five Sc4  $\delta_2 = 2$  criteria: the control surface RMSE in  $\mathcal{R}$  below 0.9 times BART-PP's, the all-spike map in  $\mathcal{R}$  above 0.5,  $\mathcal{B}_c$  in  $\mathcal{R}$  above 0.7, and the mean of  $g$  within 0.25 of the truth in each region, as reported in Section 3.2.

A secondary version of Sc4, Sc4' in Table 2 and Web Table S1, places the shift outside  $\mathcal{R}' = \{X_7 \leq 2\}$ , a covariate with negligible effect, so that a split of  $g$  on it is driven by the discrepancy alone. Results for the null covariate  $X_7$  are similar to those for  $X_5$ : at  $\delta_2 = 2$  the map  $\bar{\mathcal{B}}_c$  is 0.64 over the RCT control profiles in the compatible region and 0.05 outside it, and empirical ATE bias is near zero because the errors on the two sides of the boundary have opposite signs and the estimand averages over both; at  $\delta_2 = 1$  LRC-BART (100) borrows too much outside the compatible region, with ATE bias  $-0.041$ , RMSE 0.223 against 0.207 for BART-PP, and coverage 0.89, on both  $X_5$  and  $X_7$ .

Theorem 2 collects Theorem A2(a),(b) and Theorem A3 of Web Appendix C. Part (a) gives a conditional half-detection threshold determined by the spike scale, the RCT standard error of the leaf, the slab scale and the mixing weight. The simulation study includes shifts at and below this scale. Part (b) establishes conditional consistency of the displayed map: as the RCT controls in a region accumulate,  $\mathcal{B}_c$  converges to the truth about whether the local discrepancy is below  $c$ , irrespective of the limiting spike indicators. Part (c) is the reason we display  $\mathcal{B}_c$  and not the all-spike probability: the spike is a Gaussian component of scale  $\tau_0$ , not a point mass, so a discrepancy of that order is consistent with it and the spike indicator can remain uncertain in the limit even when the RCT data identify whether the discrepancy is below the threshold used by  $\mathcal{B}_c$ . At fixed scales and large local counts, the bound in (c) can make the limiting spike probability small when the discrepancy is large relative to the spike scale. In the fitted model  $\tau_0^2$  is learned, so this limit does not guarantee finite-sample abstention; Section 3 examines shifts near  $d_{1/2}$ . Proposition

A2 extends (b) to the  $H_g$ -tree ensemble with fixed partitions, provided the true discrepancy is representable on those partitions, with the map converging cell by cell as the RCT count in a cell grows. These results are conditional on the partitions of the  $g$  trees: we have no statement about the posterior over the partitions themselves, hence none about  $\mathcal{B}_c$  under the full posterior, since the two-ensemble decomposition with a spike-and-slab discrepancy prior lies outside the available posterior-concentration theory for BART [20]. The leaf-level results also treat  $f$  as fixed, whereas with  $f$  estimated the residual in a  $g$  leaf carries the RWD posterior error of  $f$  at the RCT profiles as an additive perturbation of  $d$ . The separation of shifted from compatible regions by  $\mathcal{B}_c$  under the full posterior is shown empirically in Section 3.

We evaluate additional surface errors over RCT control profiles and over the subsets with  $|X_5 - 2| > 0.5$ , using the first 200 paired replicates. These differ from Table 2, which uses all RCT profiles. At Sc4  $\delta_2 = 2$ , the control-profile RMSE in  $\mathcal{R}$  is 0.751 for LRC-BART and 0.704 for BART-PP, a difference of 0.047 with paired standard error 0.005. Away from the boundary it is 0.747 versus 0.723, with difference 0.024 and standard error 0.005. Boundary exclusion reduces this loss but does not remove it. At Sc4  $\delta_2 = 1$ , the far-region difference in  $\mathcal{R}$  is close to zero. Control-profile errors also place LRC-BART slightly behind BART-PP in Sc1 and Sc5, whereas the all-profile definition shows an Sc1 advantage and an Sc5 tie.

| Scenario  | Control profiles | LRC-BART | BART-PP | Difference | Paired SE |
|-----------|------------------|----------|---------|------------|-----------|
| Sc1       | $X_5 > 2$        | 0.688    | 0.682   | 0.006      | 0.005     |
| Sc1       | $X_5 \leq 2$     | 0.717    | 0.711   | 0.006      | 0.004     |
| Sc1       | $X_5 > 2.5$      | 0.701    | 0.696   | 0.005      | 0.006     |
| Sc1       | $X_5 < 1.5$      | 0.733    | 0.726   | 0.007      | 0.005     |
| Sc4 (0.5) | $X_5 > 2$        | 0.695    | 0.685   | 0.010      | 0.004     |
| Sc4 (0.5) | $X_5 \leq 2$     | 0.740    | 0.719   | 0.021      | 0.004     |
| Sc4 (0.5) | $X_5 > 2.5$      | 0.713    | 0.704   | 0.009      | 0.005     |
| Sc4 (0.5) | $X_5 < 1.5$      | 0.746    | 0.719   | 0.028      | 0.005     |
| Sc4 (1)   | $X_5 > 2$        | 0.713    | 0.705   | 0.008      | 0.004     |
| Sc4 (1)   | $X_5 \leq 2$     | 0.772    | 0.744   | 0.028      | 0.005     |
| Sc4 (1)   | $X_5 > 2.5$      | 0.731    | 0.729   | 0.002      | 0.005     |
| Sc4 (1)   | $X_5 < 1.5$      | 0.774    | 0.748   | 0.026      | 0.006     |
| Sc4 (2)   | $X_5 > 2$        | 0.751    | 0.704   | 0.047      | 0.005     |
| Sc4 (2)   | $X_5 \leq 2$     | 0.780    | 0.746   | 0.034      | 0.006     |
| Sc4 (2)   | $X_5 > 2.5$      | 0.747    | 0.723   | 0.024      | 0.005     |
| Sc4 (2)   | $X_5 < 1.5$      | 0.770    | 0.748   | 0.022      | 0.006     |
| Sc5 (2)   | $X_5 > 2$        | 0.709    | 0.688   | 0.020      | 0.004     |
| Sc5 (2)   | $X_5 \leq 2$     | 0.736    | 0.713   | 0.023      | 0.005     |
| Sc5 (2)   | $X_5 > 2.5$      | 0.722    | 0.699   | 0.023      | 0.005     |
| Sc5 (2)   | $X_5 < 1.5$      | 0.749    | 0.723   | 0.025      | 0.006     |

For the regional ESS in Table 2, we standardize the control mean to all RCT profiles in each of  $\{X_5 > 2\}$  and  $\{X_5 \leq 2\}$ . We use its RWD-only posterior variance and the posterior leaf proportions, spike indicators and scales to evaluate the realized ESS within each region. In Sc1 and Sc5 the partition is retained for comparison; it does not define a compatible region in Sc5. Using an average over the untruncated scale prior, the prior pointwise ESS, about 2.1–2.3 across regions and scenarios, describes external support before the discrepancy is learned and does not distinguish compatible from conflicting regions.

#### G.4 Full simulation tables

Table S1: Gaussian outcomes, two-arm design, all scenarios and methods,  $R = 500$ . Columns as in Table 1, plus bias and RMSE of the RCT-standardized treated mean. MAP ( $N$ ) is the meta-analytic predictive prior calibrated to prior ESS  $N$ ; LRC-BART ( $N$ ) targets prior ESS  $N$  and LRC-BART ( $p$ ) the fraction  $p$  of the ESS ceiling; LRC-BART ( $w = 0$ ) fixes every leaf in the slab; LRC-BART ( $H_g = 10$ ) doubles the discrepancy ensemble.

| Scenario | Method                  | ATE    |       |       |      |      | Treated mean |       | Control mean |       |
|----------|-------------------------|--------|-------|-------|------|------|--------------|-------|--------------|-------|
|          |                         | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  | Bias         | RMSE  |
| Sc1      | LM-NP                   | −0.002 | 0.341 | 0.342 | 0.94 | 0.83 | −0.001       | 0.145 | 0.001        | 0.250 |
|          | LM-CP                   | 0.009  | 0.243 | 0.236 | 0.96 | 0.99 | −0.000       | 0.145 | −0.009       | 0.175 |
|          | LM-PP                   | −0.002 | 0.347 | 0.337 | 0.94 | 0.85 | −0.000       | 0.145 | 0.001        | 0.243 |
|          | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|          | BART-CP                 | 0.008  | 0.145 | 0.142 | 0.96 | 1.00 | 0.005        | 0.109 | −0.003       | 0.096 |
|          | BART-PP                 | 0.006  | 0.207 | 0.205 | 0.94 | 1.00 | 0.005        | 0.110 | −0.001       | 0.170 |
|          | PSCL                    | 0.005  | 0.269 | 0.261 | 0.95 | 0.97 | 0.000        | 0.148 | −0.005       | 0.171 |
|          | MAP (100)               | 0.006  | 0.292 | 0.265 | 0.97 | 0.95 | 0.000        | 0.146 | −0.005       | 0.174 |
|          | MAP (75)                | 0.004  | 0.301 | 0.274 | 0.97 | 0.94 | −0.000       | 0.146 | −0.004       | 0.178 |
|          | MAP (50)                | 0.003  | 0.311 | 0.285 | 0.97 | 0.92 | −0.000       | 0.146 | −0.004       | 0.186 |
|          | HierLM                  | −0.003 | 0.338 | 0.341 | 0.93 | 0.85 | −0.001       | 0.145 | 0.003        | 0.249 |
|          | C-BART                  | 0.007  | 0.155 | 0.150 | 0.95 | 1.00 | 0.005        | 0.108 | −0.002       | 0.106 |
|          | LRC-BART ( $w = 0$ )    | 0.009  | 0.189 | 0.205 | 0.92 | 1.00 | 0.005        | 0.110 | −0.004       | 0.169 |
|          | LRC-BART (100)          | 0.009  | 0.171 | 0.172 | 0.94 | 1.00 | 0.005        | 0.109 | −0.004       | 0.134 |
|          | LRC-BART (75)           | 0.008  | 0.174 | 0.176 | 0.94 | 1.00 | 0.005        | 0.110 | −0.003       | 0.136 |
|          | LRC-BART (50)           | 0.009  | 0.177 | 0.181 | 0.94 | 1.00 | 0.005        | 0.110 | −0.004       | 0.142 |
|          | LRC-BART (0.9)          | 0.011  | 0.171 | 0.171 | 0.95 | 1.00 | 0.005        | 0.109 | −0.005       | 0.132 |
|          | LRC-BART (0.5)          | 0.009  | 0.175 | 0.178 | 0.93 | 1.00 | 0.005        | 0.109 | −0.004       | 0.139 |
|          | LRC-BART (0.25)         | 0.008  | 0.180 | 0.184 | 0.94 | 1.00 | 0.005        | 0.109 | −0.002       | 0.147 |
|          | LRC-BART ( $H_g = 10$ ) | 0.009  | 0.171 | 0.169 | 0.94 | 1.00 | 0.005        | 0.109 | −0.004       | 0.131 |
| Sc2      | LM-NP                   | −0.016 | 0.320 | 0.324 | 0.95 | 0.87 | −0.000       | 0.140 | 0.016        | 0.241 |
|          | LM-CP                   | 0.379  | 0.286 | 0.487 | 0.73 | 0.99 | 0.000        | 0.140 | −0.379       | 0.445 |
|          | LM-PP                   | 0.010  | 0.340 | 0.331 | 0.95 | 0.84 | 0.000        | 0.140 | −0.010       | 0.248 |
|          | BART-NP                 | −0.008 | 0.201 | 0.219 | 0.93 | 1.00 | 0.001        | 0.115 | 0.009        | 0.180 |
|          | BART-CP                 | 0.058  | 0.177 | 0.195 | 0.91 | 1.00 | 0.002        | 0.115 | −0.056       | 0.151 |
|          | BART-PP                 | 0.007  | 0.208 | 0.204 | 0.96 | 1.00 | 0.002        | 0.115 | −0.005       | 0.163 |
|          | PSCL                    | 0.024  | 0.321 | 0.292 | 0.97 | 0.91 | −0.005       | 0.141 | −0.029       | 0.205 |
|          | MAP (100)               | 0.332  | 0.360 | 0.491 | 0.83 | 0.97 | −0.005       | 0.149 | −0.337       | 0.424 |
|          | MAP (75)                | 0.250  | 0.363 | 0.445 | 0.89 | 0.93 | −0.006       | 0.149 | −0.256       | 0.368 |
|          | MAP (50)                | 0.177  | 0.365 | 0.412 | 0.92 | 0.88 | −0.006       | 0.149 | −0.183       | 0.324 |
|          | HierLM                  | −0.011 | 0.318 | 0.321 | 0.95 | 0.88 | −0.000       | 0.140 | 0.011        | 0.238 |
|          | C-BART                  | 0.042  | 0.168 | 0.186 | 0.91 | 1.00 | 0.001        | 0.114 | −0.041       | 0.141 |
|          | LRC-BART ( $w = 0$ )    | 0.008  | 0.190 | 0.204 | 0.94 | 1.00 | 0.002        | 0.114 | −0.005       | 0.165 |
|          | LRC-BART (100)          | 0.023  | 0.181 | 0.192 | 0.94 | 1.00 | 0.002        | 0.114 | −0.021       | 0.148 |
|          | LRC-BART (75)           | 0.022  | 0.182 | 0.192 | 0.93 | 1.00 | 0.002        | 0.114 | −0.021       | 0.150 |
|          | LRC-BART (50)           | 0.023  | 0.181 | 0.193 | 0.93 | 1.00 | 0.002        | 0.115 | −0.021       | 0.149 |
|          | LRC-BART (0.9)          | 0.019  | 0.183 | 0.193 | 0.93 | 1.00 | 0.002        | 0.114 | −0.017       | 0.151 |
|          | LRC-BART (0.5)          | 0.016  | 0.186 | 0.197 | 0.93 | 1.00 | 0.002        | 0.114 | −0.013       | 0.156 |
|          | LRC-BART (0.25)         | 0.012  | 0.188 | 0.200 | 0.94 | 1.00 | 0.002        | 0.114 | −0.010       | 0.160 |
|          | LRC-BART ( $H_g = 10$ ) | 0.022  | 0.181 | 0.190 | 0.93 | 1.00 | 0.001        | 0.114 | −0.021       | 0.147 |
|          | LM-NP                   | 0.003  | 0.359 | 0.352 | 0.95 | 0.81 | 0.015        | 0.151 | 0.011        | 0.255 |

*continued on the next page*

Web Table S1, continued

| Scenario        | Method                  | ATE    |       |       |      |      | Treated mean |       | Control mean |       |
|-----------------|-------------------------|--------|-------|-------|------|------|--------------|-------|--------------|-------|
|                 |                         | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  | Bias         | RMSE  |
|                 | LM-CP                   | 0.888  | 0.264 | 0.927 | 0.10 | 1.00 | 0.015        | 0.150 | -0.873       | 0.893 |
|                 | LM-PP                   | 0.041  | 0.353 | 0.342 | 0.95 | 0.85 | 0.015        | 0.150 | -0.026       | 0.240 |
|                 | BART-NP                 | -0.004 | 0.284 | 0.282 | 0.94 | 0.94 | 0.020        | 0.131 | 0.024        | 0.216 |
|                 | BART-CP                 | 0.846  | 0.206 | 0.871 | 0.01 | 1.00 | 0.020        | 0.131 | -0.825       | 0.839 |
|                 | BART-PP                 | 0.027  | 0.271 | 0.270 | 0.93 | 0.96 | 0.020        | 0.131 | -0.008       | 0.200 |
|                 | PSCL                    | 0.675  | 0.277 | 0.728 | 0.31 | 1.00 | 0.009        | 0.148 | -0.665       | 0.690 |
|                 | MAP (100)               | 0.363  | 0.360 | 0.504 | 0.82 | 0.97 | 0.011        | 0.148 | -0.352       | 0.431 |
|                 | MAP (75)                | 0.280  | 0.363 | 0.453 | 0.88 | 0.93 | 0.010        | 0.149 | -0.270       | 0.371 |
|                 | MAP (50)                | 0.207  | 0.365 | 0.415 | 0.92 | 0.91 | 0.010        | 0.149 | -0.197       | 0.323 |
|                 | HierLM                  | 0.033  | 0.356 | 0.348 | 0.95 | 0.84 | 0.015        | 0.151 | -0.018       | 0.251 |
|                 | C-BART                  | 0.280  | 0.292 | 0.431 | 0.81 | 0.99 | 0.020        | 0.131 | -0.259       | 0.375 |
|                 | LRC-BART ( $w = 0$ )    | 0.024  | 0.272 | 0.271 | 0.93 | 0.96 | 0.020        | 0.132 | -0.003       | 0.201 |
|                 | LRC-BART (100)          | 0.042  | 0.276 | 0.278 | 0.93 | 0.96 | 0.021        | 0.131 | -0.021       | 0.209 |
|                 | LRC-BART (75)           | 0.043  | 0.276 | 0.278 | 0.93 | 0.96 | 0.021        | 0.131 | -0.022       | 0.208 |
|                 | LRC-BART (50)           | 0.045  | 0.276 | 0.278 | 0.93 | 0.97 | 0.020        | 0.131 | -0.025       | 0.208 |
|                 | LRC-BART (0.9)          | 0.039  | 0.275 | 0.276 | 0.93 | 0.96 | 0.020        | 0.132 | -0.019       | 0.206 |
|                 | LRC-BART (0.5)          | 0.042  | 0.277 | 0.277 | 0.93 | 0.96 | 0.020        | 0.131 | -0.022       | 0.208 |
|                 | LRC-BART (0.25)         | 0.047  | 0.276 | 0.279 | 0.92 | 0.96 | 0.020        | 0.131 | -0.026       | 0.209 |
|                 | LRC-BART ( $H_g = 10$ ) | 0.054  | 0.276 | 0.280 | 0.93 | 0.96 | 0.021        | 0.132 | -0.034       | 0.210 |
| Sc3, $\rho = 0$ | LM-NP                   | 0.005  | 0.366 | 0.355 | 0.96 | 0.80 | 0.016        | 0.152 | 0.011        | 0.256 |
|                 | LM-CP                   | 1.037  | 0.256 | 1.067 | 0.03 | 1.00 | 0.017        | 0.152 | -1.020       | 1.035 |
|                 | LM-PP                   | 0.043  | 0.356 | 0.346 | 0.95 | 0.85 | 0.016        | 0.152 | -0.027       | 0.241 |
|                 | BART-NP                 | -0.002 | 0.292 | 0.289 | 0.94 | 0.92 | 0.022        | 0.134 | 0.024        | 0.219 |
|                 | BART-CP                 | 0.984  | 0.202 | 1.003 | 0.00 | 1.00 | 0.022        | 0.134 | -0.962       | 0.971 |
|                 | BART-PP                 | 0.029  | 0.275 | 0.273 | 0.94 | 0.96 | 0.023        | 0.135 | -0.007       | 0.197 |
|                 | PSCL                    | 0.704  | 0.269 | 0.753 | 0.26 | 1.00 | 0.012        | 0.149 | -0.693       | 0.713 |
|                 | MAP (100)               | 0.363  | 0.360 | 0.504 | 0.82 | 0.97 | 0.011        | 0.148 | -0.352       | 0.431 |
|                 | MAP (75)                | 0.280  | 0.363 | 0.453 | 0.88 | 0.93 | 0.010        | 0.149 | -0.270       | 0.371 |
|                 | MAP (50)                | 0.207  | 0.365 | 0.415 | 0.92 | 0.91 | 0.010        | 0.149 | -0.197       | 0.323 |
|                 | HierLM                  | 0.039  | 0.362 | 0.354 | 0.96 | 0.82 | 0.016        | 0.152 | -0.023       | 0.253 |
|                 | C-BART                  | 0.226  | 0.300 | 0.380 | 0.87 | 0.98 | 0.023        | 0.134 | -0.204       | 0.311 |
|                 | LRC-BART ( $w = 0$ )    | 0.023  | 0.280 | 0.275 | 0.94 | 0.96 | 0.023        | 0.134 | -0.000       | 0.201 |
|                 | LRC-BART (100)          | 0.041  | 0.284 | 0.280 | 0.94 | 0.96 | 0.023        | 0.134 | -0.018       | 0.206 |
|                 | LRC-BART (75)           | 0.042  | 0.283 | 0.281 | 0.94 | 0.96 | 0.023        | 0.134 | -0.019       | 0.207 |
|                 | LRC-BART (50)           | 0.045  | 0.283 | 0.282 | 0.94 | 0.96 | 0.023        | 0.134 | -0.022       | 0.208 |
|                 | LRC-BART (0.9)          | 0.037  | 0.282 | 0.277 | 0.94 | 0.96 | 0.023        | 0.134 | -0.015       | 0.203 |
|                 | LRC-BART (0.5)          | 0.041  | 0.282 | 0.280 | 0.94 | 0.96 | 0.023        | 0.134 | -0.018       | 0.205 |
|                 | LRC-BART (0.25)         | 0.045  | 0.283 | 0.279 | 0.94 | 0.97 | 0.023        | 0.134 | -0.022       | 0.205 |
|                 | LRC-BART ( $H_g = 10$ ) | 0.055  | 0.283 | 0.282 | 0.94 | 0.97 | 0.023        | 0.134 | -0.032       | 0.206 |
|                 | LM-NP                   | 0.008  | 0.360 | 0.350 | 0.96 | 0.80 | 0.017        | 0.151 | 0.009        | 0.252 |
|                 | LM-CP                   | 0.899  | 0.265 | 0.937 | 0.08 | 1.00 | 0.017        | 0.151 | -0.882       | 0.901 |
|                 | LM-PP                   | 0.043  | 0.353 | 0.343 | 0.95 | 0.84 | 0.017        | 0.151 | -0.027       | 0.241 |
|                 | BART-NP                 | 0.001  | 0.284 | 0.280 | 0.95 | 0.94 | 0.021        | 0.132 | 0.020        | 0.213 |
|                 | BART-CP                 | 0.847  | 0.205 | 0.872 | 0.01 | 1.00 | 0.021        | 0.133 | -0.826       | 0.838 |
|                 | BART-PP                 | 0.029  | 0.271 | 0.267 | 0.94 | 0.96 | 0.021        | 0.132 | -0.008       | 0.197 |
|                 | PSCL                    | 0.678  | 0.277 | 0.731 | 0.30 | 1.00 | 0.010        | 0.149 | -0.668       | 0.692 |

continued on the next page

Web Table S1, continued

| Scenario                      | Method                  | ATE    |       |       |      |      | Treated mean |       | Control mean |       |
|-------------------------------|-------------------------|--------|-------|-------|------|------|--------------|-------|--------------|-------|
|                               |                         | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  | Bias         | RMSE  |
|                               | MAP (100)               | 0.363  | 0.360 | 0.504 | 0.82 | 0.97 | 0.011        | 0.148 | -0.352       | 0.431 |
|                               | MAP (75)                | 0.280  | 0.363 | 0.453 | 0.88 | 0.93 | 0.010        | 0.149 | -0.270       | 0.371 |
|                               | MAP (50)                | 0.207  | 0.365 | 0.415 | 0.92 | 0.91 | 0.010        | 0.149 | -0.197       | 0.323 |
|                               | HierLM                  | 0.038  | 0.356 | 0.349 | 0.96 | 0.82 | 0.017        | 0.151 | -0.021       | 0.250 |
|                               | C-BART                  | 0.277  | 0.292 | 0.430 | 0.82 | 0.99 | 0.021        | 0.133 | -0.256       | 0.371 |
|                               | LRC-BART ( $w = 0$ )    | 0.021  | 0.273 | 0.266 | 0.95 | 0.96 | 0.021        | 0.133 | -0.001       | 0.196 |
|                               | LRC-BART (100)          | 0.040  | 0.276 | 0.275 | 0.94 | 0.96 | 0.021        | 0.132 | -0.020       | 0.204 |
|                               | LRC-BART (75)           | 0.042  | 0.276 | 0.274 | 0.94 | 0.97 | 0.021        | 0.132 | -0.021       | 0.203 |
|                               | LRC-BART (50)           | 0.044  | 0.277 | 0.274 | 0.95 | 0.97 | 0.021        | 0.132 | -0.023       | 0.204 |
|                               | LRC-BART (0.9)          | 0.039  | 0.276 | 0.273 | 0.95 | 0.97 | 0.021        | 0.133 | -0.018       | 0.202 |
|                               | LRC-BART (0.5)          | 0.043  | 0.276 | 0.274 | 0.95 | 0.97 | 0.021        | 0.132 | -0.021       | 0.203 |
|                               | LRC-BART (0.25)         | 0.046  | 0.277 | 0.275 | 0.94 | 0.97 | 0.021        | 0.132 | -0.025       | 0.205 |
|                               | LRC-BART ( $H_g = 10$ ) | 0.053  | 0.277 | 0.276 | 0.94 | 0.97 | 0.021        | 0.132 | -0.032       | 0.206 |
| Sc4, $X_5$ , $\delta_2 = 0.5$ | LM-NP                   | -0.002 | 0.341 | 0.342 | 0.94 | 0.83 | -0.001       | 0.145 | 0.001        | 0.250 |
|                               | LM-CP                   | -0.176 | 0.243 | 0.295 | 0.90 | 0.92 | -0.000       | 0.145 | 0.176        | 0.249 |
|                               | LM-PP                   | -0.007 | 0.348 | 0.338 | 0.94 | 0.83 | -0.000       | 0.145 | 0.006        | 0.243 |
|                               | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|                               | BART-CP                 | -0.164 | 0.145 | 0.217 | 0.78 | 1.00 | 0.005        | 0.109 | 0.169        | 0.195 |
|                               | BART-PP                 | 0.000  | 0.208 | 0.203 | 0.95 | 1.00 | 0.005        | 0.110 | 0.005        | 0.168 |
|                               | PSCL                    | -0.120 | 0.270 | 0.287 | 0.95 | 0.92 | 0.000        | 0.148 | 0.121        | 0.209 |
|                               | MAP (100)               | -0.115 | 0.296 | 0.294 | 0.95 | 0.88 | 0.000        | 0.146 | 0.116        | 0.212 |
|                               | MAP (75)                | -0.101 | 0.306 | 0.297 | 0.95 | 0.88 | -0.000       | 0.146 | 0.100        | 0.209 |
|                               | MAP (50)                | -0.082 | 0.316 | 0.303 | 0.96 | 0.88 | -0.000       | 0.146 | 0.082        | 0.210 |
|                               | HierLM                  | -0.014 | 0.338 | 0.342 | 0.94 | 0.84 | -0.001       | 0.145 | 0.014        | 0.249 |
|                               | C-BART                  | -0.109 | 0.159 | 0.190 | 0.90 | 1.00 | 0.005        | 0.108 | 0.115        | 0.162 |
|                               | LRC-BART ( $w = 0$ )    | 0.007  | 0.190 | 0.203 | 0.93 | 1.00 | 0.005        | 0.110 | -0.001       | 0.169 |
|                               | LRC-BART (100)          | -0.043 | 0.180 | 0.200 | 0.92 | 1.00 | 0.005        | 0.109 | 0.048        | 0.168 |
|                               | LRC-BART (75)           | -0.040 | 0.181 | 0.197 | 0.93 | 1.00 | 0.005        | 0.110 | 0.045        | 0.165 |
|                               | LRC-BART (50)           | -0.031 | 0.182 | 0.194 | 0.94 | 1.00 | 0.005        | 0.110 | 0.036        | 0.160 |
|                               | LRC-BART (0.9)          | -0.043 | 0.180 | 0.200 | 0.93 | 1.00 | 0.005        | 0.109 | 0.048        | 0.169 |
|                               | LRC-BART (0.5)          | -0.035 | 0.182 | 0.196 | 0.93 | 1.00 | 0.005        | 0.109 | 0.040        | 0.163 |
|                               | LRC-BART (0.25)         | -0.022 | 0.184 | 0.195 | 0.93 | 1.00 | 0.005        | 0.109 | 0.028        | 0.161 |
|                               | LRC-BART ( $H_g = 10$ ) | -0.048 | 0.179 | 0.198 | 0.91 | 1.00 | 0.005        | 0.109 | 0.053        | 0.167 |
| Sc4, $X_5$ , $\delta_2 = 1$   | LM-NP                   | -0.002 | 0.341 | 0.342 | 0.94 | 0.83 | -0.001       | 0.145 | 0.001        | 0.250 |
|                               | LM-CP                   | -0.361 | 0.244 | 0.433 | 0.67 | 0.74 | -0.000       | 0.145 | 0.361        | 0.403 |
|                               | LM-PP                   | -0.012 | 0.349 | 0.339 | 0.95 | 0.83 | -0.000       | 0.145 | 0.011        | 0.245 |
|                               | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|                               | BART-CP                 | -0.338 | 0.146 | 0.368 | 0.35 | 1.00 | 0.005        | 0.109 | 0.343        | 0.357 |
|                               | BART-PP                 | -0.002 | 0.209 | 0.207 | 0.95 | 1.00 | 0.005        | 0.110 | 0.007        | 0.173 |
|                               | PSCL                    | -0.246 | 0.271 | 0.358 | 0.87 | 0.80 | 0.000        | 0.148 | 0.247        | 0.300 |
|                               | MAP (100)               | -0.224 | 0.306 | 0.361 | 0.91 | 0.75 | 0.000        | 0.146 | 0.224        | 0.292 |
|                               | MAP (75)                | -0.190 | 0.317 | 0.352 | 0.92 | 0.75 | -0.000       | 0.146 | 0.190        | 0.274 |
|                               | MAP (50)                | -0.152 | 0.328 | 0.346 | 0.94 | 0.77 | -0.000       | 0.146 | 0.152        | 0.260 |
|                               | HierLM                  | -0.025 | 0.338 | 0.343 | 0.94 | 0.83 | -0.001       | 0.145 | 0.024        | 0.250 |
|                               | C-BART                  | -0.207 | 0.173 | 0.273 | 0.73 | 1.00 | 0.005        | 0.108 | 0.212        | 0.258 |
|                               | LRC-BART ( $w = 0$ )    | 0.006  | 0.191 | 0.204 | 0.93 | 1.00 | 0.005        | 0.110 | -0.000       | 0.171 |

continued on the next page

Web Table S1, continued

| Scenario                    | Method                  | ATE    |       |       |      |      | Treated mean |       | Control mean |       |
|-----------------------------|-------------------------|--------|-------|-------|------|------|--------------|-------|--------------|-------|
|                             |                         | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  | Bias         | RMSE  |
|                             | LRC-BART (100)          | -0.041 | 0.195 | 0.223 | 0.89 | 1.00 | 0.005        | 0.109 | 0.046        | 0.194 |
|                             | LRC-BART (75)           | -0.041 | 0.194 | 0.221 | 0.90 | 1.00 | 0.005        | 0.110 | 0.046        | 0.191 |
|                             | LRC-BART (50)           | -0.036 | 0.193 | 0.213 | 0.90 | 1.00 | 0.005        | 0.110 | 0.042        | 0.184 |
|                             | LRC-BART (0.9)          | -0.041 | 0.196 | 0.224 | 0.90 | 1.00 | 0.005        | 0.109 | 0.047        | 0.194 |
|                             | LRC-BART (0.5)          | -0.037 | 0.194 | 0.218 | 0.90 | 1.00 | 0.005        | 0.109 | 0.042        | 0.187 |
|                             | LRC-BART (0.25)         | -0.032 | 0.192 | 0.208 | 0.92 | 1.00 | 0.005        | 0.109 | 0.037        | 0.178 |
|                             | LRC-BART ( $H_g = 10$ ) | -0.046 | 0.195 | 0.222 | 0.89 | 1.00 | 0.005        | 0.109 | 0.051        | 0.195 |
| Sc4, $X_5$ , $\delta_2 = 2$ | LM-NP                   | -0.002 | 0.341 | 0.342 | 0.94 | 0.83 | -0.001       | 0.145 | 0.001        | 0.250 |
|                             | LM-CP                   | -0.732 | 0.247 | 0.771 | 0.16 | 0.21 | -0.000       | 0.145 | 0.732        | 0.755 |
|                             | LM-PP                   | -0.023 | 0.354 | 0.343 | 0.95 | 0.81 | -0.000       | 0.145 | 0.022        | 0.249 |
|                             | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|                             | BART-CP                 | -0.685 | 0.149 | 0.701 | 0.00 | 0.56 | 0.005        | 0.109 | 0.690        | 0.698 |
|                             | BART-PP                 | -0.006 | 0.209 | 0.205 | 0.96 | 1.00 | 0.005        | 0.110 | 0.011        | 0.170 |
|                             | PSCL                    | -0.498 | 0.273 | 0.562 | 0.54 | 0.44 | 0.000        | 0.148 | 0.498        | 0.528 |
|                             | MAP (100)               | -0.377 | 0.333 | 0.495 | 0.78 | 0.46 | 0.000        | 0.146 | 0.378        | 0.437 |
|                             | MAP (75)                | -0.296 | 0.344 | 0.451 | 0.84 | 0.54 | -0.000       | 0.146 | 0.296        | 0.380 |
|                             | MAP (50)                | -0.219 | 0.352 | 0.415 | 0.92 | 0.60 | -0.000       | 0.146 | 0.219        | 0.333 |
|                             | HierLM                  | -0.044 | 0.339 | 0.345 | 0.94 | 0.81 | -0.001       | 0.145 | 0.043        | 0.253 |
|                             | C-BART                  | -0.227 | 0.197 | 0.391 | 0.66 | 0.90 | 0.005        | 0.108 | 0.232        | 0.383 |
|                             | LRC-BART ( $w = 0$ )    | 0.004  | 0.192 | 0.207 | 0.92 | 1.00 | 0.005        | 0.110 | 0.001        | 0.173 |
|                             | LRC-BART (100)          | 0.000  | 0.189 | 0.202 | 0.92 | 1.00 | 0.005        | 0.109 | 0.005        | 0.168 |
|                             | LRC-BART (75)           | 0.001  | 0.190 | 0.201 | 0.93 | 1.00 | 0.005        | 0.110 | 0.004        | 0.167 |
|                             | LRC-BART (50)           | -0.002 | 0.190 | 0.201 | 0.92 | 1.00 | 0.005        | 0.110 | 0.008        | 0.168 |
|                             | LRC-BART (0.9)          | -0.000 | 0.189 | 0.202 | 0.93 | 1.00 | 0.005        | 0.109 | 0.006        | 0.170 |
|                             | LRC-BART (0.5)          | -0.001 | 0.190 | 0.202 | 0.92 | 1.00 | 0.005        | 0.109 | 0.006        | 0.168 |
|                             | LRC-BART (0.25)         | -0.001 | 0.191 | 0.204 | 0.93 | 1.00 | 0.005        | 0.109 | 0.007        | 0.168 |
|                             | LRC-BART ( $H_g = 10$ ) | -0.004 | 0.188 | 0.199 | 0.93 | 1.00 | 0.005        | 0.109 | 0.009        | 0.166 |
| Sc4, $X_7$ , $\delta_2 = 1$ | LM-NP                   | -0.002 | 0.341 | 0.342 | 0.94 | 0.83 | -0.001       | 0.145 | 0.001        | 0.250 |
|                             | LM-CP                   | -0.363 | 0.244 | 0.434 | 0.66 | 0.72 | -0.000       | 0.145 | 0.363        | 0.404 |
|                             | LM-PP                   | -0.012 | 0.348 | 0.337 | 0.94 | 0.83 | -0.000       | 0.145 | 0.012        | 0.244 |
|                             | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|                             | BART-CP                 | -0.340 | 0.146 | 0.369 | 0.35 | 1.00 | 0.005        | 0.109 | 0.345        | 0.359 |
|                             | BART-PP                 | -0.001 | 0.209 | 0.206 | 0.96 | 1.00 | 0.005        | 0.110 | 0.006        | 0.172 |
|                             | PSCL                    | -0.246 | 0.269 | 0.358 | 0.87 | 0.81 | 0.000        | 0.148 | 0.246        | 0.301 |
|                             | MAP (100)               | -0.222 | 0.308 | 0.360 | 0.91 | 0.75 | 0.000        | 0.146 | 0.222        | 0.291 |
|                             | MAP (75)                | -0.189 | 0.318 | 0.352 | 0.92 | 0.74 | -0.000       | 0.146 | 0.189        | 0.274 |
|                             | MAP (50)                | -0.152 | 0.328 | 0.347 | 0.94 | 0.77 | -0.000       | 0.146 | 0.151        | 0.260 |
|                             | HierLM                  | 0.001  | 0.338 | 0.341 | 0.93 | 0.85 | -0.001       | 0.145 | -0.001       | 0.249 |
|                             | C-BART                  | -0.206 | 0.173 | 0.276 | 0.73 | 1.00 | 0.005        | 0.108 | 0.211        | 0.258 |
|                             | LRC-BART ( $w = 0$ )    | 0.009  | 0.191 | 0.206 | 0.92 | 1.00 | 0.005        | 0.110 | -0.004       | 0.172 |
|                             | LRC-BART (100)          | -0.038 | 0.195 | 0.224 | 0.90 | 1.00 | 0.005        | 0.109 | 0.043        | 0.196 |
|                             | LRC-BART (75)           | -0.035 | 0.194 | 0.220 | 0.91 | 1.00 | 0.005        | 0.110 | 0.041        | 0.190 |
|                             | LRC-BART (50)           | -0.031 | 0.193 | 0.217 | 0.90 | 1.00 | 0.005        | 0.110 | 0.037        | 0.187 |
|                             | LRC-BART (0.9)          | -0.040 | 0.195 | 0.223 | 0.90 | 1.00 | 0.005        | 0.109 | 0.045        | 0.194 |
|                             | LRC-BART (0.5)          | -0.034 | 0.193 | 0.218 | 0.91 | 1.00 | 0.005        | 0.109 | 0.039        | 0.188 |
|                             | LRC-BART (0.25)         | -0.025 | 0.191 | 0.212 | 0.91 | 1.00 | 0.005        | 0.109 | 0.031        | 0.180 |

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Web Table S1, continued

| Scenario                    | Method                  | ATE    |       |       |      |      | Treated mean |       | Control mean |       |
|-----------------------------|-------------------------|--------|-------|-------|------|------|--------------|-------|--------------|-------|
|                             |                         | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  | Bias         | RMSE  |
| Sc4, $X_7$ , $\delta_2 = 2$ | LRC-BART ( $H_g = 10$ ) | -0.044 | 0.195 | 0.221 | 0.89 | 1.00 | 0.005        | 0.109 | 0.049        | 0.192 |
|                             | LM-NP                   | -0.002 | 0.341 | 0.342 | 0.94 | 0.83 | -0.001       | 0.145 | 0.001        | 0.250 |
|                             | LM-CP                   | -0.735 | 0.246 | 0.773 | 0.16 | 0.20 | -0.000       | 0.145 | 0.735        | 0.758 |
|                             | LM-PP                   | -0.023 | 0.352 | 0.339 | 0.95 | 0.81 | -0.000       | 0.145 | 0.023        | 0.247 |
|                             | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|                             | BART-CP                 | -0.685 | 0.148 | 0.701 | 0.00 | 0.54 | 0.005        | 0.109 | 0.690        | 0.698 |
|                             | BART-PP                 | -0.002 | 0.209 | 0.206 | 0.95 | 1.00 | 0.005        | 0.110 | 0.007        | 0.173 |
|                             | PSCL                    | -0.498 | 0.271 | 0.562 | 0.53 | 0.45 | 0.000        | 0.148 | 0.498        | 0.528 |
|                             | MAP (100)               | -0.358 | 0.338 | 0.486 | 0.80 | 0.48 | 0.000        | 0.146 | 0.358        | 0.424 |
|                             | MAP (75)                | -0.284 | 0.347 | 0.446 | 0.88 | 0.54 | -0.000       | 0.146 | 0.283        | 0.374 |
|                             | MAP (50)                | -0.212 | 0.353 | 0.413 | 0.92 | 0.59 | -0.000       | 0.146 | 0.211        | 0.331 |
|                             | HierLM                  | 0.005  | 0.338 | 0.341 | 0.93 | 0.85 | -0.001       | 0.145 | -0.005       | 0.249 |
|                             | C-BART                  | -0.218 | 0.199 | 0.383 | 0.69 | 0.91 | 0.005        | 0.108 | 0.223        | 0.374 |
|                             | LRC-BART ( $w = 0$ )    | 0.007  | 0.192 | 0.209 | 0.92 | 1.00 | 0.005        | 0.110 | -0.002       | 0.176 |
|                             | LRC-BART (100)          | 0.001  | 0.189 | 0.203 | 0.93 | 1.00 | 0.005        | 0.109 | 0.004        | 0.171 |
|                             | LRC-BART (75)           | 0.004  | 0.189 | 0.202 | 0.93 | 1.00 | 0.005        | 0.110 | 0.001        | 0.169 |
|                             | LRC-BART (50)           | 0.002  | 0.190 | 0.205 | 0.93 | 1.00 | 0.005        | 0.110 | 0.003        | 0.173 |
|                             | LRC-BART (0.9)          | 0.001  | 0.189 | 0.204 | 0.93 | 1.00 | 0.005        | 0.109 | 0.004        | 0.171 |
|                             | LRC-BART (0.5)          | 0.003  | 0.190 | 0.203 | 0.93 | 1.00 | 0.005        | 0.109 | 0.002        | 0.169 |
|                             | LRC-BART (0.25)         | 0.002  | 0.190 | 0.204 | 0.92 | 1.00 | 0.005        | 0.109 | 0.004        | 0.172 |
|                             | LRC-BART ( $H_g = 10$ ) | -0.002 | 0.189 | 0.204 | 0.92 | 1.00 | 0.005        | 0.109 | 0.007        | 0.170 |
| Sc5, $\delta_2 = 1$         | LM-NP                   | -0.002 | 0.341 | 0.342 | 0.94 | 0.83 | -0.001       | 0.145 | 0.001        | 0.250 |
|                             | LM-CP                   | -0.734 | 0.244 | 0.771 | 0.15 | 0.20 | -0.000       | 0.145 | 0.734        | 0.755 |
|                             | LM-PP                   | -0.022 | 0.347 | 0.338 | 0.95 | 0.82 | -0.000       | 0.145 | 0.022        | 0.244 |
|                             | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|                             | BART-CP                 | -0.687 | 0.147 | 0.702 | 0.00 | 0.55 | 0.005        | 0.109 | 0.692        | 0.699 |
|                             | BART-PP                 | -0.004 | 0.208 | 0.205 | 0.95 | 1.00 | 0.005        | 0.110 | 0.010        | 0.171 |
|                             | PSCL                    | -0.498 | 0.269 | 0.562 | 0.53 | 0.45 | 0.000        | 0.148 | 0.498        | 0.527 |
|                             | MAP (100)               | -0.345 | 0.341 | 0.480 | 0.82 | 0.49 | 0.000        | 0.146 | 0.345        | 0.417 |
|                             | MAP (75)                | -0.275 | 0.349 | 0.444 | 0.88 | 0.54 | -0.000       | 0.146 | 0.275        | 0.370 |
|                             | MAP (50)                | -0.207 | 0.355 | 0.412 | 0.92 | 0.60 | -0.000       | 0.146 | 0.206        | 0.329 |
|                             | HierLM                  | -0.021 | 0.338 | 0.342 | 0.94 | 0.84 | -0.001       | 0.145 | 0.021        | 0.249 |
|                             | C-BART                  | -0.244 | 0.198 | 0.373 | 0.68 | 0.94 | 0.005        | 0.108 | 0.249        | 0.362 |
|                             | LRC-BART ( $w = 0$ )    | 0.007  | 0.190 | 0.205 | 0.92 | 1.00 | 0.005        | 0.110 | -0.002       | 0.170 |
|                             | LRC-BART (100)          | -0.005 | 0.192 | 0.210 | 0.91 | 1.00 | 0.005        | 0.109 | 0.010        | 0.176 |
|                             | LRC-BART (75)           | -0.005 | 0.192 | 0.210 | 0.90 | 1.00 | 0.005        | 0.110 | 0.010        | 0.177 |
|                             | LRC-BART (50)           | -0.008 | 0.192 | 0.209 | 0.92 | 1.00 | 0.005        | 0.110 | 0.013        | 0.176 |
|                             | LRC-BART (0.9)          | -0.002 | 0.192 | 0.208 | 0.92 | 1.00 | 0.005        | 0.109 | 0.007        | 0.176 |
|                             | LRC-BART (0.5)          | -0.006 | 0.192 | 0.211 | 0.92 | 1.00 | 0.005        | 0.109 | 0.011        | 0.178 |
|                             | LRC-BART (0.25)         | -0.011 | 0.192 | 0.211 | 0.92 | 1.00 | 0.005        | 0.109 | 0.016        | 0.179 |
|                             | LRC-BART ( $H_g = 10$ ) | -0.009 | 0.191 | 0.207 | 0.92 | 1.00 | 0.005        | 0.109 | 0.014        | 0.176 |
|                             | LM-NP                   | -0.002 | 0.341 | 0.342 | 0.94 | 0.83 | -0.001       | 0.145 | 0.001        | 0.250 |
|                             | LM-CP                   | -1.478 | 0.247 | 1.497 | 0.00 | 0.51 | -0.000       | 0.145 | 1.478        | 1.489 |
|                             | LM-PP                   | -0.042 | 0.347 | 0.339 | 0.94 | 0.80 | -0.000       | 0.145 | 0.042        | 0.246 |
|                             | BART-NP                 | 0.004  | 0.203 | 0.220 | 0.91 | 1.00 | 0.005        | 0.109 | 0.001        | 0.185 |
|                             | BART-CP                 | -1.390 | 0.152 | 1.398 | 0.00 | 0.72 | 0.005        | 0.109 | 1.395        | 1.399 |

continued on the next page

Web Table S1, continued

| Scenario | Method                  | ATE    |       |       |      |      | Treated mean |       | Control mean |       |
|----------|-------------------------|--------|-------|-------|------|------|--------------|-------|--------------|-------|
|          |                         | Bias   | SD    | RMSE  | Cov  | Pow  | Bias         | RMSE  | Bias         | RMSE  |
|          | BART-PP                 | -0.012 | 0.208 | 0.205 | 0.95 | 1.00 | 0.005        | 0.110 | 0.017        | 0.171 |
|          | PSCL                    | -1.002 | 0.269 | 1.035 | 0.04 | 0.04 | 0.000        | 0.148 | 1.002        | 1.017 |
|          | MAP (100)               | -0.395 | 0.348 | 0.525 | 0.78 | 0.41 | 0.000        | 0.146 | 0.395        | 0.465 |
|          | MAP (75)                | -0.279 | 0.355 | 0.452 | 0.89 | 0.50 | -0.000       | 0.146 | 0.278        | 0.377 |
|          | MAP (50)                | -0.187 | 0.357 | 0.406 | 0.93 | 0.60 | -0.000       | 0.146 | 0.186        | 0.319 |
|          | HierLM                  | -0.038 | 0.338 | 0.344 | 0.94 | 0.82 | -0.001       | 0.145 | 0.037        | 0.251 |
|          | C-BART                  | -0.316 | 0.194 | 0.650 | 0.76 | 0.93 | 0.005        | 0.108 | 0.322        | 0.648 |
|          | LRC-BART ( $w = 0$ )    | 0.006  | 0.189 | 0.203 | 0.93 | 1.00 | 0.005        | 0.110 | -0.001       | 0.169 |
|          | LRC-BART (100)          | -0.001 | 0.190 | 0.203 | 0.92 | 1.00 | 0.005        | 0.109 | 0.005        | 0.171 |
|          | LRC-BART (75)           | -0.001 | 0.190 | 0.205 | 0.93 | 1.00 | 0.005        | 0.110 | 0.006        | 0.172 |
|          | LRC-BART (50)           | -0.001 | 0.190 | 0.203 | 0.93 | 1.00 | 0.005        | 0.110 | 0.006        | 0.169 |
|          | LRC-BART (0.9)          | -0.001 | 0.190 | 0.205 | 0.92 | 1.00 | 0.005        | 0.109 | 0.006        | 0.171 |
|          | LRC-BART (0.5)          | -0.001 | 0.190 | 0.204 | 0.93 | 1.00 | 0.005        | 0.109 | 0.006        | 0.171 |
|          | LRC-BART (0.25)         | 0.000  | 0.189 | 0.203 | 0.92 | 1.00 | 0.005        | 0.109 | 0.005        | 0.169 |
|          | LRC-BART ( $H_g = 10$ ) | -0.007 | 0.190 | 0.201 | 0.92 | 1.00 | 0.005        | 0.109 | 0.012        | 0.169 |

Table S2: Survival outcomes, two-arm design, all scenarios and methods,  $R = 500$ . Columns as in Table 3, with the mean posterior SD and the bias of the standardized control median on the log scale (Ctrl bias).

| Scenario | Method               | Median ratio |       |       |      |      |           | RMST ratio, $t^* = 3$ |       |       |      |      |
|----------|----------------------|--------------|-------|-------|------|------|-----------|-----------------------|-------|-------|------|------|
|          |                      | Bias         | SD    | RMSE  | Cov  | Pow  | Ctrl bias | Bias                  | SD    | RMSE  | Cov  | Pow  |
| Sc1      | AFT-NP               | 0.233        | 0.713 | 0.737 | 0.95 | 0.12 | 0.072     | -0.020                | 0.147 | 0.152 | 0.92 | 0.09 |
|          | AFT-CP               | 0.165        | 0.427 | 0.428 | 0.95 | 0.29 | -0.007    | -0.012                | 0.094 | 0.089 | 0.95 | 0.22 |
|          | AFT-PP               | 0.261        | 0.719 | 0.737 | 0.95 | 0.12 | 0.052     | -0.009                | 0.144 | 0.142 | 0.94 | 0.10 |
|          | BART-NP              | 0.260        | 0.417 | 0.511 | 0.92 | 0.46 | -0.036    | 0.131                 | 0.151 | 0.206 | 0.87 | 0.47 |
|          | BART-CP              | 0.064        | 0.238 | 0.232 | 0.96 | 0.63 | -0.019    | -0.011                | 0.074 | 0.072 | 0.95 | 0.35 |
|          | BART-PP              | 0.032        | 0.350 | 0.345 | 0.95 | 0.25 | 0.016     | -0.023                | 0.103 | 0.106 | 0.94 | 0.17 |
|          | HierAFT              | 0.246        | 0.701 | 0.733 | 0.94 | 0.12 | 0.060     | -0.008                | 0.148 | 0.152 | 0.92 | 0.11 |
|          | C-BART               | 0.034        | 0.268 | 0.248 | 0.97 | 0.42 | -0.007    | 0.003                 | 0.085 | 0.079 | 0.95 | 0.32 |
|          | LRC-BART ( $w = 0$ ) | 0.050        | 0.329 | 0.351 | 0.94 | 0.33 | 0.006     | -0.001                | 0.101 | 0.108 | 0.94 | 0.25 |
|          | LRC-BART (100)       | 0.045        | 0.305 | 0.312 | 0.95 | 0.34 | -0.001    | 0.003                 | 0.095 | 0.097 | 0.94 | 0.28 |
|          | LRC-BART (75)        | 0.048        | 0.308 | 0.312 | 0.96 | 0.36 | -0.001    | 0.003                 | 0.096 | 0.097 | 0.95 | 0.28 |
|          | LRC-BART (50)        | 0.047        | 0.312 | 0.318 | 0.96 | 0.35 | 0.001     | 0.003                 | 0.097 | 0.099 | 0.95 | 0.27 |
|          | LRC-BART (0.9)       | 0.042        | 0.298 | 0.304 | 0.96 | 0.38 | -0.001    | 0.002                 | 0.093 | 0.095 | 0.95 | 0.30 |
|          | LRC-BART (0.5)       | 0.048        | 0.305 | 0.309 | 0.96 | 0.35 | -0.002    | 0.004                 | 0.095 | 0.097 | 0.94 | 0.28 |
|          | LRC-BART (0.25)      | 0.047        | 0.311 | 0.320 | 0.95 | 0.35 | 0.001     | 0.003                 | 0.097 | 0.099 | 0.94 | 0.28 |
| Sc2      | AFT-NP               | 0.175        | 0.672 | 0.672 | 0.96 | 0.12 | 0.102     | -0.024                | 0.131 | 0.131 | 0.93 | 0.09 |
|          | AFT-CP               | 1.519        | 0.996 | 1.831 | 0.55 | 0.83 | -0.207    | 0.261                 | 0.153 | 0.304 | 0.57 | 0.82 |
|          | AFT-PP               | 0.389        | 0.763 | 0.818 | 0.95 | 0.19 | 0.020     | 0.050                 | 0.149 | 0.154 | 0.95 | 0.20 |
|          | BART-NP              | 0.251        | 0.427 | 0.495 | 0.91 | 0.43 | -0.042    | 0.110                 | 0.137 | 0.177 | 0.88 | 0.45 |
|          | BART-CP              | 0.194        | 0.330 | 0.383 | 0.93 | 0.56 | -0.048    | 0.037                 | 0.092 | 0.100 | 0.94 | 0.43 |

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Sc2

Web Table S2, continued

| Scenario           | Method               | Median ratio |       |       |      |      |           | RMST ratio, $t^* = 3$ |       |       |      |      |
|--------------------|----------------------|--------------|-------|-------|------|------|-----------|-----------------------|-------|-------|------|------|
|                    |                      | Bias         | SD    | RMSE  | Cov  | Pow  | Ctrl bias | Bias                  | SD    | RMSE  | Cov  | Pow  |
| Sc3, $\rho = -0.5$ | BART-PP              | 0.036        | 0.360 | 0.335 | 0.95 | 0.24 | 0.017     | -0.002                | 0.101 | 0.097 | 0.95 | 0.21 |
|                    | HierAFT              | 0.233        | 0.679 | 0.695 | 0.97 | 0.14 | 0.069     | 0.001                 | 0.136 | 0.134 | 0.95 | 0.12 |
|                    | C-BART               | 0.138        | 0.300 | 0.331 | 0.93 | 0.55 | -0.041    | 0.043                 | 0.090 | 0.104 | 0.92 | 0.49 |
|                    | LRC-BART ( $w = 0$ ) | 0.028        | 0.328 | 0.332 | 0.93 | 0.29 | 0.009     | 0.004                 | 0.096 | 0.099 | 0.94 | 0.24 |
|                    | LRC-BART (100)       | 0.061        | 0.322 | 0.322 | 0.94 | 0.34 | -0.007    | 0.018                 | 0.096 | 0.100 | 0.93 | 0.31 |
|                    | LRC-BART (75)        | 0.065        | 0.322 | 0.323 | 0.94 | 0.34 | -0.008    | 0.019                 | 0.096 | 0.100 | 0.93 | 0.32 |
|                    | LRC-BART (50)        | 0.061        | 0.322 | 0.322 | 0.94 | 0.34 | -0.007    | 0.018                 | 0.095 | 0.100 | 0.94 | 0.31 |
|                    | LRC-BART (0.9)       | 0.058        | 0.324 | 0.325 | 0.94 | 0.33 | -0.006    | 0.017                 | 0.096 | 0.100 | 0.94 | 0.30 |
|                    | LRC-BART (0.5)       | 0.051        | 0.326 | 0.322 | 0.95 | 0.31 | -0.002    | 0.015                 | 0.096 | 0.099 | 0.94 | 0.29 |
|                    | LRC-BART (0.25)      | 0.042        | 0.327 | 0.327 | 0.94 | 0.31 | 0.002     | 0.011                 | 0.096 | 0.100 | 0.94 | 0.28 |
|                    | AFT-NP               | 0.222        | 0.757 | 0.806 | 0.95 | 0.10 | 0.126     | -0.025                | 0.140 | 0.142 | 0.93 | 0.08 |
|                    | AFT-CP               | 3.017        | 1.311 | 3.271 | 0.03 | 1.00 | -0.320    | 0.483                 | 0.159 | 0.507 | 0.04 | 1.00 |
|                    | AFT-PP               | 0.438        | 0.802 | 0.909 | 0.95 | 0.18 | 0.009     | 0.054                 | 0.153 | 0.157 | 0.95 | 0.19 |
|                    | BART-NP              | 0.188        | 0.439 | 0.555 | 0.91 | 0.32 | -0.041    | 0.064                 | 0.150 | 0.172 | 0.92 | 0.24 |
|                    | BART-CP              | 1.953        | 0.596 | 2.063 | 0.00 | 1.00 | -0.302    | 0.499                 | 0.145 | 0.523 | 0.02 | 1.00 |
| Sc3, $\rho = 0$    | BART-PP              | 0.047        | 0.387 | 0.441 | 0.92 | 0.23 | -0.007    | -0.000                | 0.124 | 0.134 | 0.92 | 0.15 |
|                    | HierAFT              | 0.321        | 0.778 | 0.857 | 0.95 | 0.12 | 0.070     | 0.005                 | 0.146 | 0.143 | 0.95 | 0.10 |
|                    | C-BART               | 0.627        | 0.668 | 0.948 | 0.84 | 0.45 | -0.094    | 0.156                 | 0.158 | 0.236 | 0.81 | 0.43 |
|                    | LRC-BART ( $w = 0$ ) | 0.094        | 0.477 | 0.494 | 0.95 | 0.17 | 0.034     | 0.015                 | 0.131 | 0.135 | 0.94 | 0.16 |
|                    | LRC-BART (100)       | 0.147        | 0.502 | 0.539 | 0.95 | 0.18 | 0.018     | 0.030                 | 0.136 | 0.144 | 0.94 | 0.19 |
|                    | LRC-BART (75)        | 0.147        | 0.501 | 0.531 | 0.94 | 0.19 | 0.017     | 0.031                 | 0.136 | 0.142 | 0.94 | 0.19 |
|                    | LRC-BART (50)        | 0.150        | 0.499 | 0.534 | 0.95 | 0.19 | 0.016     | 0.032                 | 0.135 | 0.143 | 0.94 | 0.19 |
|                    | LRC-BART (0.9)       | 0.145        | 0.501 | 0.539 | 0.94 | 0.19 | 0.018     | 0.030                 | 0.135 | 0.143 | 0.94 | 0.20 |
|                    | LRC-BART (0.5)       | 0.147        | 0.501 | 0.540 | 0.95 | 0.17 | 0.018     | 0.031                 | 0.135 | 0.144 | 0.94 | 0.19 |
|                    | LRC-BART (0.25)      | 0.151        | 0.501 | 0.539 | 0.95 | 0.19 | 0.016     | 0.032                 | 0.136 | 0.144 | 0.94 | 0.19 |
|                    | AFT-NP               | 0.231        | 0.771 | 0.832 | 0.94 | 0.10 | 0.128     | -0.025                | 0.141 | 0.145 | 0.92 | 0.09 |
|                    | AFT-CP               | 3.527        | 1.395 | 3.769 | 0.01 | 1.00 | -0.342    | 0.553                 | 0.158 | 0.573 | 0.01 | 1.00 |
|                    | AFT-PP               | 0.440        | 0.806 | 0.919 | 0.94 | 0.19 | 0.008     | 0.054                 | 0.153 | 0.158 | 0.95 | 0.19 |
|                    | BART-NP              | 0.186        | 0.443 | 0.571 | 0.90 | 0.30 | -0.041    | 0.058                 | 0.152 | 0.173 | 0.92 | 0.21 |
|                    | BART-CP              | 2.308        | 0.631 | 2.412 | 0.00 | 1.00 | -0.323    | 0.588                 | 0.148 | 0.607 | 0.01 | 1.00 |
| Sc3, $\rho = 0.5$  | BART-PP              | 0.041        | 0.387 | 0.451 | 0.91 | 0.20 | -0.006    | -0.005                | 0.124 | 0.137 | 0.92 | 0.15 |
|                    | HierAFT              | 0.337        | 0.795 | 0.890 | 0.95 | 0.13 | 0.070     | 0.006                 | 0.147 | 0.147 | 0.95 | 0.11 |
|                    | C-BART               | 0.576        | 0.670 | 0.909 | 0.88 | 0.40 | -0.081    | 0.139                 | 0.159 | 0.222 | 0.85 | 0.38 |
|                    | LRC-BART ( $w = 0$ ) | 0.096        | 0.489 | 0.510 | 0.95 | 0.15 | 0.038     | 0.014                 | 0.133 | 0.137 | 0.94 | 0.17 |
|                    | LRC-BART (100)       | 0.155        | 0.516 | 0.558 | 0.94 | 0.17 | 0.020     | 0.031                 | 0.138 | 0.146 | 0.94 | 0.18 |
|                    | LRC-BART (75)        | 0.156        | 0.516 | 0.553 | 0.94 | 0.17 | 0.019     | 0.031                 | 0.138 | 0.145 | 0.93 | 0.18 |
|                    | LRC-BART (50)        | 0.156        | 0.514 | 0.554 | 0.94 | 0.18 | 0.019     | 0.031                 | 0.137 | 0.145 | 0.93 | 0.19 |
|                    | LRC-BART (0.9)       | 0.148        | 0.511 | 0.554 | 0.94 | 0.17 | 0.021     | 0.029                 | 0.136 | 0.144 | 0.93 | 0.18 |
|                    | LRC-BART (0.5)       | 0.151        | 0.515 | 0.552 | 0.95 | 0.18 | 0.020     | 0.030                 | 0.137 | 0.145 | 0.94 | 0.19 |
|                    | LRC-BART (0.25)      | 0.156        | 0.515 | 0.557 | 0.94 | 0.18 | 0.019     | 0.031                 | 0.138 | 0.146 | 0.93 | 0.18 |
|                    | AFT-NP               | 0.226        | 0.760 | 0.819 | 0.94 | 0.11 | 0.127     | -0.024                | 0.140 | 0.144 | 0.93 | 0.09 |
|                    | AFT-CP               | 3.038        | 1.318 | 3.285 | 0.03 | 1.00 | -0.321    | 0.486                 | 0.160 | 0.509 | 0.04 | 1.00 |
|                    | AFT-PP               | 0.441        | 0.803 | 0.914 | 0.95 | 0.18 | 0.009     | 0.054                 | 0.153 | 0.157 | 0.94 | 0.19 |
|                    | BART-NP              | 0.195        | 0.440 | 0.574 | 0.90 | 0.31 | -0.043    | 0.066                 | 0.151 | 0.176 | 0.92 | 0.23 |
|                    | BART-CP              | 1.957        | 0.597 | 2.069 | 0.01 | 1.00 | -0.302    | 0.499                 | 0.145 | 0.522 | 0.02 | 1.00 |
|                    | BART-PP              | 0.046        | 0.387 | 0.445 | 0.91 | 0.21 | -0.007    | -0.000                | 0.124 | 0.134 | 0.92 | 0.15 |

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Sc3,  $\rho = 0.5$

Web Table S2, continued

| Scenario                   | Method               | Median ratio |       |       |      |      |           | RMST ratio, $t^* = 3$ |       |       |      |      |
|----------------------------|----------------------|--------------|-------|-------|------|------|-----------|-----------------------|-------|-------|------|------|
|                            |                      | Bias         | SD    | RMSE  | Cov  | Pow  | Ctrl bias | Bias                  | SD    | RMSE  | Cov  | Pow  |
| Sc4, $X_5, \delta_2 = 0.5$ | HierAFT              | 0.327        | 0.781 | 0.872 | 0.95 | 0.14 | 0.071     | 0.006                 | 0.146 | 0.146 | 0.94 | 0.12 |
|                            | C-BART               | 0.617        | 0.662 | 0.948 | 0.85 | 0.44 | -0.091    | 0.154                 | 0.158 | 0.236 | 0.82 | 0.42 |
|                            | LRC-BART ( $w = 0$ ) | 0.096        | 0.476 | 0.503 | 0.94 | 0.16 | 0.033     | 0.016                 | 0.131 | 0.137 | 0.93 | 0.17 |
|                            | LRC-BART (100)       | 0.152        | 0.501 | 0.552 | 0.93 | 0.19 | 0.016     | 0.032                 | 0.136 | 0.146 | 0.92 | 0.20 |
|                            | LRC-BART (75)        | 0.153        | 0.502 | 0.551 | 0.94 | 0.19 | 0.016     | 0.033                 | 0.136 | 0.146 | 0.93 | 0.19 |
|                            | LRC-BART (50)        | 0.156        | 0.502 | 0.545 | 0.94 | 0.20 | 0.014     | 0.034                 | 0.136 | 0.144 | 0.93 | 0.20 |
|                            | LRC-BART (0.9)       | 0.146        | 0.501 | 0.549 | 0.94 | 0.18 | 0.018     | 0.031                 | 0.136 | 0.146 | 0.94 | 0.19 |
|                            | LRC-BART (0.5)       | 0.150        | 0.502 | 0.552 | 0.93 | 0.18 | 0.017     | 0.031                 | 0.136 | 0.146 | 0.94 | 0.19 |
|                            | LRC-BART (0.25)      | 0.156        | 0.501 | 0.548 | 0.94 | 0.20 | 0.014     | 0.034                 | 0.136 | 0.146 | 0.93 | 0.21 |
|                            | AFT-NP               | 0.233        | 0.713 | 0.737 | 0.95 | 0.12 | 0.072     | -0.020                | 0.147 | 0.152 | 0.92 | 0.09 |
|                            | AFT-CP               | -0.123       | 0.351 | 0.347 | 0.92 | 0.09 | 0.083     | -0.083                | 0.086 | 0.116 | 0.85 | 0.06 |
|                            | AFT-PP               | 0.228        | 0.714 | 0.712 | 0.95 | 0.10 | 0.062     | -0.023                | 0.141 | 0.139 | 0.94 | 0.08 |
|                            | BART-NP              | 0.260        | 0.417 | 0.511 | 0.92 | 0.46 | -0.036    | 0.131                 | 0.151 | 0.206 | 0.87 | 0.47 |
|                            | BART-CP              | -0.168       | 0.205 | 0.252 | 0.86 | 0.17 | 0.053     | -0.088                | 0.068 | 0.109 | 0.74 | 0.06 |
|                            | BART-PP              | 0.011        | 0.354 | 0.339 | 0.96 | 0.21 | 0.023     | -0.040                | 0.102 | 0.107 | 0.94 | 0.12 |
| Sc4, $X_5, \delta_2 = 1$   | HierAFT              | 0.219        | 0.690 | 0.713 | 0.95 | 0.11 | 0.068     | -0.015                | 0.147 | 0.150 | 0.92 | 0.10 |
|                            | C-BART               | -0.094       | 0.253 | 0.247 | 0.94 | 0.21 | 0.033     | -0.048                | 0.081 | 0.089 | 0.92 | 0.13 |
|                            | LRC-BART ( $w = 0$ ) | 0.047        | 0.329 | 0.354 | 0.94 | 0.32 | 0.007     | -0.011                | 0.099 | 0.107 | 0.93 | 0.24 |
|                            | LRC-BART (100)       | 0.001        | 0.308 | 0.317 | 0.95 | 0.29 | 0.014     | -0.022                | 0.094 | 0.100 | 0.93 | 0.19 |
|                            | LRC-BART (75)        | 0.005        | 0.308 | 0.317 | 0.95 | 0.29 | 0.014     | -0.021                | 0.094 | 0.100 | 0.93 | 0.20 |
|                            | LRC-BART (50)        | 0.014        | 0.312 | 0.322 | 0.95 | 0.30 | 0.011     | -0.018                | 0.095 | 0.099 | 0.93 | 0.22 |
|                            | LRC-BART (0.9)       | -0.004       | 0.305 | 0.313 | 0.95 | 0.28 | 0.015     | -0.023                | 0.093 | 0.099 | 0.92 | 0.19 |
|                            | LRC-BART (0.5)       | 0.004        | 0.307 | 0.315 | 0.95 | 0.28 | 0.013     | -0.021                | 0.094 | 0.099 | 0.93 | 0.20 |
|                            | LRC-BART (0.25)      | 0.015        | 0.312 | 0.324 | 0.95 | 0.30 | 0.011     | -0.018                | 0.095 | 0.100 | 0.93 | 0.20 |
|                            | AFT-NP               | 0.233        | 0.713 | 0.737 | 0.95 | 0.12 | 0.072     | -0.020                | 0.147 | 0.152 | 0.92 | 0.09 |
|                            | AFT-CP               | -0.357       | 0.289 | 0.446 | 0.75 | 0.05 | 0.193     | -0.144                | 0.079 | 0.163 | 0.57 | 0.05 |
|                            | AFT-PP               | 0.191        | 0.703 | 0.691 | 0.95 | 0.10 | 0.075     | -0.039                | 0.137 | 0.139 | 0.92 | 0.07 |
|                            | BART-NP              | 0.260        | 0.417 | 0.511 | 0.92 | 0.46 | -0.036    | 0.131                 | 0.151 | 0.206 | 0.87 | 0.47 |
|                            | BART-CP              | -0.367       | 0.176 | 0.401 | 0.50 | 0.04 | 0.142     | -0.156                | 0.062 | 0.168 | 0.32 | 0.07 |
|                            | BART-PP              | -0.012       | 0.361 | 0.339 | 0.97 | 0.17 | 0.033     | -0.059                | 0.101 | 0.113 | 0.91 | 0.07 |
| Sc5, $\delta_2 = 1$        | HierAFT              | 0.193        | 0.681 | 0.700 | 0.94 | 0.10 | 0.078     | -0.023                | 0.145 | 0.149 | 0.91 | 0.09 |
|                            | C-BART               | -0.202       | 0.251 | 0.305 | 0.84 | 0.08 | 0.076     | -0.093                | 0.080 | 0.120 | 0.79 | 0.04 |
|                            | LRC-BART ( $w = 0$ ) | 0.040        | 0.331 | 0.353 | 0.94 | 0.31 | 0.010     | -0.023                | 0.097 | 0.107 | 0.92 | 0.19 |
|                            | LRC-BART (100)       | -0.011       | 0.321 | 0.340 | 0.93 | 0.24 | 0.024     | -0.037                | 0.096 | 0.108 | 0.90 | 0.16 |
|                            | LRC-BART (75)        | -0.007       | 0.322 | 0.337 | 0.93 | 0.24 | 0.022     | -0.036                | 0.096 | 0.107 | 0.90 | 0.17 |
|                            | LRC-BART (50)        | -0.001       | 0.322 | 0.333 | 0.94 | 0.24 | 0.020     | -0.034                | 0.096 | 0.106 | 0.91 | 0.17 |
|                            | LRC-BART (0.9)       | -0.018       | 0.321 | 0.342 | 0.93 | 0.24 | 0.027     | -0.040                | 0.096 | 0.110 | 0.90 | 0.16 |
|                            | LRC-BART (0.5)       | -0.013       | 0.322 | 0.338 | 0.94 | 0.24 | 0.024     | -0.038                | 0.096 | 0.108 | 0.91 | 0.17 |
|                            | LRC-BART (0.25)      | 0.001        | 0.321 | 0.338 | 0.93 | 0.26 | 0.020     | -0.034                | 0.095 | 0.106 | 0.91 | 0.17 |
|                            | AFT-NP               | 0.233        | 0.713 | 0.737 | 0.95 | 0.12 | 0.072     | -0.020                | 0.147 | 0.152 | 0.92 | 0.09 |
|                            | AFT-CP               | -0.720       | 0.191 | 0.741 | 0.18 | 0.39 | 0.510     | -0.244                | 0.069 | 0.253 | 0.09 | 0.40 |
|                            | AFT-PP               | 0.193        | 0.702 | 0.695 | 0.94 | 0.09 | 0.075     | -0.029                | 0.141 | 0.141 | 0.93 | 0.08 |
|                            | BART-NP              | 0.260        | 0.417 | 0.511 | 0.92 | 0.46 | -0.036    | 0.131                 | 0.151 | 0.206 | 0.87 | 0.47 |
|                            | BART-CP              | -0.703       | 0.119 | 0.712 | 0.00 | 0.64 | 0.402     | -0.256                | 0.054 | 0.262 | 0.01 | 0.65 |
|                            | BART-PP              | -0.029       | 0.352 | 0.333 | 0.95 | 0.17 | 0.037     | -0.041                | 0.107 | 0.110 | 0.93 | 0.11 |
|                            | HierAFT              | 0.204        | 0.683 | 0.701 | 0.94 | 0.10 | 0.073     | -0.018                | 0.146 | 0.150 | 0.92 | 0.09 |

Sc5,  $\delta_2 = 1$ 

continued on the next page

Web Table S2, continued

| Scenario | Method               | Median ratio |       |       |      |      |           | RMST ratio, $t^* = 3$ |       |       |      |      |
|----------|----------------------|--------------|-------|-------|------|------|-----------|-----------------------|-------|-------|------|------|
|          |                      | Bias         | SD    | RMSE  | Cov  | Pow  | Ctrl bias | Bias                  | SD    | RMSE  | Cov  | Pow  |
|          | C-BART               | -0.233       | 0.288 | 0.405 | 0.77 | 0.13 | 0.120     | -0.085                | 0.099 | 0.143 | 0.77 | 0.11 |
|          | LRC-BART ( $w = 0$ ) | 0.053        | 0.329 | 0.353 | 0.94 | 0.32 | 0.005     | 0.006                 | 0.103 | 0.110 | 0.93 | 0.26 |
|          | LRC-BART (100)       | 0.032        | 0.328 | 0.353 | 0.93 | 0.30 | 0.013     | 0.001                 | 0.104 | 0.112 | 0.93 | 0.26 |
|          | LRC-BART (75)        | 0.030        | 0.328 | 0.348 | 0.94 | 0.29 | 0.013     | 0.000                 | 0.104 | 0.111 | 0.93 | 0.25 |
|          | LRC-BART (50)        | 0.026        | 0.326 | 0.350 | 0.94 | 0.30 | 0.014     | -0.001                | 0.104 | 0.111 | 0.93 | 0.24 |
|          | LRC-BART (0.9)       | 0.029        | 0.327 | 0.350 | 0.94 | 0.31 | 0.014     | 0.001                 | 0.104 | 0.111 | 0.94 | 0.26 |
|          | LRC-BART (0.5)       | 0.028        | 0.326 | 0.349 | 0.94 | 0.30 | 0.013     | 0.000                 | 0.104 | 0.111 | 0.93 | 0.25 |
|          | LRC-BART (0.25)      | 0.028        | 0.325 | 0.351 | 0.94 | 0.30 | 0.013     | 0.000                 | 0.103 | 0.112 | 0.93 | 0.25 |

## G.5 Application

We harmonize the covariate labels before constructing the model matrix. The non-Hispanic category is labelled differently in the two sources, so the original coding creates two non-Hispanic dummy columns. For the 29 non-Hispanic trial patients, the registry non-Hispanic indicator takes the value used for the registry’s 13 Hispanic patients, while a trial-only indicator has no variation in registry controls. Predictions then depend on extrapolation and prior regularization in each model as well as the other covariates. Race was recorded with four levels and Asian as reference although the cohort table pools Asian into Other/Unknown. Harmonization gives seven columns: age, male, Black, Other/Unknown, Hispanic, high-risk cytogenetics and ASCT after induction. The original coding remains a sensitivity analysis. Its ESS ceilings are 12.6 for PFS and 14.6 for OS, compared with 63.7 and 36.7 under harmonization; a target of 30 is capped under the original coding.

We construct the registry cohort from 797 patients by excluding 7 with negative survival times, those without a recorded induction start, 49 transplanted on or before induction, and 1 on elotuzumab with lenalidomide and dexamethasone. Under the original covariate coding the reference analyses of the earlier version give the following comparisons under the same scripts (posterior medians: complete-pooling AFT 1.03 and 0.90, hierarchical AFT 1.24 and 1.25, AFT-BART 1.06 and 0.91 for PFS and OS, against 1.03, 0.90, 1.21, 1.18, 1.06 and 0.90), which permits a coding sensitivity comparison of the reference rows in Table 5; the MAP-AFT-BART rows cannot be rerun because their sampler is not in the repository.

Table S3: The ESS ladder, Gaussian outcomes,  $R = 500$ : absolute targets  $N_{\text{target}} \in \{50, 75, 100\}$  and fractional targets 0.25, 0.5, 0.9 of the ceiling  $\sigma_1^2/V_\mu^f$ . Prior ESS is  $\text{ESS}_0$  at the calibrated scale, realized ESS its posterior counterpart, ceiling the mean ceiling, and capped the fraction of replicates in which the target was at or above the ceiling and the calibration returned the point-mass limit. Prior ESS is a block average and can exceed the whole-chain ceiling by a few percent when capped (Section 3.1).

| Scenario                 | Target          | Bias   | RMSE  | Cov  | Prior ESS | Realized ESS | Ceiling | Capped |
|--------------------------|-----------------|--------|-------|------|-----------|--------------|---------|--------|
| Sc1                      | LRC-BART (50)   | 0.009  | 0.181 | 0.94 | 42.7      | 28.5         | 128     | 0.00   |
|                          | LRC-BART (75)   | 0.008  | 0.176 | 0.94 | 62.1      | 40.6         | 127     | 0.04   |
|                          | LRC-BART (100)  | 0.009  | 0.172 | 0.94 | 82.1      | 52.5         | 127     | 0.20   |
|                          | LRC-BART (0.25) | 0.008  | 0.184 | 0.94 | 28.0      | 18.7         | 127     | 0.00   |
|                          | LRC-BART (0.5)  | 0.009  | 0.178 | 0.93 | 53.6      | 34.8         | 128     | 0.00   |
|                          | LRC-BART (0.9)  | 0.011  | 0.171 | 0.95 | 90.0      | 55.7         | 127     | 0.00   |
| Sc2                      | LRC-BART (50)   | 0.023  | 0.193 | 0.93 | 24.5      | 13.5         | 23      | 1.00   |
|                          | LRC-BART (75)   | 0.022  | 0.192 | 0.93 | 24.5      | 13.5         | 23      | 1.00   |
|                          | LRC-BART (100)  | 0.023  | 0.192 | 0.94 | 24.5      | 13.5         | 23      | 1.00   |
|                          | LRC-BART (0.25) | 0.012  | 0.200 | 0.94 | 5.1       | 3.3          | 23      | 0.00   |
|                          | LRC-BART (0.5)  | 0.016  | 0.197 | 0.93 | 9.7       | 6.0          | 23      | 0.00   |
|                          | LRC-BART (0.9)  | 0.019  | 0.193 | 0.93 | 16.5      | 9.6          | 23      | 0.00   |
| Sc3, $\rho = 0$          | LRC-BART (50)   | 0.045  | 0.282 | 0.94 | 44.2      | 1.9          | 224     | 0.00   |
|                          | LRC-BART (75)   | 0.042  | 0.281 | 0.94 | 64.7      | 1.7          | 224     | 0.00   |
|                          | LRC-BART (100)  | 0.041  | 0.280 | 0.94 | 84.5      | 1.6          | 224     | 0.00   |
|                          | LRC-BART (0.25) | 0.045  | 0.279 | 0.94 | 49.4      | 1.9          | 224     | 0.00   |
|                          | LRC-BART (0.5)  | 0.041  | 0.280 | 0.94 | 94.1      | 1.6          | 224     | 0.00   |
|                          | LRC-BART (0.9)  | 0.037  | 0.277 | 0.94 | 160.7     | 1.3          | 224     | 0.00   |
| Sc4, $X_5, \delta_2 = 1$ | LRC-BART (50)   | -0.036 | 0.213 | 0.90 | 42.5      | 13.1         | 126     | 0.00   |
|                          | LRC-BART (75)   | -0.041 | 0.221 | 0.90 | 62.0      | 17.1         | 125     | 0.05   |
|                          | LRC-BART (100)  | -0.041 | 0.223 | 0.89 | 82.0      | 20.8         | 125     | 0.22   |
|                          | LRC-BART (0.25) | -0.032 | 0.208 | 0.92 | 27.6      | 9.7          | 126     | 0.00   |
|                          | LRC-BART (0.5)  | -0.037 | 0.218 | 0.90 | 52.5      | 14.8         | 125     | 0.00   |
|                          | LRC-BART (0.9)  | -0.041 | 0.224 | 0.90 | 89.4      | 21.1         | 126     | 0.00   |
| Sc4, $X_5, \delta_2 = 2$ | LRC-BART (50)   | -0.002 | 0.201 | 0.92 | 42.4      | 0.8          | 125     | 0.00   |
|                          | LRC-BART (75)   | 0.001  | 0.201 | 0.93 | 61.8      | 0.8          | 125     | 0.04   |
|                          | LRC-BART (100)  | 0.000  | 0.202 | 0.92 | 83.1      | 0.8          | 124     | 0.27   |
|                          | LRC-BART (0.25) | -0.001 | 0.204 | 0.93 | 27.4      | 0.8          | 125     | 0.00   |
|                          | LRC-BART (0.5)  | -0.001 | 0.202 | 0.92 | 52.2      | 0.8          | 125     | 0.00   |
|                          | LRC-BART (0.9)  | -0.000 | 0.202 | 0.93 | 88.5      | 0.8          | 124     | 0.00   |
| Sc5, $\delta_2 = 2$      | LRC-BART (50)   | -0.001 | 0.203 | 0.93 | 42.7      | 0.3          | 128     | 0.00   |
|                          | LRC-BART (75)   | -0.001 | 0.205 | 0.93 | 62.1      | 0.3          | 127     | 0.04   |
|                          | LRC-BART (100)  | -0.001 | 0.203 | 0.92 | 82.1      | 0.2          | 127     | 0.20   |
|                          | LRC-BART (0.25) | 0.000  | 0.203 | 0.92 | 28.0      | 0.3          | 127     | 0.00   |
|                          | LRC-BART (0.5)  | -0.001 | 0.204 | 0.93 | 53.6      | 0.3          | 128     | 0.00   |
|                          | LRC-BART (0.9)  | -0.001 | 0.205 | 0.92 | 90.0      | 0.3          | 127     | 0.00   |

Table S4: Regional and global shifts, survival outcomes (median ratio),  $R = 500$ : columns as in Table 2, with the surface RMSE, the map threshold  $c = 0.5$  and  $g$  on the log-time scale.

| Scenario  | Method               | ATE    |      | Surface RMSE  |                 | $\bar{B}_c$   |                 | Mean $\hat{g}$ |                 | ESS  |
|-----------|----------------------|--------|------|---------------|-----------------|---------------|-----------------|----------------|-----------------|------|
|           |                      | Bias   | Cov  | $\mathcal{R}$ | $\mathcal{R}^c$ | $\mathcal{R}$ | $\mathcal{R}^c$ | $\mathcal{R}$  | $\mathcal{R}^c$ |      |
| Sc4 (0.5) | BART-PP              | 0.011  | 0.96 | 0.783         | 0.838           | —             | —               | —              | —               | —    |
|           | C-BART               | −0.094 | 0.94 | 0.747         | 0.805           | 0.96          | 0.96            | −0.08          | −0.08           | 44.6 |
|           | LRC-BART ( $w = 0$ ) | 0.047  | 0.94 | 0.784         | 0.820           | 0.65          | 0.64            | −0.18          | −0.24           | 0.2  |
|           | LRC-BART (100)       | 0.001  | 0.95 | 0.764         | 0.809           | 0.79          | 0.78            | −0.15          | −0.18           | 19.2 |
| Sc4 (1)   | BART-PP              | −0.012 | 0.97 | 0.803         | 0.888           | —             | —               | —              | —               | —    |
|           | C-BART               | −0.202 | 0.84 | 0.762         | 0.920           | 0.91          | 0.90            | −0.16          | −0.17           | 44.5 |
|           | LRC-BART ( $w = 0$ ) | 0.040  | 0.94 | 0.814         | 0.861           | 0.52          | 0.46            | −0.33          | −0.53           | 0.2  |
|           | LRC-BART (100)       | −0.011 | 0.93 | 0.792         | 0.868           | 0.64          | 0.58            | −0.30          | −0.44           | 12.8 |
| Sc5 (1)   | BART-PP              | −0.029 | 0.95 | —             | 0.827           | —             | —               | —              | —               | —    |
|           | C-BART               | −0.233 | 0.77 | —             | 0.843           | —             | 0.43            | —              | −0.62           | 25.9 |
|           | LRC-BART ( $w = 0$ ) | 0.053  | 0.94 | —             | 0.810           | —             | 0.15            | —              | −0.96           | 0.3  |
|           | LRC-BART (100)       | 0.032  | 0.93 | —             | 0.807           | —             | 0.16            | —              | −0.94           | 2.0  |

Table S5: Characteristics of the merged analytic cohort: median (range) for age,  $n$  (%) otherwise. Survival is measured from the first day of treatment (EloKRd) or the start of induction (UCMM); RMST at 5 years is the Kaplan–Meier estimate.

| Characteristic         | EloKRd ( $n = 30$ ) | UCMM ( $n = 253$ ) |
|------------------------|---------------------|--------------------|
| Age, years             | 62.8 (43–81)        | 61.3 (32–97)       |
| Male                   | 21 (70%)            | 130 (51%)          |
| Race: White            | 23 (77%)            | 164 (65%)          |
| Race: Black            | 4 (13%)             | 76 (30%)           |
| Race: Other/Unknown    | 3 (10%)             | 13 (5%)            |
| Hispanic or Latino     | 1 (3%)              | 13 (5%)            |
| High-risk cytogenetics | 15 (50%)            | 58 (23%)           |
| ASCT after induction   | 9 (30%)             | 203 (80%)          |
| PFS events             | 13 (43%)            | 159 (63%)          |
| Median PFS, years      | not reached         | 5.09               |
| PFS RMST at 5 years    | 3.71                | 3.65               |
| OS events              | 10 (33%)            | 60 (24%)           |
| Median OS, years       | not reached         | not reached        |
| OS RMST at 5 years     | 4.12                | 4.57               |

The posterior of the RMST ratio under the three absolute targets (Web Figure S2) is unimodal with a largest observed draw of 2.0 for OS and one draw reaching 3.1 among 8,000 for PFS.

The lower block of Table 5 reports the earlier MAP-AFT-BART analysis, whose sampler is unavailable for a controlled comparison. Its 97.5% limits were 16 to 33 for PFS and 4.8 to 7.2 for OS. The corresponding limits under the calibrated discrepancy model are 1.23 to 1.40 and 1.04 to 1.11. Over 8,000 draws at  $N_{\text{target}} = 30$ , the PFS ratio has a 99.9th percentile of 1.6 and a maximum of 3.1. The conditional discrepancy variance is  $H_g \tau_0^2$ , with fitted mean about 0.07 on the log-time scale. These fitted distributions have shorter observed tails; the comparison does not isolate the effects of prior, model, coding and implementation. Across the absolute target ladder, point estimates are 0.97 to 0.99 for PFS and 0.90 to 0.92 for OS, compared with 1.08 to 1.10 and 0.97 to 1.00 in the earlier analysis. The original-coding sensitivity gives 1.04 for PFS. Within the current model this demonstrates a coding effect, without attributing the remaining cross-model difference to one cause.

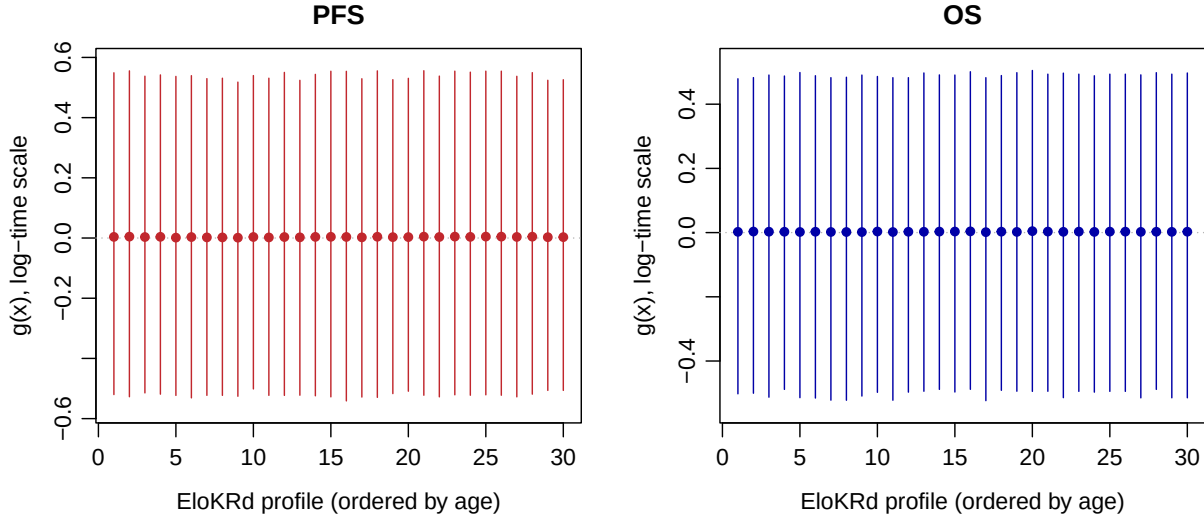


Figure S1: Posterior mean and 95% interval of the discrepancy  $g(x)$  on the log-time scale at the 30 EloKRd profiles, ordered by age, at  $N_{\text{target}} = 30$  for PFS (red) and OS (blue). In single-arm mode  $g$  is a prior draw, so the intervals are centred at zero with SD 0.26 everywhere; discrepancy uncertainty is therefore the same at every profile. Variation in external support is summarized by the local ESS map in Figure 2.

## G.6 A conditional prior-cost comparison

Consider a profile in a fixed region represented by one reached leaf in each of  $H$  trees. Hold the partitions, weight  $0 < w < 1$  and variances  $0 < \tau^2 < \lambda^2$  fixed. For a specified configuration with

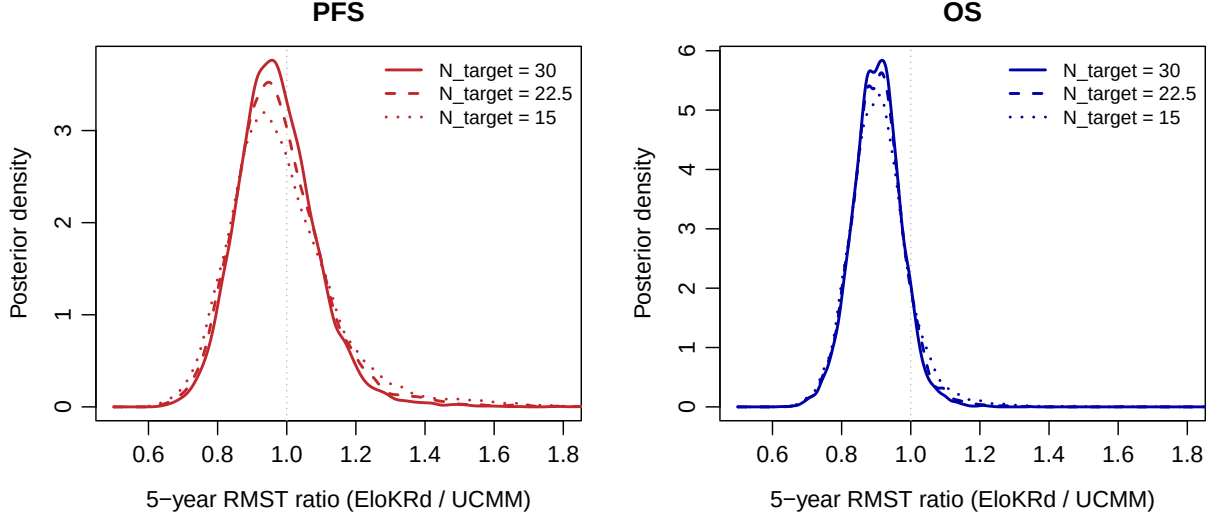


Figure S2: Posterior density of the standardized 5-year RMST ratio for PFS and OS under LRC-BART at  $N_{\text{target}} = 30, 22.5$  and  $15$  (harmonized coding,  $H_f = 10$ ,  $w = 1$ ).

$m$  slab leaves, let  $u_h$  denote the  $H$  discrepancy contributions and require  $\sum_h u_h = \delta$ . Relative to the all-spike density normalizers, each slab adds  $c = \log\{w/(1-w)\} + \log(\lambda/\tau)$  to the negative log joint prior. The constrained minimum is

$$Q_m(\delta) = mc + \min_{\sum_h u_h = \delta} \sum_h \frac{u_h^2}{2v_h} = mc + \frac{\delta^2}{2S_m}, \quad S_m = (H-m)\tau^2 + m\lambda^2, \quad m \in \{0, \dots, H\}.$$

Here  $v_h$  is the variance of contribution  $h$ . A Lagrange multiplier gives  $u_h = \delta v_h / S_m$ , so the remaining spike contributions participate in the optimum. If  $c > 0$ , a specified configuration with  $m \geq 1$  has a smaller optimized cost than the all-spike configuration exactly when

$$\delta^2 > \frac{2mc}{(H\tau^2)^{-1} - S_m^{-1}}.$$

The allowed values of  $m$  are integers; minimizing over an unconstrained continuous value can give an infeasible allocation. If  $c \leq 0$ , the normalizing and weight penalty no longer favors the all-spike configuration even at zero discrepancy.

This calculation compares joint prior densities at constrained modes. Posterior indicator probabilities integrate over leaf parameters, sum over configurations and account for the likelihood and partition probabilities. They therefore do not follow from the ordering of  $Q_m$ . Nor does the

calculation imply that a learned regional shift is assigned to one or two leaves. In a Gaussian common-mean comparison with regional sample sizes  $n_1, n_2$  and a difference  $\delta$  between source means, optimizing the pooled mean gives a negative log-likelihood loss of  $\delta^2/[2(\sigma_1^2/n_1 + \sigma_2^2/n_2)]$ ; its information is limited by the less precise source.

The separate ensemble in (1) permits different partition and shrinkage priors for the external surface and its discrepancy. With fixed per-leaf spike variance, fewer discrepancy trees reduce the total spike variance at a profile. The slab range rule and the partitions also change with the model configuration, so this observation alone does not establish superior detection. The simulations assess the specified  $H_g = 5$  model, including the remaining regional surface loss.

## G.7 The single-arm simulation in detail

We provide further details of the single-arm study in Section 3.4. The trial contributes  $n_T$  treated subjects only, 200 for Gaussian outcomes and 30 or 200 for survival outcomes, the 300 RWD controls are the sole control source, and the estimands are standardized to the trial profiles. The comparators reduce to the complete-pooling models, the hierarchical models, BART-CP, and LRC-BART in the single-arm mode of Section 2.6 with  $w = 1$ , at the absolute target 100, at 0.9 of the ceiling, and with the spike scale at the grid minimum, which is near-complete pooling of the leaf means;  $w = 0.9$  is the slab sensitivity. Sc3 is not identifiable without an internal control and is omitted. Table 4 and Web Table S6 report the results.

Under Sc1, BART-CP, the pooled LRC-BART and the target-100 LRC-BART have Gaussian RMSE 0.16 with coverage 0.94 to 1.00, and at  $n_T = 30$  the survival RMST ratio has RMSE 0.16 to 0.19 with coverage 0.92 to 0.97.

Under Sc2, all methods have substantial bias. With the source assignment of Sc2 and no RCT controls, the RWD is the only information about the control surface at trial profiles that lie where RWD support is limited, and every method is far more biased than in the two-arm study: on the Gaussian ATE, LM-CP has bias 1.68, BART-CP 0.41 with coverage 0.75, and LRC-BART 0.42, and the survival median ratio at  $n_T = 30$  has biases of 0.8 to 1.3 for the tree models. Single-arm borrowing under measured confounding is a harder problem than the two-arm study, because no internal controls are available to reveal a discrepancy, and limited RWD overlap increases extrapolation; the rows reported for this scenario in the earlier version came from a different mechanism (Web

Appendix G.2).

For LRC-BART, uncertainty depends on the target relative to the ceiling. The single-arm ceiling is 27 to 34 in Sc2 and about 75 at  $n_T = 30$  in Sc1, so an absolute target of 100 is usually capped there. With  $w = 1$ , the resulting small discrepancy scale approximates pooling. In Sc1 at  $n_T = 200$  the ceiling is about 160, and target 100 allows appreciable discrepancy uncertainty. The prior mean of a Gaussian control outcome or control log time remains centered on the external surface. Nonlinear survival-ratio means need not remain unchanged: under Sc1 with  $n_T = 200$ , the median-ratio bias is 0.076 for the pooled LRC-BART and 0.108 for target 100. The fractional target at 0.9 of the ceiling widens intervals, with coverage 0.82 to 0.99 across the reported designs (Web Table S6).

The  $w = 0.9$  sensitivity has posterior SD 1.4 for the Gaussian ATE and 10 to 34 for the median ratio, with power 0 and coverage 1. Conditional on  $w$  and the scales, the slab contributes  $(1-w)H_g\tau_1^2$  to the pointwise prior variance of the counterfactual log-time mean. It can also change nonlinear ratio means substantially: under Sc1, the median-ratio bias at  $n_T = 200$  is 1.492 with  $w = 0.9$ , compared with 0.108 with  $w = 1$ . We therefore report  $w = 1$  for the primary single-arm analysis. HierLM and HierAFT with the stated hyperprior and no RCT controls draw the control surface from the hierarchy; their posterior SDs are 8 for the Gaussian ATE and extremely large for the median ratio (Web Appendix G.2).

Table S6: Single-arm design, all methods,  $R = 500$ . Columns as in Table 4; LRC-BART (100) and LRC-BART ( $w = 0.9$ ) use the absolute target 100, capped at the ceiling where infeasible, and LRC-BART (0.9,  $w = 0.9$ ) the fractional target with the slab.

| Design                              | Method                     | Treatment effect |       |       |      |      | Control |       |       |     |
|-------------------------------------|----------------------------|------------------|-------|-------|------|------|---------|-------|-------|-----|
|                                     |                            | Bias             | SD    | RMSE  | Cov  | Pow  | Bias    | RMSE  | Prior | ESS |
| Gaussian, $n_T = 200$<br>ATE<br>Sc1 | LM-CP                      | 0.007            | 0.259 | 0.254 | 0.95 | 0.97 | -0.003  | 0.233 | —     |     |
|                                     | HierLM                     | -0.014           | 8.239 | 0.497 | 1.00 | 0.00 | 0.018   | 0.487 | —     |     |
|                                     | BART-CP                    | 0.010            | 0.163 | 0.162 | 0.94 | 1.00 | -0.003  | 0.127 | —     |     |
|                                     | LRC-BART (pooled)          | 0.007            | 0.165 | 0.162 | 0.94 | 1.00 | 0.001   | 0.127 | 172.3 |     |
|                                     | LRC-BART (100)             | 0.006            | 0.252 | 0.161 | 1.00 | 1.00 | 0.001   | 0.127 | 81.9  |     |
|                                     | LRC-BART (0.9)             | 0.007            | 0.202 | 0.163 | 0.98 | 1.00 | 0.000   | 0.128 | 115.8 |     |
|                                     | LRC-BART ( $w = 0.9$ )     | 0.010            | 1.383 | 0.169 | 1.00 | 0.00 | -0.003  | 0.136 | 81.9  |     |
|                                     | LRC-BART (0.9, $w = 0.9$ ) | 0.006            | 1.372 | 0.169 | 1.00 | 0.00 | 0.001   | 0.133 | 116.4 |     |
| Gaussian, $n_T = 200$<br>ATE<br>Sc2 | LM-CP                      | 1.682            | 0.417 | 1.734 | 0.01 | 1.00 | -1.676  | 1.726 | —     |     |
|                                     | HierLM                     | 1.642            | 8.317 | 1.754 | 1.00 | 0.00 | -1.636  | 1.747 | —     |     |

*continued on the next page*

Web Table S6, continued

| Design                                       | Method                     | Treatment effect |          |          |      |      | Control  |          |           |
|----------------------------------------------|----------------------------|------------------|----------|----------|------|------|----------|----------|-----------|
|                                              |                            | Bias             | SD       | RMSE     | Cov  | Pow  | Bias     | RMSE     | Prior ESS |
|                                              | BART-CP                    | 0.408            | 0.310    | 0.506    | 0.75 | 1.00 | -0.399   | 0.492    | —         |
|                                              | LRC-BART (pooled)          | 0.420            | 0.316    | 0.515    | 0.75 | 1.00 | -0.412   | 0.501    | 34.8      |
|                                              | LRC-BART (100)             | 0.421            | 0.316    | 0.515    | 0.76 | 1.00 | -0.413   | 0.501    | 34.7      |
|                                              | LRC-BART (0.9)             | 0.417            | 0.416    | 0.511    | 0.89 | 0.98 | -0.409   | 0.498    | 23.1      |
|                                              | LRC-BART ( $w = 0.9$ )     | 0.416            | 1.329    | 0.511    | 1.00 | 0.00 | -0.408   | 0.498    | 34.8      |
|                                              | LRC-BART (0.9, $w = 0.9$ ) | 0.420            | 1.345    | 0.514    | 1.00 | 0.00 | -0.411   | 0.500    | 23.0      |
| Survival, $n_T = 30$<br>median ratio<br>Sc1  | AFT-CP                     | 1.540            | 4.678    | 4.444    | 0.92 | 0.15 | -0.006   | 0.204    | —         |
|                                              | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | $> 10^3$ | $> 10^3$ | —         |
|                                              | BART-CP                    | -0.013           | 0.580    | 0.482    | 0.97 | 0.05 | -0.026   | 0.103    | —         |
|                                              | LRC-BART (pooled)          | 0.182            | 0.918    | 0.686    | 0.98 | 0.02 | -0.017   | 0.102    | 80.2      |
|                                              | LRC-BART (100)             | 0.180            | 0.917    | 0.686    | 0.98 | 0.03 | -0.017   | 0.099    | 79.0      |
|                                              | LRC-BART (0.9)             | 0.222            | 1.174    | 0.711    | 0.99 | 0.01 | -0.005   | 0.133    | 51.8      |
|                                              | LRC-BART ( $w = 0.9$ )     | 1.423            | 11.199   | 1.959    | 1.00 | 0.00 | 0.318    | 0.433    | 79.3      |
|                                              | LRC-BART (0.9, $w = 0.9$ ) | 1.555            | 13.493   | 2.257    | 1.00 | 0.00 | 0.335    | 0.469    | 52.2      |
| Survival, $n_T = 30$<br>median ratio<br>Sc2  | AFT-CP                     | 20.944           | 46.014   | 51.645   | 0.19 | 0.89 | -0.441   | 0.526    | —         |
|                                              | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | $> 10^3$ | $> 10^3$ | —         |
|                                              | BART-CP                    | 0.814            | 1.109    | 1.231    | 0.92 | 0.25 | -0.199   | 0.277    | —         |
|                                              | LRC-BART (pooled)          | 1.324            | 1.859    | 1.864    | 0.95 | 0.21 | -0.207   | 0.284    | 29.2      |
|                                              | LRC-BART (100)             | 1.329            | 1.865    | 1.875    | 0.95 | 0.20 | -0.208   | 0.284    | 29.1      |
|                                              | LRC-BART (0.9)             | 1.511            | 3.461    | 2.055    | 0.97 | 0.12 | -0.180   | 0.279    | 19.3      |
|                                              | LRC-BART ( $w = 0.9$ )     | 4.243            | 28.669   | 5.305    | 1.00 | 0.00 | 0.179    | 0.537    | 28.9      |
|                                              | LRC-BART (0.9, $w = 0.9$ ) | 4.544            | 33.228   | 6.324    | 1.00 | 0.00 | 0.193    | 0.446    | 19.0      |
| Survival, $n_T = 30$<br>RMST ratio<br>Sc1    | AFT-CP                     | 0.036            | 0.178    | 0.202    | 0.90 | 0.16 | 0.079    | 0.187    | —         |
|                                              | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.311    | 0.355    | —         |
|                                              | BART-CP                    | -0.111           | 0.176    | 0.186    | 0.92 | 0.02 | -0.021   | 0.089    | —         |
|                                              | LRC-BART (pooled)          | -0.046           | 0.192    | 0.157    | 0.97 | 0.02 | 0.002    | 0.085    | 80.2      |
|                                              | LRC-BART (100)             | -0.046           | 0.192    | 0.157    | 0.97 | 0.02 | 0.002    | 0.084    | 79.0      |
|                                              | LRC-BART (0.9)             | -0.040           | 0.218    | 0.156    | 0.98 | 0.01 | 0.002    | 0.085    | 51.8      |
|                                              | LRC-BART ( $w = 0.9$ )     | 0.080            | 0.998    | 0.200    | 1.00 | 0.00 | 0.020    | 0.089    | 79.3      |
|                                              | LRC-BART (0.9, $w = 0.9$ ) | 0.086            | 1.077    | 0.204    | 1.00 | 0.00 | 0.020    | 0.090    | 52.2      |
| Survival, $n_T = 30$<br>RMST ratio<br>Sc2    | AFT-CP                     | 1.216            | 0.522    | 1.336    | 0.13 | 0.94 | -0.518   | 0.559    | —         |
|                                              | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.017    | 0.201    | —         |
|                                              | BART-CP                    | 0.149            | 0.271    | 0.286    | 0.95 | 0.14 | -0.219   | 0.260    | —         |
|                                              | LRC-BART (pooled)          | 0.234            | 0.297    | 0.339    | 0.94 | 0.19 | -0.210   | 0.251    | 29.2      |
|                                              | LRC-BART (100)             | 0.235            | 0.297    | 0.340    | 0.94 | 0.19 | -0.211   | 0.251    | 29.1      |
|                                              | LRC-BART (0.9)             | 0.256            | 0.452    | 0.358    | 0.97 | 0.13 | -0.207   | 0.248    | 19.3      |
|                                              | LRC-BART ( $w = 0.9$ )     | 0.527            | 2.689    | 0.643    | 1.00 | 0.00 | -0.185   | 0.228    | 28.9      |
|                                              | LRC-BART (0.9, $w = 0.9$ ) | 0.560            | 3.085    | 0.731    | 1.00 | 0.00 | -0.184   | 0.228    | 19.0      |
| Survival, $n_T = 200$<br>median ratio<br>Sc1 | AFT-CP                     | 0.168            | 0.470    | 0.480    | 0.95 | 0.23 | 0.002    | 0.098    | —         |
|                                              | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | $> 10^3$ | $> 10^3$ | —         |
|                                              | BART-CP                    | 0.108            | 0.273    | 0.286    | 0.95 | 0.57 | -0.026   | 0.058    | —         |
|                                              | LRC-BART (pooled)          | 0.076            | 0.272    | 0.262    | 0.97 | 0.49 | -0.018   | 0.055    | 169.5     |
|                                              | LRC-BART (100)             | 0.108            | 0.565    | 0.284    | 1.00 | 0.17 | -0.008   | 0.055    | 81.7      |
|                                              | LRC-BART (0.9)             | 0.087            | 0.350    | 0.268    | 0.99 | 0.34 | -0.014   | 0.055    | 114.0     |
|                                              | LRC-BART ( $w = 0.9$ )     | 1.492            | 15.408   | 4.227    | 1.00 | 0.00 | 0.318    | 0.402    | 81.8      |

continued on the next page

Web Table S6, continued

| Design                                       | Method                     | Treatment effect |          |          |      |      | Control  |          |           |
|----------------------------------------------|----------------------------|------------------|----------|----------|------|------|----------|----------|-----------|
|                                              |                            | Bias             | SD       | RMSE     | Cov  | Pow  | Bias     | RMSE     | Prior ESS |
| Survival, $n_T = 200$<br>median ratio<br>Sc2 | LRC-BART (0.9, $w = 0.9$ ) | 1.302            | 10.341   | 1.607    | 1.00 | 0.00 | 0.312    | 0.428    | 113.7     |
|                                              | AFT-CP                     | 9.478            | 5.162    | 10.850   | 0.01 | 1.00 | -0.407   | 0.419    | —         |
|                                              | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | $> 10^3$ | $> 10^3$ | —         |
|                                              | BART-CP                    | 1.136            | 0.778    | 1.359    | 0.60 | 0.82 | -0.189   | 0.214    | —         |
|                                              | LRC-BART (pooled)          | 1.195            | 0.851    | 1.409    | 0.64 | 0.82 | -0.196   | 0.219    | 35.9      |
|                                              | LRC-BART (100)             | 1.193            | 0.856    | 1.410    | 0.65 | 0.81 | -0.196   | 0.219    | 35.9      |
|                                              | LRC-BART (0.9)             | 1.336            | 2.306    | 1.562    | 0.82 | 0.57 | -0.181   | 0.207    | 23.9      |
|                                              | LRC-BART ( $w = 0.9$ )     | 3.935            | 26.373   | 4.611    | 1.00 | 0.00 | 0.142    | 0.396    | 36.2      |
|                                              | LRC-BART (0.9, $w = 0.9$ ) | 4.380            | 34.059   | 7.322    | 1.00 | 0.00 | 0.158    | 0.379    | 23.8      |
| Survival, $n_T = 200$<br>RMST ratio<br>Sc1   | AFT-CP                     | -0.023           | 0.101    | 0.098    | 0.94 | 0.15 | 0.074    | 0.114    | —         |
|                                              | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.305    | 0.317    | —         |
|                                              | BART-CP                    | -0.001           | 0.084    | 0.082    | 0.97 | 0.32 | -0.021   | 0.061    | —         |
|                                              | LRC-BART (pooled)          | -0.024           | 0.078    | 0.076    | 0.95 | 0.25 | 0.001    | 0.054    | 169.5     |
|                                              | LRC-BART (100)             | -0.020           | 0.121    | 0.076    | 0.99 | 0.09 | 0.003    | 0.054    | 81.7      |
|                                              | LRC-BART (0.9)             | -0.022           | 0.094    | 0.076    | 0.99 | 0.17 | 0.002    | 0.054    | 114.0     |
|                                              | LRC-BART ( $w = 0.9$ )     | 0.109            | 1.179    | 0.275    | 1.00 | 0.00 | 0.021    | 0.057    | 81.8      |
|                                              | LRC-BART (0.9, $w = 0.9$ ) | 0.097            | 0.860    | 0.138    | 1.00 | 0.00 | 0.019    | 0.057    | 113.7     |
| Survival, $n_T = 200$<br>RMST ratio<br>Sc2   | AFT-CP                     | 1.164            | 0.428    | 1.244    | 0.01 | 1.00 | -0.535   | 0.550    | —         |
|                                              | HierAFT                    | $> 10^3$         | $> 10^3$ | $> 10^3$ | 1.00 | 0.00 | 0.001    | 0.101    | —         |
|                                              | BART-CP                    | 0.305            | 0.206    | 0.361    | 0.61 | 0.77 | -0.233   | 0.261    | —         |
|                                              | LRC-BART (pooled)          | 0.284            | 0.206    | 0.336    | 0.70 | 0.74 | -0.224   | 0.250    | 35.9      |
|                                              | LRC-BART (100)             | 0.284            | 0.207    | 0.336    | 0.69 | 0.73 | -0.224   | 0.250    | 35.9      |
|                                              | LRC-BART (0.9)             | 0.301            | 0.350    | 0.353    | 0.85 | 0.50 | -0.222   | 0.248    | 23.9      |
|                                              | LRC-BART ( $w = 0.9$ )     | 0.562            | 2.419    | 0.633    | 1.00 | 0.00 | -0.200   | 0.228    | 36.2      |
|                                              | LRC-BART (0.9, $w = 0.9$ ) | 0.585            | 2.496    | 0.662    | 1.00 | 0.00 | -0.198   | 0.226    | 23.8      |